

MATH 540: ALGEBRAIC TOPOLOGY

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The following notes were taken during the Algebraic Topology course (MATH 540) at the University of Southern California in Fall 2023, taught by Professor Joseph Helfer. The textbook used was Hatcher's *Algebraic Topology*.

These notes may contain mistakes. Some of the exercises and statements are not provided with a proof. Please use them with caution and verify the information as needed.

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1 Categories

Definition 1.1. A **directed graph** is a quadruple (O, A, S, T) where O stands for the set of vertices, A stands for the set of the (names of) edges, $S, T : A \rightarrow O$ that describes where each edge $a \in A$ starts from ($S(a)$) and ends at ($T(a)$).

Definition 1.2. A **category** $\mathcal{C}$ is a pair $\mathcal{C} = (G, \circ)$ where G is a directed graph (O, A, S, T) and $\circ$ is a function assigning each pair of $f, g \in A$ s.t. $t(f) = s(g)$ a **composite** $g \circ f \in A$ with $s(g \circ f) = s(f)$ and $t(g \circ f) = t(g)$.

We have the following notations and terminologies for categories:

- O is called the **object** of $\mathcal{C}$, denoted as $Ob(\mathcal{C})$.
- $X \in \mathcal{C}$ if $X \in Ob(\mathcal{C})$.
- A is called the **morphisms** of $\mathcal{C}$, denoted as $Ar(\mathcal{C})$.
- Any $f \in Ar(\mathcal{C})$ with $S(f) = X, T(f) = Y$ is denoted as $f : X \rightarrow Y$; X is called the **domain** and Y is called the **codomain** of f .
- According to the definition, $\circ$ takes a pair of morphisms $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ and returns $g \circ f : X \rightarrow Z$.
- $\mathcal{C}(X, Y) = \{f : X \rightarrow Y \mid f \in Ar(\mathcal{C})\}$.
- As a result, $\circ$ gives a map

$$\mathcal{C}(X, Y) \times \mathcal{C}(Y, Z) \rightarrow \mathcal{C}(X, Z)$$

where $f \times g$ is sent to $g \circ f$.

- $g \circ f$ is often written as gf .

There are two more axioms that $\circ$ has to satisfy to make $\mathcal{C}$ a category:

- (i) (Associativity) $\forall h : W \rightarrow X, g : X \rightarrow Y, f : Y \rightarrow Z, hgf = h(gf) = (hg)f$.
- (ii) (Identities) $\forall Y \in \mathcal{C}, \exists 1_Y : Y \rightarrow Y$ s.t. $\forall f : X \rightarrow Y, g : Y \rightarrow Z, 1_Y f = f, g 1_Y = g$.

Corollary 1.3. $\forall X \in \mathcal{C}, 1_X$ is unique.

Exercise 1.4. Any monoid is the same thing as a category with only one object.

Proof. For any monoid M , consider the category $\mathcal{M}$ with only one object $*$.

Let $\mathcal{M}(*, *) = M$, and the composition of $a, b : * \rightarrow *$ is defined as the product of a and b on M , $ab : * \rightarrow *$.

Conversely, given any one-object category $\mathcal{C}$ and $X \in \mathcal{C}$, it is routine to verify that $\mathcal{C}(X, X)$ with the composition operation forms a monoid. $\square$

Example 1. Some examples for categories:

- (i) **Set** is a category with its objects sets and the morphisms between any two sets a function.
- (ii) **Grp** is a category with its objects groups and the morphisms between any two groups a homomorphism.
- (iii) **Top** is a category with its objects topological spaces and the morphisms between any two topological spaces a continuous function.
- (iv) **Vect_ℝ^{fd}** is a category with its objects finite-dimensional real vector spaces and the morphisms between any two vector spaces a linear map.

It is routine to verify that the above structures are indeed categories by checking the axioms (associativity and identity).

Remark 1. Define $\mathcal{U}$ to be the set of all sets under "usual" set theoretic operations. It does not contain all sets, since for example $\mathcal{U} \notin \mathcal{U}$ and $\mathcal{P}(\mathcal{U}) \notin \mathcal{U}$.

Whenever we mention **Set** or any other categories whose elements are sets, we always assume their elements all come from $\mathcal{U}$.

A set or structure is called **small** if it is an element of $\mathcal{U}$, otherwise it is called **large**. Many categories are large but there are also small ones (e.g. the one-object category).

2 Isomorphisms, Products, Initial and Terminal Objects

Definition 2.1. Let $\mathcal{C}$ be a category. $f : X \rightarrow Y$ in $\text{Ar}(\mathcal{C})$ is an **isomorphism** if $\exists g : Y \rightarrow X$ in $\text{Ar}(\mathcal{C})$ s.t. $gf = 1_X$, $fg = 1_Y$.

X and Y are said to be **isomorphic** and denoted as $X \cong Y$.

Exercise 2.2. (i) Inverse of $f \in \text{Ar}(\mathcal{C})$ is (at most) unique. It is denoted as f^{-1} .

(ii) If $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are isomorphisms in $\mathcal{C}$, then gf is also an isomorphism.

Proof. (i) Suppose $\exists g, g'$ s.t. $gf = 1_X$, $fg' = 1_Y$, then $gfg' = g = g'$ which proves the uniqueness of g .

(ii) Notice that $f^{-1}g^{-1}$ is the inverse of gf . □

Definition 2.3. Let $\mathcal{C}$ be a category. $X \in \mathcal{C}$ is **terminal** if $\exists ! f : Y \rightarrow X$, $\forall Y \in \mathcal{C}$, denoted as $!_Y : Y \rightarrow X$.

Let X is **initial** if $\exists ! f : X \rightarrow Y$, $\forall Y \in \mathcal{C}$.

Exercise 2.4. If X and X' are both terminals (initials), then $X \cong X'$.

Proof. Let $f : X \rightarrow X'$, $g : X' \rightarrow X$ to be the unique morphisms between the two objects. Consider the composites of them.

fg has to be identical to the identity map $1_{X'}$ and gf has to be the identity map 1_X . Thus, X and X' are isomorphic. □

Exercise 2.5. (i) In **Set**, every one-element (empty) set is terminal (initial).

(ii) In **Top**, every one-element (empty) topological space is terminal (initial).

(iii) In **Grp**, every one-element group is terminal and initial.

Proof. (i) If X is a one-element set, then any function from other sets Y to X would have $f(y) = x$ for all $y \in Y$, so X is terminal.

Similarly, if X is an empty set, then any function from X to Y would be empty, so X is initial.

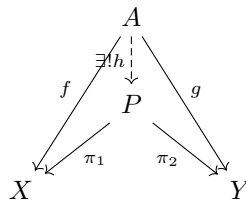
(ii) If X is a one-element topological space, then any continuous function from other topological space Y to X would have $f(y) = x$ for all $y \in Y$, so X is terminal.

Similarly, if X is an empty space, then any continuous function from X to Y would be empty, so X is initial.

(iii) If X is a one-element group, then any homomorphism from other groups Y to X would have $\varphi(y) = e$ for all $y \in Y$, so X is terminal.

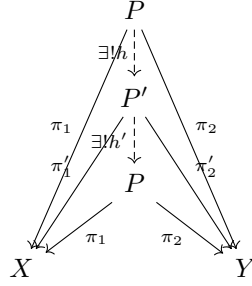
Similarly, if X is any one-element group, then any homomorphism from X to Y would be $\varphi(e) = \tilde{e}$, so X is initial. □

Definition 2.6. Given $X, Y \in \mathcal{C}$, the **product** of X and Y is a triple (P, π_1, π_2) with $P \in \mathcal{C}$, $\pi_1 : P \rightarrow X$, $\pi_2 : P \rightarrow Y$ s.t. given any other (A, f, g) with $A \in \mathcal{C}$, $f : A \rightarrow X$, $g : A \rightarrow Y$, $\exists ! h : A \rightarrow P$ s.t. $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$.



Or in other words, the two triangles (or the diagram) commute.

Theorem 2.7. *If (P, π_1, π_2) and (P', π'_1, π'_2) are both products of X and Y , then $P \cong P'$.*



Proof. By the universal property of product, we have a unique $h : P \rightarrow P'$ s.t. $\pi_1 = \pi'_1 \circ h$ and $\pi_2 = \pi'_2 \circ h$. Similarly, we have a unique $h' : P' \rightarrow P$ s.t. $\pi'_1 = \pi_1 \circ h'$ and $\pi'_2 = \pi_2 \circ h'$.

Thus, we have $\pi_1 = \pi_1 \circ h' \circ h$, indicating that $h' \circ h = 1_P$. Similarly, $h \circ h' = 1_{P'}$. By the definition of isomorphism, we have $P \cong P'$. □

- Exercise 2.8.** (i) Given $X, Y \in \mathbf{Set}$, $X \times Y$ with the projection maps form a product $(X \times Y, \pi_1, \pi_2)$ of X and Y .
 (ii) Given $X, Y \in \mathbf{Top}$, $X \times Y$ with the product topology and projection maps form a product $(X \times Y, \pi_1, \pi_2)$ of X and Y .
 (iii) Given $X, Y \in \mathbf{Grp}$, $X \times Y$ with the projection homomorphisms $\pi_1 : X \times Y \rightarrow X$, $\pi_2 : X \times Y \rightarrow Y$ form a product $(X \times Y, \pi_1, \pi_2)$ of X and Y .

Proof. We are just going to prove (i) and the other two are analogous.

Let $\pi_1 : X \times Y \rightarrow X$ be $\pi_1(x, y) = x$ and $\pi_2 : X \times Y \rightarrow Y$ be $\pi_2(x, y) = y$. Then $\forall f : A \rightarrow X$ and $g : A \rightarrow Y$, let $h : A \rightarrow X \times Y$ with $h(t) = (f(t), g(t))$, then we have $f = \pi_1 \circ h$ and $g = \pi_2 \circ h$. □

Space of The Day ($\mathbb{R}^n$ and S^n)

Remark 2. *The spaces for today are $\mathbb{R}^n$ and S^n .*

- $\mathbb{R}$ is the unique complete Archimedean ordered field. We usually use order topology or metric topology on $\mathbb{R}$.
- On the other hand, we usually use product or metric topology on $\mathbb{R}^n$.
- S^n is defined as

$$S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}.$$

- On the other hand, D^n is defined as

$$D^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}.$$

- We often use subspace topology on S^n and D^n .

Exercise 2.9. S^n is homeomorphic to

- The quotient D^n / S^{n-1} where $x \sim y \Leftrightarrow x = y$ or $x, y \in S^{n-1}$.
- The gluing $D^n \cup_{S^{n-1}} D^n = D^n \sqcup D^n / \sim$, where $(i, x) \sim (j, y) \Leftrightarrow x = y$ and $i = j$ or $x, y \in S^{n-1}$.
- The one-point compactification of $\mathbb{R}^n$, $X = \mathbb{R}^n \cup \{\infty\}$. $U \subset X$ is open if $U \subset \mathbb{R}^n$ is open or $U = X - C$ where $C \subset \mathbb{R}^n$ is compact.

Space of The Day (Manifold)

Definition 2.10. A second-countable (basis is countable) Hausdorff space X is an **n -manifold** if $\exists$ open cover $X = \bigcup_{U \in \mathcal{U}} U$ s.t. each U is homeomorphic to an open subset of $\mathbb{R}^n$.

Theorem 2.11 (Whitney Embedding Theorem). *Every manifold is homeomorphic to a subset of $\mathbb{R}^n$.*

Definition 2.12. An ***n*-chart** on a space X is (U, φ) s.t. U is open in X and $\varphi : U \rightarrow V$ is a homeomorphism, where V is an open subset of $\mathbb{R}^n$. An **atlas** $\mathcal{A}$ on X is a set of charts s.t. $X = \bigcup_{(U, \varphi) \in \mathcal{A}} U$.

Corollary 2.13. X is a manifold iff it admits an atlas.

Definition 2.14. Given two charts (U_1, φ_1) and (U_2, φ_2) , the associated **transition map** is the homeomorphism:

$$\varphi_1(U_1 \cap U_2) \rightarrow \varphi_2(U_1 \cap U_2)$$

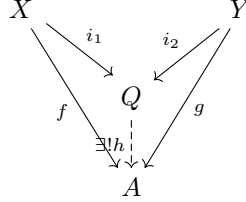
given by $\varphi_2|_{U_1 \cap U_2} \circ (\varphi_1|_{U_1 \cap U_2})^{-1}$.

Definition 2.15. A C^k -**structure** on a manifold X is an atlas $\mathcal{A}$ for which the transition maps are all C^k -**differentiable**. (If $k = \infty$, it is called **smooth**.) A C^k (**smooth**) **manifold** is a pair $(X, \mathcal{A})$, with X a manifold and $\mathcal{A}$ a C^k (**smooth**) atlas.

Exercise 2.16. S^n is a smooth manifold.

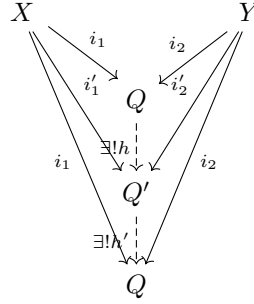
3 Coproducts and Duality

Definition 3.1. Let $\mathcal{C}$ be a category. Given $X, Y \in \mathcal{C}$, a **coproduct** of X and Y is a triple (Q, i_1, i_2) with $Q \in \mathcal{C}$, $i_1 : X \rightarrow Q$, $i_2 : Y \rightarrow Q$ s.t. given any other (A, f, g) with $A \in \mathcal{C}$, $f : X \rightarrow A$, $g : Y \rightarrow A$, $\exists! h : Q \rightarrow A$ s.t. $h \circ i_1 = f$ and $h \circ i_2 = g$.



Or in other words, the two triangles (or the diagram) commutes. h is denoted as $[f, g]$.

Theorem 3.2. If (Q, i_1, i_2) , (Q', i'_1, i'_2) are both coproducts of X and Y , then $Q \cong Q'$.



Proof. By the property of coproduct, we have a unique $h : Q \rightarrow Q'$ and a unique $h' : Q' \rightarrow Q$ s.t. $i_1 = h' \circ h \circ i_1$ so $h' \circ h = 1_Q$. Similarly, $h \circ h' = 1_{Q'}$. Thus, $Q \cong Q'$. $\square$

Exercise 3.3. (i) Given $X, Y \in \mathbf{Set}$, $X \sqcup Y = \{0\} \times X \cup \{1\} \times Y$ with inclusions i_1, i_2 give a coproduct $(X \sqcup Y, i_1, i_2)$ of X and Y in $\mathbf{Set}$.

(ii) Same in $\mathbf{Top}$ with $X \sqcup Y$ under the disjoint union topology.

(iii) In $\mathbf{Ab}$ (or $\mathbf{Vect}_{\mathbb{R}}$), given $X, Y \in \mathbf{Ab}$, the product $X \times Y$ with $i_1 : X \rightarrow X \times Y$ and $i_2 : Y \rightarrow X \times Y$ give a coproduct $(X \times Y, i_1, i_2)$ in $\mathbf{Ab}$.

Proof. We are just going to prove (i) and the other two are analogous.

Given any $f : X \rightarrow A$ and $g : Y \rightarrow A$, let $h : X \sqcup Y \rightarrow A$ be $h((i, z)) = f(z)$ if $i = 0$ and $h((i, z)) = g(z)$ if $i = 1$, then $f(z) = h((i, z)) = h(i_1(z))$ and $g(z) = h((i, z)) = h(i_2(z))$, so $f = h \circ i_1$ and $g = h \circ i_2$. $\square$

The coproduct for $\mathbf{Grp}$, on the other hand, is not that trivial. To find it, we have the following definition:

Definition 3.4. Let $G, H \in \mathbf{Grp}$. The **free product** $G * H$ is defined as the set by

(i) A **word** in $G * H$ is a finite sequence $s_1 \dots s_n$ with $s_i \in G$ or $s_i \in H$ (assume G and H are disjoint).

(ii) An **elementary reduction** of a word $s_1 \dots s_n = s'_1 \dots s'_n$ obtained by either replacing a pair $s_i s_{i+1}$ where s_i and s_{i+1} both in G or H by their product, or removing an identity element $s_i = e_G$ or e_H .

(iii) A word w is **reduced** if no more elementary reduction can be done on it.

$G * H$ is then the set of all reduced words, with $w_1 \cdot w_2$ being the reduced form of $\overline{w_1 w_2}$.

Corollary 3.5. For any word w , $\exists! w'$ reduced that can be obtained by applying consecutive elementary reduction on w .

Proof. To prove that every word w has a reduced form w' , we induct on the length of w . If w is empty then it is clearly reduced. Otherwise, assume the claim holds for n and let w have length $n + 1$. If $s_{n+1} = e_G$ or

e_H then clearly w has the same reduced word as $s_1 \dots s_n$. Let v' be the reduced form of $s_1 s_2 \dots s_n$, suppose $s_{n+1} \in G$, then if v' ends with an element in H , then $v' * s_{n+1}$ is reduced word. Otherwise, $v' * s_{n+1}$ can be reduced by multiplying the last term in v' and s_{n+1} .

To prove that the reduced form for every word w is unique, we also induct on the length of w . If w is empty then clearly w has a unique reduced form. Suppose the claim holds for length n , then suppose w have length $n + 1$. Suppose w has two reduced form, v and v' . We know that $v \cdot (s_{n+1})^{-1}$ and $v' \cdot (s_{n+1})^{-1}$ are both the reduced forms of $s_1 \dots s_n$, therefore $v \cdot (s_{n+1})^{-1} = v' \cdot (s_{n+1})^{-1}$ and therefore $v = v'$, indicating w has a unique reduced form and therefore the claim holds for any finite words. $\square$

Exercise 3.6. (i) $G * H$ is group.

(ii) Show that the maps $i_1 : G \rightarrow G * H$ and $i_2 : H \rightarrow G * H$ by sending e_G and e_H to empty words and $i_1(g) = (g)$ and $i_2(h) = (h)$ form a coproduct $(G * H, i_1, i_2)$ of G and H (first need to show i_1, i_2 are group homomorphisms.)

Proof. (i) $G * H$ apparently admits associativity by definition. It is closed since the concatenation of any two finite sequence is always finite and so is its reduced form. The identity in $G * H$ is the empty word and the inverse of $s_1 s_2 \dots s_n$ is $(s_n)^{-1} (s_{n-1})^{-1} \dots (s_1)^{-1}$, which is also an element in $G * H$. Therefore, $G * H$ is a group.

(ii) i_1 is a group homomorphism (and so is i_2 by an analogous argument) since

$$\begin{aligned} i_1(g)i_1(h) &= gh \\ &= i_1(gh), \end{aligned}$$

and $i_1(gh) = e_{G*H}$ which is the empty word if $h = g^{-1}$. For any group homomorphism $\phi_1 : X \rightarrow A$ and $\phi_2 : Y \rightarrow A$, we define $\varphi : G * H \rightarrow A$ s.t. $\varphi(s_1 \dots s_n) = \phi_{i_1}(s_1) \dots \phi_{i_n}(s_n)$ where $i_j = 1$ if $s_j \in G$ and $i_j = 2$ if $s_j \in H$.

Apparently, $\varphi \circ i_1 = \phi_1$ and $\varphi \circ i_2 = \phi_2$. It remains to show that φ is a group homomorphism. Suppose $w = s_1 \dots s_n$ and $w' = r_1 \dots r_m$ are two reduced words in $G * H$. Let $v = t_1 \dots t_k$ be the reduced form of $w * w'$, then we have

$$\begin{aligned} \varphi(w)\varphi(w') &= \phi_{i_1}(s_1) \dots \phi_{i_n}(s_n) \phi_{j_1}(r_1) \dots \phi_{j_m}(r_m) \\ &= \varphi(v) \end{aligned}$$

by taking $\phi_{i_n}(s_n) \phi_{j_1}(r_1) = \phi_{i_n}(s_n r_1)$ if $i_n = j_1$, and following the same process for i_{n-1} and j_2 , etc., until $s_{n+1-k} r_k \neq e_G$ or e_H . Since the reduced form of $w * w'$ is equivalent to $w * w'$, we conclude that φ is a group homomorphism and therefore $(G * H, i_1, i_2)$ form a coproduct of G and H . $\square$

Definition 3.7. Let $\mathcal{C}$ be a category. Its **dual category** $\mathcal{C}^{op}$ is defined as follows:

- (i) $\text{Ob}(\mathcal{C}^{op}) = \text{Ob}(\mathcal{C})$.
- (ii) $\forall X, Y \in \mathcal{C}$, and $f \in \mathcal{C}(X, Y)$, $\exists f \in \mathcal{C}^{op}(Y, X)$. These are the only maps in $\mathcal{C}^{op}(Y, X)$.
- (iii) $\forall f : X \rightarrow Y$ and $g : Y \rightarrow Z$ in $\mathcal{C}^{op}(X, Y)$ and $\mathcal{C}^{op}(Y, Z)$, $g \circ_{\mathcal{C}^{op}} f = f \circ_{\mathcal{C}} g \in \mathcal{C}(Z, X)$.

Exercise 3.8. (i) $\mathcal{C}^{op}$ is a category.

- (ii) Show that $X \in \mathcal{C}$ is terminal iff X is initial in $\mathcal{C}^{op}$ and $X \in \mathcal{C}$ is initial iff X is terminal in $\mathcal{C}^{op}$.
- (iii) $\forall X, Y \in \mathcal{C}$, $P \in \mathcal{C}$ is a product of X and Y iff P is a coproduct of X and Y in $\mathcal{C}^{op}$. $\forall X, Y \in \mathcal{C}$, $Q \in \mathcal{C}$ is a coproduct of X and Y iff Q is a product of X and Y in $\mathcal{C}^{op}$.

Proof. (i) For any $X \in \mathcal{C}^{op}$, we have $1_X : X \rightarrow X \in \mathcal{C}^{op}(X, X)$ since $1_X \in \mathcal{C}(X, X)$. Apparently, $1_X \circ f = f$ in $\mathcal{C}^{op}$ since $f \circ 1_X = f$ in $\mathcal{C}$. Similarly, $f \circ 1_X = f$ in $\mathcal{C}^{op}$ since $1_X \circ f = f$ in $\mathcal{C}$. Therefore, every object

in $\mathcal{C}^{op}$ has an identity. $\mathcal{C}^{op}$ also holds associativity since morphisms in $\mathcal{C}$ holds associativity, and every morphism in $\mathcal{C}^{op}$ is also a morphism in $\mathcal{C}$.

- (ii) If $X \in \mathcal{C}$ is terminal, then $\forall W \in \mathcal{C}, \exists ! f : W \rightarrow X \in \mathcal{C}(W, X)$. Thus, $\forall W \in \mathcal{C}^{op}, \exists ! f : X \rightarrow W \in \mathcal{C}^{op}(X, W)$. Therefore, X is initial in $\mathcal{C}^{op}$. Similarly, if X is initial in $\mathcal{C}^{op}$, then $\forall Y \in \mathcal{C}^{op}, \exists ! f : X \rightarrow Y \in \mathcal{C}^{op}(X, Y)$. By definition, we have $\exists ! f : Y \rightarrow X \in \mathcal{C}(Y, X)$. Therefore, X is terminal in $\mathcal{C}$.

The proof for X initial in $\mathcal{C} \Leftrightarrow X$ terminal in $\mathcal{C}^{op}$ is analogous.

- (iii) Suppose (P, π_1, π_2) is a product of X and Y in $\mathcal{C}$. We have that $\forall A \in \mathcal{C}, h : A \rightarrow X, k : A \rightarrow Y, \exists ! l : A \rightarrow P$ s.t. $\pi_1 \circ l = h, \pi_2 \circ l = k$. By definition, we have $h \in \mathcal{C}^{op}(X, A), k \in \mathcal{C}^{op}(Y, A), \pi_1 \in \mathcal{C}^{op}(X, P), \pi_2 \in \mathcal{C}^{op}(Y, P)$, and $l \in \mathcal{C}^{op}(P, A)$. Additionally, $h = l \circ \pi_1$ and $k = l \circ \pi_2$. Therefore, (P, π_1, π_2) form a coproduct of X and Y . Similarly, suppose (P, π_1, π_2) is a coproduct of X and Y in $\mathcal{C}^{op}$, then for any $A \in \mathcal{C}^{op}, h : X \rightarrow A, k : Y \rightarrow A, \exists ! l : P \rightarrow A$ s.t. $h = l \circ \pi_1$ and $k = l \circ \pi_2$. By definition, we have $h \in \mathcal{C}(A, X), k \in \mathcal{C}(A, Y), \pi_1 \in \mathcal{C}(P, X), \pi_2 \in \mathcal{C}(P, Y)$, and $l \in \mathcal{C}(A, P)$. Additionally, $h = \pi_1 \circ l$ and $k = \pi_2 \circ l$. Therefore, (P, π_1, π_2) is a product of X and Y in $\mathcal{C}$.

The proof for P coproduct of X and Y in $\mathcal{C} \Leftrightarrow P$ product of X and Y in $\mathcal{C}^{op}$ is analogous. □

Space of The Day (n -torus)

Definition 3.9. The **n -torus** is the space $\mathbb{T}^n = (S^1)^n$ (with the product topology).

Exercise 3.10. $\mathbb{T}^n$ is an n -manifold.

Exercise 3.11. $\mathbb{T}^n$ is homeomorphic to the quotient by $\mathbb{R}^n$ by the equivalence relation defined by $x \sim y$ iff $\exists n \in \mathbb{Z}^n$ s.t. $x = y + n$.

Exercise 3.12. $\mathbb{T}^n$ is homeomorphic to the quotient $[0, 1]^n / \sim$ where $(x_1, \dots, x_n) \sim (y_1, \dots, y_n)$ iff $x_i = y_i$ or $x_i = 1 - y_i$ for any $1 \leq i \leq n$.

Exercise 3.13. Given any action $G \times X \rightarrow X$ on a topological space X by G , the quotient X/G is defined as $X / \sim$ where $x \sim y$ iff $\exists g \in G$ s.t. $x = gy$.

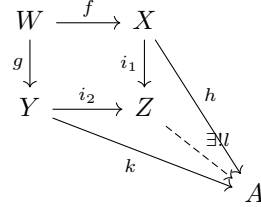
Show that the map $\mathbb{Z}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $(n, r) \rightarrow n + r$ defines a group action. The quotient is homeomorphic to $\mathbb{T}^n$.

4 Pushouts and Pullbacks

Definition 4.1. Let $\mathcal{C}$ be a category, (f, g) be a pair of arrows with a common source. A triple (Z, i_1, i_2) with $i_1 : X \rightarrow Z$, $i_2 : Y \rightarrow Z$ is called a **pushout** of f and g if it satisfies:

- (i) $i_1 \circ f = i_2 \circ g$.
- (ii) Given any other triple (A, h, k) with $h : X \rightarrow A$, $k : Y \rightarrow A$, $h \circ f = k \circ g$, $\exists ! l : Z \rightarrow A$ a.t. $h = l \circ i_1$ and $k = l \circ i_2$.

In other words, a pushout ensures that there is always a unique l to make the following diagram commute:

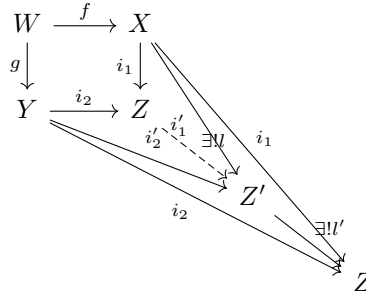


Exercise 4.2. If W is initial, then a pushout of f and g is a coproduct of X and Y .

Proof. Since W is initial, there is only one morphism from W to Z . Therefore, $i_1 \circ f = i_2 \circ g$ always holds. A triple (Z, i_1, i_2) satisfying the second condition is equivalent to the triple being a coproduct of X and Y . $\square$

Exercise 4.3. Pushouts are unique up to isomorphism.

Proof. It is routine to check that l is an isomorphism from Z to Z' (and vice versa).



$\square$

Exercise 4.4. Let $W, X, Y \in \mathbf{Set}$. $f : W \rightarrow X$, $g : W \rightarrow Y$ injective. Let Z be the quotient $(X \sqcup Y) / \sim$ with $\sim$ the least equivalent relation s.t.

$$(1, f(w)) \sim (2, g(w)), \forall w \in W.$$

Z is often called the *gluing* of X and Y along $gf^{-1} : f(W) \rightarrow g(W)$. Define $i_1 : X \rightarrow Z$ be $q \circ i_X$ and $i_2 : Y \rightarrow Z$ be $q \circ i_Y$, then (Z, i_1, i_2) is a pushout of f and g in $\mathbf{Set}$.

Proof. By definition, $i_1(x) = (1, f(w))$ if $x = f(w)$, and $i_2(y) = (2, g(w))$ if $y = g(w)$. Due to the equivalence relation, we always have $i_1 \circ f = i_2 \circ g$. Let $l : Z \rightarrow A$ be $l((i, z)) = h(z)$ if $i = 1$ and $l((i, z)) = k(z)$ if $i = 2$, then l makes the diagram commutes.

Notice that since $i_1 \circ f = i_2 \circ g$, we only need to consider (Z, i_1, i_2) as a coproduct of X and Y . $\square$

Exercise 4.5. The pushout of f and g if they are not injective is still $(X \sqcup Y) / \sim$.

Exercise 4.6. Show that the pushouts in $\mathbf{Top}$ have the same description as the pushouts in $\mathbf{Set}$.

Theorem 4.7 (The Pasting Lemma). Let $X \in \mathbf{Top}$ and $A \subset X$, $B \subset X$ are both closed or open with $X = A \cup B$, then the square of inclusions is a pushout in $\mathbf{Top}$.

$$\begin{array}{ccc} A \cap B & \hookrightarrow & A \\ \downarrow & & \downarrow \\ B & \hookrightarrow & X \end{array}$$

Exercise 4.8. Let $K, G, H \in \mathbf{Grp}$, $\varphi_1 : K \rightarrow G$, $\varphi_2 : K \rightarrow H$ be group homomorphisms, then we have the coproduct $(G * H, j_1, j_2)$ with $j_1 : G \rightarrow G * H$, $j_2 : H \rightarrow G * H$. Let $N \subset G * H$ to be the normal subgroup generated by $j_1(\varphi_1(k)) \cdot j_2(\varphi_2(k))^{-1}$, $\forall k \in K$.

Define $Z = G * H/N$, let $i_1 = q \circ j_1$, $i_2 = q \circ j_2$, then $(G * H, i_1, i_2)$ is a pushout of φ_1 and φ_2 .

Proof. Let q be the quotient map from $G * H$ to $G * H/N$. We know that $qi_1\varphi_1 = qi_2\varphi_2$ since $i_1\varphi_1(k) \cdot i_2\varphi_2(k)^{-1} \in N$.

We also know that $G * H$ is a coproduct of G and H so for any $A \in \mathbf{Grp}$ with homomorphisms $\phi_1 : G \rightarrow A$ and $\phi_2 : H \rightarrow A$, we have a unique homomorphism $\sigma : G * H \rightarrow A$ to make the diagram commute. Therefore, what we need to show is that there is a homomorphism $\delta : G * H/N \rightarrow A$ s.t. $\delta \circ q = \sigma$, which is equivalent as showing $w \cdot (w')^{-1} \in N \Leftrightarrow \sigma(w) = \sigma(w')$. We then can define $\delta(N) = e_A$ and $\delta(aN) = \sigma(a)$. Notice that if the claim holds then $aN = bN$ tells $\sigma(a)\sigma(b)^{-1} = \delta(aN)\delta(bN)^{-1} = \delta(N) = e$, so what we actually need to show is that N is in the kernel of σ . Notice that if $k \in K$, then

$$\begin{aligned} \sigma((i_1(\varphi_1(k)))(i_2(\varphi_2(k)))^{-1}) &= \sigma(i_1(\varphi_1(k)))\sigma(i_2(\varphi_2(k)))^{-1} \\ &= \phi_1(\varphi_1(k))\phi_2(\varphi_2(k))^{-1} \\ &= e_A \end{aligned}$$

since we have $\phi_1 \circ \varphi_1 = \phi_2 \circ \varphi_2$. Therefore, δ is well-defined and thus $G * H/N$ is a pushout of G and H .

$$\begin{array}{ccccc} K & \xrightarrow{\varphi_1} & G & & \\ \varphi_2 \downarrow & & \downarrow i_1 & \searrow q & \\ H & \xrightarrow{i_2} & G * H & \xrightarrow[\phi_1]{} & G * H/N \\ & \searrow \phi_2 & \searrow \sigma & \searrow \exists! \delta & \\ & & & & A \end{array}$$

□

Definition 4.9. Let $\mathcal{C}$ be a category, (f, g) be a pair of arrows with a common target Z . A triple (P, π_1, π_2) with $\pi_1 : P \rightarrow X$, $\pi_2 : P \rightarrow Y$ is called a **pullback** of f and g if it satisfies:

- (i) $f \circ \pi_1 = g \circ \pi_2$.
- (ii) Given any other triple (A, h, k) with $h : A \rightarrow X$, $k : A \rightarrow Y$, $f \circ h = g \circ k$, $\exists! l : A \rightarrow P$ a.t. $h = \pi_1 \circ l$ and $k = \pi_2 \circ l$.

In other words, a pullout ensures that there is always a unique l to make the following diagram commutes:

$$\begin{array}{ccccc} A & & & & \\ \searrow h & & \searrow \pi_1 & & \\ & P & \xrightarrow{\pi_1} & X & \\ \downarrow k & & \downarrow \pi_2 & & \downarrow f \\ & Y & \xrightarrow{g} & Z & \end{array}$$

Space of The Day (Connected Sum)

Definition 4.10. Given two n -manifold M_1 and M_2 , their **connected sum** $M_1 \# M_2$ is obtained by removing the interior of a small n -ball from each and gluing them together along the resulting boundary.

More precisely, given embeddings $\varphi_i : D^n \rightarrow M_i$, $M_1 \# M_2$ is defined to be the quotient

$$(M_1 - \varphi_1(\text{Int}(D^n))) \sqcup (M_2 - \varphi_2(\text{Int}(D^n))) / \sim$$

where $(1, x) \sim (2, y)$ iff $i = j$ and $x = y$ or $i \neq j$ and $x = \varphi_1(z)$, $y = \varphi_2(z)$ for some $z \in S^{n-1} \subset D^n$.

Exercise 4.11. $M_1 \# M_2$ is also a manifold.

Definition 4.12. The connected sum of g 2-tori, $\sum_g = \mathbb{T}^2 \# \dots \# \mathbb{T}^2$, is called the **orientable surface of genus g** .

Theorem 4.13. Every orientable compact surface is homeomorphic to $\sum_g$ for exactly one g .

5 Functors

Definition 5.1. Let $\mathcal{C}, \mathcal{D}$ be categories. A **functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of:

- (i) A function $F : \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$.
- (ii) A function $F : \mathcal{C}(X, Y) \rightarrow \mathcal{D}(FX, FY)$, which satisfies:
- (iii) $F(1_X) = 1_{FX}$, $\forall X \in \mathcal{C}$,
- (iv) $F(g \circ f) = F(g) \circ Ff$, $\forall f \in \mathcal{C}(X, Y), g \in \mathcal{C}(Y, Z)$.

Example 2. If $\mathcal{C}, \mathcal{D}$ both consist of a single object (so they both represent a monoid), then a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a monoid homomorphism.

Definition 5.2. Given $F : \mathcal{C} \rightarrow \mathcal{D}, G : \mathcal{D} \rightarrow \mathcal{E}$, their composite is the functor $G \circ F : \mathcal{C} \rightarrow \mathcal{E}$ s.t

- (i) $(G \circ F)(X) = G(FX)$, $\forall X \in \mathcal{C}$.
- (ii) $(G \circ F)(f) = G(Ff)$, $\forall f \in \mathcal{C}(X, Y)$.

Definition 5.3. **CAT** is the category of small categories where $\mathbf{CAT}(\mathcal{C}, \mathcal{D})$ is the set of functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and the composition is simply the composition of functors.

Exercise 5.4. Check that $G \circ F$ is a functor and **CAT** is a category.

Definition 5.5. For each of **Top**, **Ab**, **Grp**, **Rng**, etc., $\exists U : \mathcal{C} \rightarrow \mathbf{Set}$ in which $U(X)$ is the underlying set of X , and for each $f : X \rightarrow Y$, $U(f)$ is the underlying function. U is called the **forgetful functor**.

Exercise 5.6. Show that U is a functor.

Exercise 5.7. The followings can be made into functors $F : \mathcal{C} \rightarrow \mathcal{D}$:

- $\mathcal{C} = \mathbf{Top}$ and $\mathcal{D} = \mathbf{Set}$. FX is the set of connected components of X .
- $\mathcal{C} = \mathbf{Set}$ and $\mathcal{D} = \mathbf{Top}$. FX is X with discrete topology.
- $\mathcal{C} = \mathbf{Set}$ and $\mathcal{D} = \mathbf{Top}$. FX is X with indiscrete topology.
- $\mathcal{C} = \mathbf{Rng}$ and $\mathcal{D} = \mathbf{Grp}$. $F(R)$ is $GLn(R)$.
- $\mathcal{C} = \mathbf{Grp}$ and $\mathcal{D} = \mathbf{Ab}$. $F(G)$ is the Abelianization of G (G/G').
- $\mathcal{C} = \mathbf{Graph}$ and $\mathcal{D} = \mathbf{Top}$. $F(G)$ is the geometric realization of G .
- $\mathcal{C} = \mathbf{CAT}$ and $\mathcal{D} = \mathbf{Graph}$. $F(G)$ is the underlying graph of the category $\mathcal{C}$.

Proof. Consider $GLn : \mathbf{Rng} \rightarrow \mathbf{Grp}$. $\forall \varphi : R \rightarrow S$, let $GLn(\varphi) = \varphi^*$.

We define $\varphi^* : GLn_R \rightarrow GLn_S$ by $[r_{ij}] \rightarrow [\varphi(r_{ij})]$, then we have $GLn(1_R) = 1_{GLn(R)}$.

Given $\varphi : R \rightarrow S, \phi : S \rightarrow T$, we define $\phi^* \circ \varphi^* = (\phi \circ \varphi)^*$. Now we verify that φ^* is a group homomorphism.

$$\begin{aligned} \varphi^*(A \cdot B)_{ij} &= \varphi((A \cdot B)_{ij}) \\ &= \varphi\left(\sum A_{ik} B_{kj}\right) \\ &= \sum \varphi(A_{ik}) \varphi(B_{kj}) \\ &= \sum \varphi^*(A)_{ik} \varphi^*(B)_{kj}. \end{aligned}$$

Therefore, $\varphi^*(A \cdot B) = \varphi^*(A) \cdot \varphi^*(B)$ and φ^* is a group homomorphism, so GLn is a functor. $\square$

Space of The Day (Graph)

Definition 5.8. A **graph** G is defined by (G_0, G_1, S, T) , where G_0, G_1 are sets, and $S, T : G_1 \rightarrow G_0$.

Definition 5.9. The **geometric realization** of G is the space $|G| = G_0 \sqcup (G_1 \times [0, 1]) / \sim$, where $\sim$ is the equivalence relation generated by $(s(e), 0) \sim ((e, 0), 1)$ and $(t(e), 0) \sim ((e, 1), 1)$.

6 Natural Transformations

Example 3. Let $\mathcal{C}$ be the terminal category (trivial monoid). A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is an object $FX \in \mathcal{D}$. Therefore, $[\mathcal{C}, \mathcal{D}] \cong \mathcal{D}$.

Let $\mathcal{C}$ be the discrete category with two objects. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a pair of objects $FX, FY \in \mathcal{D}$. Therefore, $[\mathcal{C}, \mathcal{D}] \cong \mathcal{D} \times \mathcal{D}$. In this case, a natural transformation $\alpha : F \rightarrow G$ is a pair of morphisms $\alpha_X, \alpha_Y : FX \rightarrow GX, FY \rightarrow GY$.

Let $\mathcal{C}$ be the category with two objects and one non-identity morphism. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a pair of objects $FX, FY \in \mathcal{D}$ with $Ff : FX \rightarrow FY$. A natural transformation $\alpha : F \rightarrow G$ is a pair of morphisms $\alpha_X, \alpha_Y : FX \rightarrow GX, FY \rightarrow GY$ s.t.

$$\begin{array}{ccc} FX & \xrightarrow{Ff} & FY \\ \downarrow \alpha_X & & \downarrow \alpha_Y \\ GX & \xrightarrow{Gf} & GY \end{array}$$

the square commutes.

Definition 6.1. Given functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a **natural transformation** $\alpha : F \rightarrow G$ consists of morphisms $\alpha_X : FX \rightarrow GX$, for every $X \in \mathcal{C}$, s.t. for any $X, Y \in \mathcal{C}$, $f : X \rightarrow Y$, the above square commutes.

Definition 6.2. Given $F, G, H : \mathcal{C} \rightarrow \mathcal{D}$ and natural transformations $\alpha : F \rightarrow G$ and $\beta : G \rightarrow H$, the composite $\beta \circ \alpha$ is a natural transformation containing $\beta_X \circ \alpha_X$, for any $X \in \mathcal{C}$.

Exercise 6.3. Verify that $\beta \circ \alpha$ is a natural transformation.

Corollary 6.4. $[\mathcal{C}, \mathcal{D}]$ is a category with the above composition of natural transformations.

Exercise 6.5. Check that the followings are natural transformation: $F \rightarrow G$.

- $\mathcal{C} = \mathbf{CRng}$, $\mathcal{D} = \mathbf{Grp}$, $F = GL_n$, $G = \mathbb{R} \setminus \{0\}$, $\alpha = \det$.
- $\mathcal{C} = \mathbf{Top}$, $\mathcal{D} = \mathbf{Top}$, $F = 1_{\mathbf{Top}}$, $G = \text{connected components (with quotient topology)}$, $\alpha = X \rightarrow [X]$.
- $\mathcal{C} = \mathbf{Grp}$, $\mathcal{D} = \mathbf{Grp}$, $F = 1_{\mathbf{Grp}}$, $G = \text{Abelianization}$, $\alpha = G \rightarrow G/G'$.

Space of The Day (CW Complex)

Definition 6.6. **CW complexes**, also named **Cell complexes**, are defined inductively.

A 0-complex is a discrete space, and a 1-complex is a graph. For $n \geq 2$, we have the following pushout square that glues n -disks onto X_n :

$$\begin{array}{ccc} \bigsqcup S^n & \xrightarrow{i} & \bigsqcup D^{n+1} \\ \downarrow [\varphi_\alpha|_{S^n}] & & \downarrow [\varphi_\alpha] \\ X_n & \xrightarrow{i} & X_{n+1} \end{array}$$

The maps $\varphi_\alpha : D^n \rightarrow X_n$ are called the **attaching maps**. X_n is called the **n -skeleton** of X .

Exercise 6.7. For $n \geq 1$, S^n has 2 cell complex structures, one with one 0-cell and one n -cell, and the other with 2 k -cells, for any $0 \leq k \leq n$.

Exercise 6.8. A 2-torus has a cell complex structure with 1 0-cell, 2 1-cells, and 1 2-cell.

7 Diagrams and Limits

Definition 7.1. Fix categories J and $\mathcal{C}$. A **diagram** of shape J (typically a small category) in $\mathcal{C}$ is a functor $D : J \rightarrow \mathcal{C}$. The category of diagrams of shape J in $\mathcal{C}$ is the category $[J, \mathcal{C}]$.

Example 4. • Suppose J is the category with monoid homomorphisms:

$$\begin{array}{ccc} 0 & \xrightarrow{01} & 1 \\ \downarrow 02 & \searrow 03 & \downarrow 13 \\ 2 & \xrightarrow{23} & 3 \end{array}$$

then a diagram of shape J in $\mathcal{C}$ is just a commutative square in $\mathcal{C}$:

$$\begin{array}{ccc} D(0) & \xrightarrow{D(01)} & D(1) \\ \downarrow D(02) & & \downarrow D(13) \\ D(2) & \xrightarrow{D(23)} & D(3) \end{array}$$

• Suppose instead J is like

$$\begin{array}{ccc} & 2 & \xrightarrow{23} 3 \\ 02 \uparrow & & \nearrow 03' \\ 0 & & \\ \downarrow 01 & \searrow 03 & \\ 1 & \xrightarrow{13} & 3 \end{array}$$

Then a diagram of shape J in $\mathcal{C}$ only need to satisfy that $(13) \circ (01) = (03)$ and $(23) \circ (02) = (03')$.

Definition 7.2. Given a diagram D of shape J in $\mathcal{C}$. A **cone** over D is a pair of (X, ν) with $X \in \mathcal{C}$ and ν a natural transformation: $\Delta_X \rightarrow D$, where

$$\Delta_X : J \rightarrow \mathcal{C}, \Delta_X(a) = X \text{ and } \Delta_X(f) = 1_X, \text{ for any } a \in J \text{ and } f \in \text{Ar}(J).$$

Explicitly, ν is a family of morphisms $\nu_a : X \rightarrow D(a)$ for any $s.t.$ the square commutes:

$$\begin{array}{ccc} \Delta_X(a) & \xrightarrow{1_X} & \Delta_X(b) \\ \downarrow \nu_a & & \downarrow \nu_b \\ D(a) & \xrightarrow{D(f)} & D(b) \end{array}$$

for any $a, b \in J$ and $f \in J(a, b)$.

Example 5. • If $J = (* \rightarrow *)$, then $D = (A, B)$ and a cone over D (X, ν) is an object $X \in \mathcal{C}$ with two morphisms $\pi_1 : X \rightarrow A$ and $\pi_2 : X \rightarrow B$.

• If $J = (* \rightarrow * \leftarrow *)$, then $D = (A, B, C)$ with $f : A \rightarrow C$ and $g : B \rightarrow C$. A cone (X, ν) is an object $X \in \mathcal{C}$ with morphisms $\pi_1 : X \rightarrow A$, $\pi_2 : X \rightarrow B$, and $\pi_3 : X \rightarrow C$ s.t. $f \circ \pi_1 = g \circ \pi_2 = \pi_3$.

Definition 7.3. A cone (X, ν) is a **limit cone** ($X = \lim D$) if it satisfies that:

$$\text{given any cone } (X', \nu') \text{ over } D, \exists ! h : X' \rightarrow X \text{ s.t. } \nu' = \nu \circ h.$$

Exercise 7.4. • If $J = \emptyset$, then a limit of a diagram of shape J is a terminal object.

• If $J = (* \rightarrow *)$, then a limit of a diagram of shape J is a product of A and B .

• If $J = (* \rightarrow * \leftarrow *)$, then a limit of a diagram of shape J is a pullback of f and g .

Exercise 7.5. If $(X, \nu), (X', \nu')$ are limits of D , then $X \cong X'$.

Definition 7.6. A **co-cone** over D is a pair (X, ν) where $X \in \mathcal{C}$, ν a transformation $D \rightarrow \Delta_X$.

Definition 7.7. A co-cone (X, ν) is a **colimit cone** ($X = \text{colim } D$) if it satisfies:

given any co-cone (X', ν') over D , $\exists! h : X \rightarrow X'$ s.t. $\nu = \nu' \circ h$.

- Exercise 7.8.**
- If $J = \emptyset$, then a limit of a diagram of shape J is an initial object.
 - If $J = (* \quad *)$, then a colimit of a diagram of shape J is a coproduct of A and B .
 - If $J = (* \rightarrow * \leftarrow *)$, then a colimit of a diagram of shape J is a pushout of f and g .

Example 6. A diagram of a discrete category in $\mathcal{C}$ is a collection of objects $\{X_\alpha\}_{\alpha \in J} \subset \mathcal{C}$. A limit/colimit of such a diagram is called a J -indexed product/coproduct.

Example 7. Given a collection of spaces $\{X_\alpha\}_{\alpha \in J}$, let $\prod X_\alpha$ be the product with the product topology, $\pi_\alpha : \prod X_\alpha \rightarrow X_\alpha$. Then $(\prod X_\alpha, \pi)$ forms a J -indexed product in **Top**.

Similarly, $(\bigsqcup_{\alpha \in J} X_\alpha, i)$ forms a J -indexed coproduct in **Top**.

Space of The Day (Wedge Product)

Definition 7.9. A **pointed space** is a pair (X, x) with $x \in X$, $X \in \mathbf{Top}$.

Definition 7.10. Given two pointed spaces (X, x_0) and (Y, y_0) , the **Wedge product** $X \vee Y$ is obtained by the pushout:

$$\begin{array}{ccc} * & \xrightarrow{x_0} & X \\ \downarrow y_0 & & \downarrow \\ Y & \longrightarrow & X \vee Y \end{array}$$

which is exactly the space $X \sqcup Y / \sim$ with $(1, x_0) \sim (2, y_0)$. The Wedge product is clearly again a pointed space.

Exercise 7.11. Let $\mathbf{Top}_*$ be the category of pointed spaces in which a morphism $f : (X, x_0) \rightarrow (Y, y_0)$ is a continuous map with $f(x_0) = y_0$. Composition of morphisms is the ordinary composition of maps. Show the following:

- $\mathbf{Top}_*$ is a category.
- Show that the natural inclusion $i_1 : (X, x_0) \rightarrow X \vee Y$ and $i_2 : (Y, y_0) \rightarrow X \vee Y$ gives a coproduct of (X, x_0) and (Y, y_0) in $\mathbf{Top}_*$.

8 The Fundamental Group

Definition 8.1. Let $X \in \mathbf{Top}$. A **path** in X from x_0 to x_1 is a continuous map $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x_0$ and $\gamma(1) = x_1$. $\gamma : x_0 \rightsquigarrow x_1$ denotes such a path.

A **loop** in X is $\gamma : x_0 \rightsquigarrow x_0$. The set of loops at x_0 in X is denoted as $\Omega_{x_0}X$.

Exercise 8.2. Fix $b_0 = (1, 0) \in S^1$. Show that there is a bijection between $\Omega_{x_0}X$ and the set of pointed maps $(S^1, b_0) \rightarrow (X, x_0)$.

Proof. Given any pointed map $f : (S^1, b_0) \rightarrow (X, x_0)$, let $\gamma : [0, 1] \rightarrow X$ with $\gamma(t) = f((\cos 2\pi t, \sin 2\pi t))$. Then γ is a loop at x_0 . $\square$

Definition 8.3. Given paths $r_{01} : x_0 \rightsquigarrow x_1$ and $r_{12} : x_1 \rightsquigarrow x_2$, their composite is the path $r_{02} : x_0 \rightsquigarrow x_2$ defined as

$$r_{02}(t) = \begin{cases} r_{01}(2t), & 0 \leq t \leq \frac{1}{2} \\ r_{12}(2t - 1), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

which is continuous directly from the pasting lemma, and thus also a path.

Definition 8.4. Given two maps $f_0, f_1 : A \rightarrow X$ in $\mathbf{Top}$, a **homotopy** $h : f_0 \rightarrow f_1$ is a continuous family of maps $\{h_t : A \rightarrow X \mid 0 \leq t \leq 1\}$ s.t.

- (i) $h_0(a) = f_0(a)$, $h_1(a) = f_1(a)$, for any $a \in A$.
- (ii) The map $[0, 1] \times A \rightarrow X$ defined by $H(t, a) = h_t(a)$ is continuous.

We say that f_0 and f_1 are **homotopic**, denoted as $f_0 \sim f_1$.

Definition 8.5. Given subspace $B \subset A$, assuming that $f_0|_B = f_1|_B$, then a homotopy h from f_0 to f_1 is a **homotopy relative to B** if $h_t|_B = f_0|_B = f_1|_B$ for all $t \in [0, 1]$.

If $B = \{0, 1\}$ then h is called a **homotopy relative to endpoints**.

Definition 8.6. A subset $K \subset \mathbb{R}^n$ is **convex** if whenever $x_0, x_1 \in K$, the set $\{(1-t)x_0 + tx_1 \mid t \in [0, 1]\}$ is a subset of K .

Exercise 8.7. If $K \subset \mathbb{R}^n$ is convex, any two maps $f, g : X \rightarrow K$ are homotopic for any space X .

Proof. Let $h_t(x) = (1-t)f(x) + t(g(x))$. $\square$

Definition 8.8. If X, x_0 is a pointed space, the **fundamental group** $\pi_1(X, x_0)$ is $\Omega_{x_0}X / \sim$, where two loops are equivalent if they are homotopic relative to endpoints.

The group operation is $[\gamma_0] \cdot [\gamma_1] = [\gamma_0 \cdot \gamma_1]$ where the composition is defined in **Definition 8.3**.

Exercise 8.9. The above operation is well defined. (i.e., $\gamma_0 \cdot \gamma_1 \sim \gamma'_0 \cdot \gamma'_1$ if $\gamma_i \sim \gamma'_i$.)

Exercise 8.10. Given $f, g : X \rightarrow Y$, $h : Y \rightarrow Z$. If $f \sim g$ then $hf \sim hg$.

Similarly, if f, g agree on $A \subset X$ and $f \sim g$ relative to A then $hf \sim hg$ relative to A .

In particular, given any path $\gamma : x_0 \rightsquigarrow x_1$, a reparametrization $\gamma' = \gamma \circ \varphi$, where $\varphi : [0, 1] \rightarrow [0, 1]$ is a homeomorphism with $\varphi(0) = 0$ and $\varphi(1) = 1$, satisfies that $\gamma \sim \gamma'$.

Exercise 8.11. Prove the followings:

- (i) The operation in $\pi_1(X, x_0)$ is associative.
- (ii) Prove that given $\gamma : x_0 \rightsquigarrow x_1$, $c_{x_0} \cdot \gamma \sim \gamma \sim \gamma \cdot c_{x_1}$.
- (iii) Prove that given $\gamma : x_0 \rightsquigarrow x_1$, the **reversed path** $\bar{\gamma} : x_1 \rightsquigarrow x_0$ where $\bar{\gamma}(t) = \gamma(1-t)$ satisfies that $\gamma \cdot \bar{\gamma} \sim c_{x_0}$ and $\bar{\gamma} \cdot \gamma \sim c_{x_1}$.
- (iv) Prove that $\pi_1(X, x_0)$ is a group.

Space of The Day (Covering Space)

Definition 8.12. Given $f : X \rightarrow Y$ continuous. An open set $U \subset Y$ is **evenly covered** by f , if $\exists$ homeomorphism $i : f^{-1}(U) \rightarrow U \times I$ (I discrete), s.t. the following graph commutes:

$$\begin{array}{ccc} f^{-1}(U) & \xrightarrow{i} & U \times I \\ & \searrow & \downarrow \pi_1 \\ & f|_{f^{-1}(U)} & U \end{array}$$

In other word, every disjoint component of $f^{-1}(U)$ is homeomorphic to U .

Definition 8.13. f is a **covering map** if Y is a union of open evenly covered sets.

Definition 8.14. A **covering space** of Y is a space X with a covering map $f : X \rightarrow Y$.

Theorem 8.15. (i) $\mathbb{R}$ is a covering space of S^1 .

(ii) $\mathbb{R}^2$ is a covering space of $\mathbb{T}^2$.

(iii) Any cover of a graph is a graph.

(iv) Any cover of a CW-complex is a CW-complex.

(v) Each open cell is evenly covered.

Exercise 8.16. Find a degree d covering function $f : S^1 \rightarrow S^1$ for each $d \geq 1$.

Space of The Day (Contractible Space)

Definition 8.17. $X \in \mathbf{Top}$ is **contractible** if $1_X : X \rightarrow X$ is homotopic to $c_{x_0} : X \rightarrow X$ for some $x_0 \in X$.

Definition 8.18. $K \subset \mathbb{R}^n$ is **star-shaped** if $\exists x_0 \in K$ s.t.

$$\forall x \in K, \{(1-t)x_0 + tx \mid 0 \leq t \leq 1\} \subset K.$$

Exercise 8.19. X convex implies that it is star-shaped, and X is star-shaped implies that it is contractible.

Exercise 8.20. $\pi_1(K, x_0) \cong \mathbf{1}$, $\pi_1(\mathbb{R}^n, x_0) \cong \mathbf{1}$, where $\mathbf{1}$ is the trivial group.

9 The Fundamental Groupoid

Definition 9.1. A **groupoid** is a category where all the morphisms are isomorphisms. We have a category $\mathbf{Gpd}$ for groupoids, which is a (full) subcategory of $\mathbf{Cat}$.

Remark 3. Any one-object groupoid is a group. If $\mathbf{G}$ is a groupoid, then $\mathbf{G}(x, x)$ is a group for any $x \in \mathbf{G}$.

Exercise 9.2. Given $x, y \in \mathbf{G}$ and a morphism $f : x \rightarrow y$, the map $\varphi : \mathbf{G}(x, x) \rightarrow \mathbf{G}(y, y)$ by sending $\varphi(u)$ to $f \circ u \circ f^{-1}$ is a group isomorphism.

Exercise 9.3. If $\mathbf{G}, \mathbf{H}$ are groupoids. $F, G : \mathbf{G} \rightarrow \mathbf{H}$ functors, then any natural transformation $\alpha : F \rightarrow G$ is a natural isomorphism.

Definition 9.4. Given a space X , its **fundamental groupoid** $\Pi_1(X)$ is a groupoid with $Ob \Pi_1(X) = X$ and for $x, y \in \Pi_1(X)$, $\Pi_1(X)(x, y) = \Omega_{x,y}(X) / \sim$ where $\Omega_{x,y}(X)$ is the set of paths $\{\gamma : x \rightsquigarrow y\}$ and $\sim$ is the homotopy equivalence. The composition of morphisms is defined as $[\gamma_{12}] \circ [\gamma_{01}] = [\gamma_{01} \cdot \gamma_{12}]$.

Exercise 9.5. Prove that $\Pi_1(X)$ is a groupoid.

Exercise 9.6. Given a path $\gamma : x_0 \rightarrow x_1$, prove that the map $\varphi : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ with $\varphi([\alpha]) = [\bar{\gamma} \cdot \alpha \cdot \gamma]$ is a group isomorphism.

Corollary 9.7. If X is path-connected, then $\pi_1(X, x)$ is independent of the choice of x up to isomorphism (i.e. $\pi_1(X, x_0) \cong \pi_1(X, x_1)$ for any $x_0, x_1 \in X$). In this case $\pi_1(X)$ is used to denote $\pi_1(X, x)$.

Definition 9.8. A groupoid $\mathbf{G}$ is **connected** if $\mathbf{G}(x, y) \neq \emptyset$ for any $x, y \in \mathbf{G}$. If $\mathbf{G}$ is connected then $\mathbf{G}(x, x)$ is independent of the choice of x up to isomorphism.

Definition 9.9. A space X is **simply connected** if $\Pi_1(X)(x, y)$ has a single element for any $x, y \in X$. In other words, X is path-connected and any two paths γ_1, γ_2 between the same two vertices are homotopic.

Corollary 9.10. X is simply connected iff $\pi_1(X, x_0)$ is trivial for some $x_0 \in X$ and X is path-connected.

Space of The Day (Presentation Complex)

Definition 9.11. Let Γ be a graph with vertices $\Gamma_0 \subset \Gamma$ and edges Γ_1 , with each edge $e \in \Gamma_1$ having $s(e)$, $t(e) \in \Gamma$, and an embedding $\varphi_e : [0, 1] \rightarrow \Gamma$.

A **combinatorial path** in Γ is a sequence of edges $e_1^{\pm 1} \dots e_n^{\pm 1}$, each taken with a sign such that $t(e_i^+) = s(e_{i+1}^+)$ or $t(e_i^+) = t(e_{i+1}^-)$.

This gives rise to a path $\gamma : [0, 1] \rightarrow \Gamma$ with $\gamma(t) = \varphi_{e_i}(nt - i)$ for $t \in [i/n, (i+1)/n]$.

A **combinatorial loop** is defined analogously.

Definition 9.12. Given combinatorial loops $\gamma_1, \dots, \gamma_k$, we can form a 2-simplex by gluing a disk along each loop γ_i . The resulted space is called a **combinatorial 2-simplex**.

Definition 9.13. The **free group** on n generators $\mathbb{F}_n$ is the set of all reduced words on n letters, $a_1, \dots, a_n$, and their formal inverses $\bar{a}_1, \dots, \bar{a}_n$, with multiplication given by concatenation (and reduction).

It can alternatively be defined as $\mathbb{Z} * \mathbb{Z} \dots * \mathbb{Z}$.

Exercise 9.14. $\mathbb{F}_n$ is defined by the universal property that there is an n -letter subset S with inclusion map $i : S \rightarrow \mathbb{F}_n$, such that for any group homomorphism $\phi : S \rightarrow G$, there is a unique group homomorphism $\bar{\phi} : \mathbb{F}_n \rightarrow G$ such that $\bar{\phi}i = \phi$.

$$\begin{array}{ccc}
 \mathbb{F} & \xrightarrow{\bar{\phi}} & G \\
 \uparrow i & \nearrow \phi & \\
 S & &
 \end{array}$$

Definition 9.15. By the universal property of free group, any generating set $\{a_1, \dots, a_n\}$ of a group G induces a surjective group homomorphism $\varphi : \mathbb{F}_n \rightarrow G$.

If the kernel of φ is generated by $R_1, \dots, R_m$, then we say $\{R_i\}$ are the **defining relations** of G , and we have an isomorphism:

$$G \cong \mathbb{F}_n / \langle R_1, \dots, R_m \rangle.$$

The group is denoted as $\langle a_1, \dots, a_n \mid R_1, \dots, R_m \rangle$, which is called a **group presentation**.

Example 8. • $\mathbb{Z}^2 \cong \langle a, b \mid aba^{-1}b^{-1} \rangle$.

- $C_n \cong \langle a \mid a^n \rangle$.
- $D_n \cong \langle a, b \mid a^n, abab \rangle$.
- $\mathbb{F} \cong \langle a, b \mid \rangle$.

Definition 9.16. Given a group G and its presentation $\langle a_1, \dots, a_n \mid R_1, \dots, R_m \rangle$, its **presentation complex** is defined as the combinatorial 2-complex X_G , whose underlying graph is $\bigvee_{i=1}^n S^1$, denoted $a_1, \dots, a_n$, with a disk attached along the paths $R_1, \dots, R_m$.

Exercise 9.17. $\mathbb{T}^2 \cong X_G$ where $G \cong \langle a, b \mid aba^{-1}b^{-1} \rangle$.

Definition 9.18. The **Klein bottle** is the space X_G where $G \cong \langle a, b \mid aba^{-1}b \rangle$, it is non-orientable.

Theorem 9.19. $\sum_g \cong X_G$, where $G \cong \langle a_1, b_1, \dots, a_g, b_g \mid a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1} \rangle$. This is actually a compact 2-manifold.

Theorem 9.20. $\pi_1(X_G) \cong G$.

10 Functoriality of π_1

Definition 10.1. Given a pointed map $f : (X, x) \rightarrow (Y, y)$, we define a group homomorphism $f_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$ by the formula $f_*([\gamma]) = [f \circ \gamma]$ for $\gamma \in \pi_1(X, x)$.

As a result, any pointed map (morphism in **Top**_{*}) can be turned into a respective group homomorphism (morphism in **Grp**). Therefore, we can define π_1 as a functor between the categories.

Definition 10.2. $\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Grp}$ is a functor with $\pi_1(X, x) = \pi_1(X, x)$ and $\pi_1(f) = f_*$.

Definition 10.3. Similarly, we can define $f_* : \Pi_1(X) \rightarrow \Pi_1(Y)$ as a functor (morphism between **Gpd**) with $f_*(x) = f(x)$ and $f_*([\gamma]) = [f \circ \gamma]$.

As a result, $\Pi_1 : \mathbf{Top} \rightarrow \mathbf{Gpd}$ can be seen as a functor.

Remark 4. The category **Gpd**_{*} is defined as a groupoid **G** with an object $x \in \mathbf{G}$. We have $\Pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Gpd}_*$ with $\Pi_1(X, x) = \Pi_1(X)(x, x)$ and $\Pi_1(f) = f_*$ is a functor. Then we have π_1 is the composite of Π_1 and $i : \mathbf{Gpd}_* \rightarrow \mathbf{Grp}$.

Corollary 10.4. If X, Y are path-connected spaces and $f : X \rightarrow Y$ is a homeomorphism, then $\pi_1(X) \cong \pi_1(Y)$. More importantly, if $\pi_1(X) \not\cong \pi_1(Y)$, then $X \not\cong Y$.

Theorem 10.5. The fundamental groups of the following path-connected spaces are isomorphic to

- $X = K \subset \mathbb{R}^n$, $\pi_1(X) \cong \mathbf{1}$.
- $X = S^1$, $\pi_1(X) \cong \mathbb{Z}$.
- $X = S^n$, $\pi_1(X) \cong \mathbf{1}$.
- $X = \mathbb{T}^n$, $\pi_1(X) \cong \mathbb{Z}^n$.
- $X = S^1 \vee S^1$, $\pi_1(X) \cong \mathbb{Z} * \mathbb{Z}$.
- $X = \sum_g$, $g > 1$, $\pi_1(X)$ is a non-Abelian group with $(\pi_1(\sum_g))_{ab} \cong \mathbb{Z}^n$.
- $X = \mathbb{R}^2 - \{0\}$, $\pi_1(X) \cong \mathbb{Z}$.
- $X = \mathbb{R}^n - \{0\}$, $\pi_1(X) \cong \mathbf{1}$.

Space of The Day (Cayley Complex)

Definition 10.6. Given a group G and its presentation $\langle g_1, \dots, g_n \mid R_1, \dots, R_m \rangle$, the **Cayley graph** Γ_G is defined as the graph with vertices G and an edge between $g \in G$ and $g \cdot g_i$ for each generator g_i .

Therefore, each word of the generators gives a combinatorial path in Γ_G starting at any vertex.

Exercise 10.7. There is a free action from G on Γ_G with the quotient $\bigvee_{i=1}^n S^1$.

Definition 10.8. We now form the **Cayley complex** $\tilde{X}_G$, which has the underlying graph Γ_G and a disk glued along each relation R_i starting at each $g \in G$.

Exercise 10.9. There is a free action from G on $\tilde{X}_G$ with the quotient homeomorphic to X_G .

Exercise 10.10. $\mathbb{R}^2 \cong \tilde{X}_G$ where $G = \langle a, b \mid aba^{-1}b^{-1} \rangle$.

Theorem 10.11. $\tilde{X}_G$ is simply connected.

11 Homotopy Equivalence

Theorem 11.1. *Given two spaces $X, Y \in \mathbf{Top}$, the group homomorphisms $(pr_1)_*$ and $(pr_2)_*$ induced by the projection maps $pr_1 : (X \times Y, (x, y)) \rightarrow (X, x)$ and $pr_2 : (X \times Y, (x, y)) \rightarrow (Y, y)$ induce an isomorphism $\langle (pr_1)_*, (pr_2)_* \rangle : \pi_1(X \times Y, (x, y)) \rightarrow \pi_1(X, x) \times \pi_1(Y, y)$.*

In other words, $\pi_1(X \times Y, (x, y))$ behave like the product of $\pi_1(X, x)$ and $\pi_1(Y, y)$.

Corollary 11.2. *Assume $\pi_1(S^1) \cong \mathbb{Z}$, then $\pi_1(\mathbb{T}^n) \cong \mathbb{Z}^n$.*

Proposition 11.3. *If $f, g : (X, x_0) \rightarrow (Y, y_0)$ are homotopic relative to x_0 , then we have $f_* = g_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$. In other words, $f\gamma \simeq g\gamma$ where $\gamma \in \pi_1(X, x_0)$.*

Proof. Suppose $H : X \times I \rightarrow Y$ is the homotopy between f and g , then $H' : I \times I \rightarrow Y$ defined as $H'(i, t) = H(\gamma(i), t)$ is the homotopy between $f\gamma$ and $g\gamma$ relative to endpoints.

Therefore, $f_* = g_*$. □

Exercise 11.4. (a) *Given $f : X \rightarrow Y, g, h : Y \rightarrow Z, gf \simeq hf$ if $g \simeq h$.*

(b) *Similarly, if g and h are homotopic relative to $f(A)$ for some $A \in X$, then $gf \simeq hf$ relative to $f(A)$.*

Proof. The ideas of the proofs are similar. $H' : X \times I \rightarrow Z$ defined as $H'(x, t) = H(f(x), t)$ is a homotopy between gf and hf . The homotopy is relative to A if H is relative to $f(A)$. □

Theorem 11.5. *Given continuous maps $f, g : X \rightarrow Y$, we have induced functors f_* and g_* between $\Pi_1(X)$ and $\Pi_1(Y)$. Given homotopy $H : f \rightarrow g$, we have an **induced natural transformation** $h_* : f_* \rightarrow g_*$, where $(h_*)_x = \gamma_x = H(x, t) : f(x) \rightsquigarrow g(x)$.*

Proof. We have a homotopy $h' : I \times I \rightarrow Y$ from $f \circ \gamma_0$ to $g \circ \gamma_0$, where γ_0 is a path from x to y in X . Explicitly, we can define $h'_t(i) = h(\gamma_0(i), t)$ where h is the homotopy between f and g .

We want to show that $(f \circ \gamma_0) \cdot \gamma_y \sim \gamma_x \cdot (g \circ \gamma_0)$. Notice that we have two paths γ_1, γ_2 from I to $I \times I$, between $(0, 0)$ and $(1, 1)$, namely

$$\gamma_1 = \begin{cases} (2t, 0) & 0 \leq t \leq \frac{1}{2} \\ (1, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

and

$$\gamma_2 = \begin{cases} (0, 2t) & 0 \leq t \leq \frac{1}{2} \\ (2t - 1, 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

We know they are homotopic rel endpoints because $I \times I$ is convex and thus contractible.

Now notice that $\gamma'_1 := f \circ \gamma_0 \cdot \gamma_y = h' \circ \gamma_1$ and $\gamma'_2 := \gamma_x \cdot (g \circ \gamma_0) = h' \circ \gamma_2$ are paths from I to Y , between $f(x)$ and $g(y)$. Explicitly, they are

$$\gamma'_1 = \begin{cases} f\gamma_0(2t) & 0 \leq t \leq \frac{1}{2} \\ h(y, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

and

$$\gamma'_2 = \begin{cases} h(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ g\gamma_0(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

They are homotopic rel endpoints because they are composite of h' and homotopic paths, so we have $[\gamma'_1] = [\gamma'_2]$, so $(\gamma_x)_{x \in X}$ is a natural transformation. □

Theorem 11.6. *Given two functors $F, G : \mathbf{G} \rightarrow \mathbf{H}$ between groupoids, and a natural transformation $\alpha : F \rightarrow G$, then the following diagram commutes for any $x \in X$.*

$$\begin{array}{ccc} \mathbf{G}(x, x) & \xrightarrow{F} & \mathbf{H}(F(x), F(x)) \\ & \searrow G & \downarrow (\alpha_x)_* \\ & & \mathbf{H}(G(x), G(x)) \end{array}$$

Proof. Notice that we have $\alpha_x(Ff) = (Gf)\alpha_x$ for any $x \in \mathbf{G}$ and $f : x \rightarrow x$.

Therefore, $\alpha_x F f \alpha_x^{-1} = Gf$, so $(\alpha_x)_* F = G$. □

Theorem 11.7. Suppose $f, g : X \rightarrow Y$ are homotopic via H . Fix $x_0 \in X$, and let $\gamma_{x_0} = H(x_0, t) : f(x_0) \rightsquigarrow g(x_0)$, then the following diagram commutes.

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{f_*} & \pi_1(Y, f(x_0)) \\ & \searrow g_* & \downarrow (\gamma_{x_0})_* \\ & & \pi_1(Y, g(x_0)) \end{array}$$

Proof. Let f_*, g_* be the functors from $\Pi_1(X)$ to $\Pi_1(Y)$, γ_{x_0} be the induced natural transformation, then we get the above result from the previous two theorems. □

Definition 11.8. A map $f : X \rightarrow Y$ is a **homotopy equivalence** if $\exists g : Y \rightarrow X$ s.t. $g \circ f \sim 1_X$, and $f \circ g \sim 1_Y$.

We say X and Y are **homotopy equivalent** and write $X \simeq Y$.

Theorem 11.9. If $f : X \rightarrow Y$ is a homotopy equivalence, then $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ is a group isomorphism for any $x_0 \in X$.

Exercise 11.10. (a) X is contractible iff it is homotopy equivalent to a point.

(b) X is simply connected if X is contractible.

Proof. (a) If X is contractible, then $1_X \sim c_{x_0}$ for some $x_0 \in X$. We thus have $f : X \rightarrow \{y\}$ and $g : \{y\} \rightarrow X$ with $g(y) = x_0$. f is a homotopy equivalence because $gf = c_{x_0} \sim 1_X$.

On the other hand, if $X \simeq \{y\}$, then $\exists g : \{y\} \rightarrow X$ that is a homotopy equivalence, and $g(y) = x_0$ for some $x_0 \in X$. Therefore, $1_X \sim c_{x_0}$ so X is contractible.

(b) We can directly prove the statement as we have $X \simeq \{y\}$ from a). Therefore, $\pi_1(X, x) \cong \pi_1(Y, y_0)$ because we have $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ and $g_* : \pi_1(Y, y_0) \rightarrow \pi_1(X, x_0)$ group homomorphisms, and $g_* f_* = (gf)_* = 1_* : \pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$ which is an isomorphism. Therefore, f_* and g_* are group isomorphisms so $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$ which is trivial. □

Definition 11.11. A **retraction** of a space X onto a space Y is continuous map $r : X \rightarrow Y$ s.t. $r|_Y = 1_Y$. We then say Y is a **retract** of X . In other words, r is the left inverse of $i : Y \rightarrow X$ (i.e. $ri = 1_Y$).

Definition 11.12. r is a **deformation retraction** if $ir \sim 1_X$ relative to Y (i.e. $\exists H : Y \times I \rightarrow X$ s.t. $H(y, t) = y$, for any $y \in Y, t \in I$.)

Example 9. Any space Y is a retract of $Y \sqcup Y$ but not a deformation retract.

Exercise 11.13. Show that S^n is a retract of $\mathbb{R}^{n+1} - \{0\}$.

Proof. Take $H((x), t) = (1-t)(\frac{x}{\sqrt{\sum x_i^2}}) + tx$ □

Definition 11.14. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is an **equivalence of categories** if $\exists G : \mathcal{D} \rightarrow \mathcal{C}$ s.t. $GF \cong 1_{\mathcal{C}}$ and $FG \cong 1_{\mathcal{D}}$.

An **equivalence of groupoids** is just a functor F being an equivalence of categories.

Theorem 11.15. *If $F : \mathbf{G} \rightarrow \mathbf{H}$ is an equivalence of groupoids, $F : \mathbf{G}(x, x) \rightarrow \mathbf{H}(F(x), F(x))$ is a group isomorphism.*

Proof. Let G be the inverse of F , then we have a natural isomorphism $\alpha : GF \rightarrow 1_G$. Thus, the following diagram commutes:

$$\begin{array}{ccc} \mathbf{G}(x, x) & \xrightarrow{F} \mathbf{H}(F(x), F(x)) & \xrightarrow{G} \mathbf{G}(GF(x), GF(x)) \\ & \searrow 1_G & \downarrow \alpha_x \\ & & \mathbf{G}(x, x) \end{array}$$

Therefore, $G : \mathbf{H}(F(x), F(x)) \rightarrow \mathbf{G}(GF(x), GF(x))$ is surjective. Similarly applying the argument to G , we have $G : \mathbf{H}(F(x), F(x)) \rightarrow \mathbf{G}(GF(x), GF(x))$ is surjective. Therefore, $G : \mathbf{H}(y, y) \rightarrow \mathbf{G}(G(y), G(y))$ is a group isomorphism and similarly, $F : \mathbf{G}(x, x) \rightarrow \mathbf{H}(F(x), F(x))$ is a group isomorphism. $\square$

Corollary 11.16. (a) *If $f : X \rightarrow Y$ is a homotopy equivalence, then $f_* : \Pi_1(X) \rightarrow \Pi_1(Y)$ is an equivalence of categories.*

(b) *If $f : X \rightarrow Y$ is a homotopy equivalence, then $f_* : \pi_1(X, x) \rightarrow \pi_1(Y, f(x))$ is a group isomorphism.*

(c) *If X, Y are path connected then $\pi_1(X) \cong \pi_1(Y)$.*

Proof. (a) As f is a homotopy equivalence, there is a map $g : Y \rightarrow X$ with $gf \sim 1_X$.

Therefore, $(gf)_* = g_*f_* \cong 1_*$ so f_* is an equivalence of categories.

(b) This is a direct result from the above theorem.

(c) This follows directly from the part above. $\square$

Corollary 11.17. $\pi_1(S^n) \cong \pi_1(\mathbb{R}^{n+1} - \{0\})$.

Space of the Day (Real/Complex Projective Spaces)

Definition 11.18. *The n -th real projective space, $\mathbb{R}P^n$, is defined as $(\mathbb{R}P^{n+1} - \{0\}) / \sim$, where $v \sim w$ iff there exists a non-zero $a \in \mathbb{R}$ s.t. $v = aw$.*

Particularly, when $n = 2$, $\mathbb{R}P^2$ is called the real projective plane.

Exercise 11.19. *Show that $\mathbb{R}P^n$ is homeomorphic to the quotient S^n/C_2 by the antipodal action of C_2 . Additionally, $\mathbb{R}P^1 \cong S^1$.*

Proof. Notice that we can deform all the equivalence class of $\mathbb{R}P^n$ onto S^n without impacting the equivalence relation. Also, notice that two points on S^n are equivalent iff they are antipodal (i.e., on the same line passing the origin). The conclusion follows. $\square$

Exercise 11.20. *Show that the quotient $q : S^n \rightarrow \mathbb{R}P^n$ is a covering map.*

Proof. For any $x \in S^n$, let $p(x) = x$ or $-x$ depending on whether x_0 is positive or not (and if $x_0 = 0$, then we look at the value of x_1). Then, for any neighborhood of $p(x)$, we get two open sets in S^n , one is the identical neighborhood of x in S^n and the other is the symmetric neighborhood of $-x$. As long as the two neighborhoods are small enough they are going to be disjoint. Thus, by definition, we have p a covering map. $\square$

Exercise 11.21. *Show that $\mathbb{R}P^n$ is homeomorphic to $D^n / \sim$, where $\sim$ is generated by $x \sim -x$ for $x \in S^{n-1}$.*

Proof. We simply consider the upper hemisphere of S^n with the equivalence relation on the equator of S^n (i.e., the boundary S^{n-1} of D^n). $\square$

Exercise 11.22. Show that $\mathbb{R}P^n$ is a smooth manifold.

Proof. Given S^n is a manifold, we are able to prove $\mathbb{R}P^n$ is a manifold as it is being covered by S^n . Given an atlas U_α of S^n , we have its images an open cover of $\mathbb{R}P^n$ and each set in the cover is homeomorphic to an open subset of $\mathbb{R}^n$. Therefore, the open cover forms an atlas for $\mathbb{R}P^n$, and we have $\mathbb{R}P^n$ a manifold. $\square$

Exercise 11.23. Show that $\mathbb{R}P^n$ has a cell complex structure with a single cell in dimension $0, 1, \dots, n$.

Proof. Let $(x_0 : \dots : x_n)$ be the equivalence class of $(x_0, \dots, x_n)$ in $\mathbb{R}P^n$.

Notice that if $x_0 \neq 0$, we can have a homeomorphism $\{(x_0, \dots, x_n), x_0 \neq 0\} \rightarrow \mathbb{R}^n$ by sending $(x_0, \dots, x_n)$ to $(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})$. On the other hand, if $x_0 = 0$, then we can directly map the set $\{(x_0, \dots, x_n), x_0 = 0\}$ to $\mathbb{R}P^{n-1}$ homeomorphically.

Therefore, $\mathbb{R}P^n \cong \mathbb{R}^n \sqcup \mathbb{R}P^{n-1}$ and by induction we have that $\mathbb{R}P^n \cong \bigsqcup_{i=0}^n \mathbb{R}^i$, implying what the conclusion says. $\square$

Definition 11.24. The *n-th complex projective space*, $\mathbb{C}P^n$, is similarly defined as $(\mathbb{C}P^{n+1} - \{0\})/\sim$, where $v \sim w$ iff there exists a non-zero $a \in \mathbb{C}$ s.t. $v = aw$.

Exercise 11.25. Show that $\mathbb{C}P^n \cong S^{2n+1}/S^1$, where S^1 is considered as a subset of $\mathbb{C}$ and acts on S^{2n+1} by multiplication.

Exercise 11.26. Show that $\mathbb{C}P^n$ is a smooth manifold.

Proof. Let $U_i = \{(x_0, \dots, x_n), x_i \neq 0\}$, then $\mathbb{C}P^n = \bigcup U_i$. Notice that by the above map, $U_i \cong \mathbb{C}^n \cong \mathbb{R}^{2n}$, so $\bigcup U_i$ form an atlas of $\mathbb{C}P^n$, proving that it is a manifold. $\square$

Exercise 11.27. Show that $\mathbb{C}P^n$ has a cell complex structure with a single cell in dimension $0, 2, \dots, 2n$.

Proof. The idea is the same as what we did for real projective spaces. $\square$

Exercise 11.28. $\mathbb{C}P^1 \cong S^2$.

12 π_1 and Covering Space

Theorem 12.1. $\pi_1(X, x_0) \cong \mathbb{Z}$.

Proof. Let $x_0 = (1, 0)$ and consider the covering map $p : \mathbb{R} \rightarrow S^1$ that sends t to $e^{2\pi it}$. Clearly, $p(\tilde{x}_0) = x_0$ iff $\tilde{x}_0 \in \mathbb{Z}$. For any loop γ at x_0 , let $\tilde{\gamma}$ be the unique path $I \rightarrow \mathbb{R}$ that satisfies $p \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(0) = \tilde{x}_0$. (The theorem used here will be proved later)

Since $p \circ \tilde{\gamma} = \gamma$, we have $p \circ \tilde{\gamma}(1) = x_0$ so $\tilde{\gamma}(1) \in \mathbb{Z}$. We thus obtain a map $\varphi : \Omega_{x_0} S^1 \rightarrow \mathbb{Z}$ with $\varphi(\gamma) = \tilde{\gamma}(1) \in \mathbb{Z}$. Intuitively, $\tilde{\gamma}(1) - \tilde{\gamma}(0)$ is the number of times γ goes around S^1 .

We will then show that if $\gamma \sim \gamma'$, then $\tilde{\gamma} \sim \tilde{\gamma}'$ and $\tilde{\gamma}(1) = \tilde{\gamma}'(1)$.

Thus, $\bar{\varphi} : \pi_1(S^1, x_0) \rightarrow \mathbb{Z}$ induced by φ is well-defined.

$\bar{\varphi}$ is a surjective group homomorphism since $\bar{\varphi}(e^{2\pi i \text{int}}) = n$, and it is injective because $\mathbb{R}$ is contractible, so if $\varphi(\gamma) = \varphi(\gamma')$, then $\tilde{\gamma}$ and $\tilde{\gamma}'$ are paths starting from $\tilde{x}_0$ and end at $\varphi(\gamma) + \tilde{x}_0$, so they are homotopic. Thus, $\gamma = p \circ \tilde{\gamma} \sim p \circ \tilde{\gamma}' = \gamma'$. $\square$

Definition 12.2. Given a covering map $p : \hat{X} \rightarrow X$, a **deck transformation** of $(\hat{X}, p)$ is a homeomorphism $f : \hat{X} \rightarrow \hat{X}$ such that $p \circ f = p$.

Remark 5. Since the deck transformations are closed under compositions and have inverses, they form a group, denoted as $\text{Aut}(\hat{X}, p)$ or $\text{Aut}(\hat{X}/X)$.

Definition 12.3. A covering map $p : \tilde{X} \rightarrow X$ is a **universal cover** if $\tilde{X}$ is simply connected. Notice that if X has a universal cover, then X is path-connected because the image of a path-connected space under a continuous map is still path-connected.

Theorem 12.4. If $p : \tilde{X} \rightarrow X$ is a universal cover and X is locally path-connected, then $\pi_1(X) \cong \text{Aut}(\tilde{X}/X)$.

Theorem 12.5. If G acts freely on a space $\hat{X}$ and the quotient map $q : \hat{X} \rightarrow \hat{X}/G = X$ is a covering map, then $\text{Aut}(\tilde{X}/X) \cong G$.

Corollary 12.6. We have

- $\text{Aut}(\mathbb{R}/S^1) \cong \mathbb{Z}$.
- $\text{Aut}(\mathbb{R}^n/\mathbb{T}^n) \cong \mathbb{Z}^n$.
- $\text{Aut}(S^n/\mathbb{R}P^n) \cong \mathbb{Z}/2\mathbb{Z}$.
- $\text{Aut}(S^3/L(p, q)) \cong \mathbb{Z}/q\mathbb{Z}$.
- $\text{Aut}(\tilde{X}_G/X_G) \cong G$.

Definition 12.7. Given a map $p : \hat{X} \rightarrow X$ and a map $f : Y \rightarrow X$, a **lift** of f to $\hat{X}$ (along p) is a map $\hat{f} : Y \rightarrow \hat{X}$ s.t. $p \circ \hat{f} = f$. Similarly, a **lift** at x_0 is a point $\hat{x}_0 \in \hat{X}$ s.t. $p(\hat{x}_0) = x_0$.

Theorem 12.8. Let $p : \hat{X} \rightarrow X$ be a covering map, then:

- (i) for any path $\gamma : x_0 \rightsquigarrow x_1$ in X there is a unique lift $\hat{\gamma} : \hat{x}_0 \rightsquigarrow \hat{x}_1$ of γ ;
- (ii) for $\gamma_0, \gamma_1 : x_0 \rightsquigarrow x_1$ be two paths in X , let $\hat{\gamma}_0, \hat{\gamma}_1$ be their unique lifts starting at $\hat{x}_0$, then if $\gamma_0 \sim \gamma_1$ relative to endpoints, then $\hat{\gamma}_0(1) = \hat{\gamma}_1(1)$ and $\hat{\gamma}_0 \sim \hat{\gamma}_1$.

Proof. (i) Notice that the claim holds if $\gamma(I) \subset U$ where U is an evenly covered open set, since there would be $V \in \hat{X}$ s.t. $\hat{x}_0 \in V$ and $p|_V : V \rightarrow U$ is a homeomorphism, so we can take $\hat{\gamma} = p|_V^{-1} \circ \gamma$.

Now let $U_1, \dots, U_m$ be an open cover of $\gamma(I)$ consists of evenly open covered sets (possible by compactness). Consider their inverses, we then get an open cover of I . By the Lebesgue Number Lemma, we can find some $0 \leq a_0 < \dots < a_n \leq 1$ s.t. $\gamma([a_i, a_{i+1}]) \subset U_j$ for some j .

Next, we can prove by induction that there is a unique lift of $\gamma|_{[0, a_1]}$ and any $\gamma|_{[a_i, a_{i+1}]}$. Pasting all of those together we get a unique lift of γ .

- (ii) Let $h : I \times I \rightarrow X$ be the homotopy from γ_0 to γ_1 relative to endpoints. We have a unique lift of h , called $\hat{h}$, where for each fixed x , $\hat{h}_t(x) : I \rightarrow \hat{X}$ is the unique lift of $h_t(x) : \gamma_0(x) \rightsquigarrow \gamma_1(x)$ starting at $\hat{\gamma}_0(x)$. We thus conclude that $\hat{h}_t(1) = \hat{\gamma}_0(1) = \hat{\gamma}_1(1)$ for any t . (Notice that the lift of the constant map at $\gamma_0(1) = x_1$ starting at $\hat{\gamma}_0(1)$ is unique, and the constant map at $\hat{\gamma}_0(1)$ is such a lift. Therefore, we conclude that $\hat{h}_t(1) = \hat{\gamma}_0(1)$ for any t .)

Now we need to show that this uniquely determined $\hat{h}$ with $\hat{h}_0 = \gamma_0$, $\hat{h}_1 = \gamma_1$, and $\hat{h}_t(0) = \hat{x}_0$ and $\hat{h}_t(1) = \hat{\gamma}_0(1)$ is continuous.

We use Lebesgue number lemma on $I \times I$ s.t. each sub-square S has its image under f , $f(S)$, lying in an evenly covered open set. As a result, the lift of $f(S)$ is uniquely determined so it has to be matching $\hat{h}(S)$. Therefore, $f = \hat{h}$ and since f is continuous by the pasting lemma, $\hat{h}$ is also continuous. $\square$

Remark 6. From now on, if $\gamma : x_0 \rightsquigarrow x_1$ then $\hat{\gamma}$ is the unique lift of γ starting at $\hat{x}_0$. Sometimes we use $\hat{\gamma} = l_{\hat{x}_0}\gamma$.

Exercise 12.9. Given $\gamma_{01} : x_0 \rightsquigarrow x_1$ and $\gamma_{12} : x_1 \rightsquigarrow x_2$ in X and $\hat{x}_0 \in \hat{X}$, with $\hat{x}_1 = l_{\hat{x}_0}\gamma_{01}(1)$, then

- (i) $l_{\hat{x}_0}\gamma_{01} \cdot l_{\hat{x}_1}\gamma_{12} = l_{\hat{x}_0}(\gamma_{01} \cdot \gamma_{12})$.
- (ii) $l_{\hat{x}_1}\overline{\gamma_{01}} = \overline{(l_{\hat{x}_0}\gamma_{01})}$.

Theorem 12.10. Any covering map $p : \hat{X} \rightarrow X$ induces an injection $p_* : \pi_1(\hat{X}, \hat{x}_0) \rightarrow \pi_1(X, x_0)$ for any $x_0 \in X$ and any lift $\hat{x}_0$ of x_0 .

Proof. Suppose $p \circ \hat{f} \sim 1_{x_0}$. Then we have $\hat{f} \sim \hat{1}_{x_0} = 1_{\hat{x}_0}$, since the lift of 1_{x_0} starting at $\hat{x}_0$ is unique. Therefore, $[f] = [1_{x_0}]$ which is the identity in $\pi_1(\hat{X}, \hat{x}_0)$, so p_* is injective. $\square$

Space of The Day (Lens Space)

Definition 12.11. Fix coprime integers p and q and consider the action C_q on $S^3 \subset \mathbb{C}^2$ generated by the map $(z_1, z_2) \rightarrow (e^{2\pi i/q} z_1, e^{2p\pi i/q} z_2)$.

The quotient S^3/C_q is called the **Lens Space**, denoted as $L(p; q)$.

Exercise 12.12. Show that the action is free.

Exercise 12.13. Prove that $q : S^3 \rightarrow L(p; q)$ is a covering map. Therefore, $\pi_1(L(p; q)) \cong \mathbb{Z}/q\mathbb{Z}$.

Exercise 12.14. $L(p; q)$ is a manifold.

Exercise 12.15. $L(p, 2) \cong \mathbb{RP}^3$.

13 The Lifting Criterion

Lemma 13.1. *Given a pointed covering map $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ and $\gamma \in \Omega_{x_0}(X)$, the lift $\hat{\gamma}$ is a loop at $\hat{x}_0$ iff $[\gamma] \in p_*(\pi_1(\hat{X}, \hat{x}_0))$.*

Proof. It is clear that $[\gamma] \in p_*(\pi_1(\hat{X}, \hat{x}_0))$ if $\hat{\gamma}$ is a loop at $\hat{x}_0$ since $\gamma = p_*(\hat{\gamma})$.

Suppose $\gamma \sim p \circ k$ where k is a loop at $\hat{x}_0$, then we have $\hat{\gamma} \sim k$ and $\hat{\gamma}(1) = k(1) = \hat{x}_0$ by the homotopy lifting property and the fact that $\hat{\gamma}(0) = k(0)$. So $\hat{\gamma}$ is a loop at $\hat{x}_0$. $\square$

Theorem 13.2. *Let Y be a connected and locally path connected space. Let $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ be the pointed covering map and $f : (Y, y_0) \rightarrow (X, x_0)$ be a pointed map. Then f has at most one pointed lift $\hat{f}$, and it exists iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\hat{X}_0, \hat{x}_0))$.*

Proof. (i) Since Y is path connected, there exists $\gamma : y_0 \rightarrow y$ a path for any y . Therefore, $f \circ \gamma : x_0 \rightarrow f(y)$ is a path in X and it has a unique lift $l_{\hat{x}_0}(f \circ \gamma)$. Therefore, $\hat{f}(y) = l_{\hat{x}_0}(f \circ \gamma)(1)$ so it is (at most) uniquely determined.

(ii) Assume that $\hat{f}$ exists. We have that $f \circ \gamma$ is a loop at x_0 and $\hat{f} \circ \gamma$ a loop at $\hat{x}_0$. Since $f \circ \gamma = p \circ \hat{f} \circ \gamma$, we have $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\hat{X}_0, \hat{x}_0))$.

(iii) Suppose that $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\hat{X}_0, \hat{x}_0))$. We want to show that our definition of $\hat{f}$ is well-defined. The key is to show that $\hat{f}(y)$ is independent of the choice of path from y_0 to y .

Suppose that γ_1 and γ_2 are two paths from y_0 to y . Then we have $\gamma_1 \cdot \overline{\gamma_2}$ a loop at y_0 . Therefore, letting $\alpha_1 = f \circ \gamma_1$ and $\alpha_2 = f \circ \gamma_2$ we get $\alpha_1 \cdot \overline{\alpha_2} = f \circ (\gamma_1 \cdot \overline{\gamma_2})$. By assumption, we have $[f \circ (\gamma_1 \cdot \overline{\gamma_2})] = [\alpha_1 \cdot \overline{\alpha_2}] \in p_*(\pi_1(\hat{X}_0, \hat{x}_0))$.

Let $\hat{x}_1 = l_{\hat{x}_0}\alpha_1(1)$, then, by the above lemma, we get that $l_{\hat{x}_0}\alpha_1 \cdot \overline{\alpha_2} = l_{\hat{x}_0}\alpha_1 \cdot l_{\hat{x}_1}\overline{\alpha_2}$ a loop at $\hat{x}_0$. Therefore, $l_{\hat{x}_0}\alpha_1(1) = l_{\hat{x}_1}\overline{\alpha_2}(0) = \overline{l_{\hat{x}_0}\alpha_2}(1) = l_{\hat{x}_0}\alpha_2(1)$. Therefore, $\hat{f}$ is well-defined.

(iv) We now need to show that $\hat{f}$ is continuous. Fix $y \in Y$ and a path-connected neighborhood W of y so that $f(W)$ is contained in an evenly covered open set $U \subset X$. Let V be the sheet over U containing $\hat{f}(y)$.

Let $y' \in W$, and paths $\gamma : y_0 \rightsquigarrow y$ and $\alpha : y \rightsquigarrow y'$. We then get

$$\hat{f}(y') = l_{\hat{x}_0}(f \circ (\gamma \cdot \alpha))(1) = (l_{\hat{x}_0}(f \circ \gamma) \cdot l_{\hat{f}(y)}(f \circ \alpha))(1) = l_{\hat{f}(y)}(f \circ \alpha)(1) \in V.$$

Therefore, $\hat{f}(W) \subset V$ and we have $\hat{f}|_W = (p|_V)^{-1} \circ f$ which is continuous. $\square$

Space of The Day (Simplicial Complexes)

Definition 13.3. *Given a set of points $S \subset \mathbb{R}^n$, its **convex hull** $\text{Conv}(S)$ is defined as the smallest convex set containing all points in S .*

Equivalently, it is the set of all convex linear combinations $\sum a_i s_i$ where $a_i \geq 0$ and $\sum a_i = 1$.

Definition 13.4. *A set of points $S \subset \mathbb{R}^n$ is **linearly independent** if for some $s \in S$, the set of vectors $\{s' - s \mid s' \in S - \{s\}\}$ is linearly independent.*

Exercise 13.5. *If S is a linearly independent set of points, then for every $s \in S$, the set of vectors $\{s' - s \mid s' \in S - \{s\}\}$ is linearly independent.*

Proof. Suppose that for some $s \in S$, the set of vectors $\{s' - s \mid s' \in S - \{s\}\}$ is linearly independent. This implies that

$$\sum a_i (s' - s) = 0$$

is equivalent to $a_i = 0$ for any i . Now suppose that

$$\sum b_i(s' - s) = 0,$$

then for any $\bar{s}$

$$\sum b_i(s' - s) = \sum b_i(s' - s) + b_i(s - \bar{s}) = 0,$$

indicating that $b_i = 0$ because of linearly independence. $\square$

Corollary 13.6. *S is a linearly independent set of points is equivalent to no k -element subset of S lie in a $k - 2$ -dimensional subspace of $\mathbb{R}^n$.*

Therefore, if S is linearly independent in $\mathbb{R}^n$, the $|S| \leq n + 1$.

Definition 13.7. *A **k -simplex** σ is a subset of $\mathbb{R}^n$ of the form $\text{Conv}(S)$ for a linearly independent $(k + 1)$ -element set S . The elements of S are called the **vertices** of σ .*

*A **face** of a simplex is the convex hull of a subset of its vertices.*

Definition 13.8. *The **standard n -simplex** is the convex hull of the endpoints of the $n + 1$ standard basis vectors in $\mathbb{R}^{n+1}$, where each vector starts from the origin.*

Definition 13.9. *A **simplicial complex** in $\mathbb{R}^n$ is a set K of simplices s.t.*

- (i) *If $\sigma \in K$ and σ' is a face of σ , then $\sigma' \in K$.*
- (ii) *For any $\sigma_1, \sigma_2 \in K$, $\sigma_1 \cap \sigma_2$ is a face of both of them.*

*The **dimension** of K is the maximal dimension of its simplices.*

*The **underlying space** $|K|$ is the union of the underlying spaces of the simplices σ in K .*

Corollary 13.10. *Any simplicial complex has a CW complex structure, in which the k -skeleton is the union of all k -simplices.*

Definition 13.11. *A **triangulation** of a topological space X is a pair (K, φ) where K is a simplicial complex and $\varphi : X \rightarrow |K|$ is a homeomorphism.*

Exercise 13.12. *D^n and S^n admits a triangulation.*

14 Consequences of the Lifting Criterion

Definition 14.1. A *map of covering spaces* $f : \hat{X}_1 \rightarrow \hat{X}_2$ with covers $p_i : \hat{X}_i \rightarrow X$ is a map s.t. $p_1 = p_2 \circ f$.

An *isomorphism of covering spaces* is an invertible map of covering space.

Notice that a deck transformation is an invertible map of covering space from $\hat{X}$ to itself.

Theorem 14.2. If X is locally connected, then any map of covering spaces $f : \hat{X}_1 \rightarrow \hat{X}_2$ is also a covering map.

Proof. We first show that $\forall x \in \hat{X}_2, \exists U$ connected neighborhood of $p_2(x)$ that is evenly covered by both p_1 and p_2 .

To start, we show that any open subset of an evenly covered set is evenly covered. Let U be evenly covered and $V \subset U$. We have $p^{-1}(U) = \bigcup W_\alpha$ and $W_\alpha \cong U$ for any α . Therefore, since W_α are all disjoint, we have $p^{-1}(V) = \bigcup (W_\alpha \cap p|_{W_\alpha}^{-1}(V)) = \bigcup W'_\alpha$ an union of disjoint open subset of W_α 's. Clearly, $p|_{W'_\alpha}$ is a surjective continuous map from W'_α to V . As a restriction of homeomorphism, $p|_{W'_\alpha}$ is a homeomorphism from W'_α to V for any α . Therefore, V is evenly covered.

Let U_1 and U_2 be the neighborhood of $p_2(x)$ that is evenly covered by p_1 and p_2 , respectively. We now have that $U_1 \cap U_2$ is evenly covered by both p_i , which is a direct result from the statement we showed above. By the local connectedness, there exists a connected neighborhood $U \subset U_1 \cap U_2$ that is, by the above statement again, evenly covered by both p_i .

Now suppose $V \subset \hat{X}_2$ is the unique sheets over U containing x . Notice that for any W_α that are sheet over U s.t. $f(W_\alpha) \subset V$, $f|_{W_\alpha} = p_2^{-1}|_V \circ p_1|_{W_\alpha}$ is a homeomorphism so $W_\alpha \cong V$.

If we can show that $f(W_\alpha) \subset V$ for any W_α with $f(W_\alpha) \cap V \neq \emptyset$, then we can show $\{W_\alpha\}$ evenly covers V (any W_α 's are disjoint as they are also the sheets over U).

To prove the above statement, we use the connectedness of U . We have W_α connected for any α since it is homeomorphic to U . Therefore, $f(W_\alpha)$ is connected and is a subset of $p_2^{-1}(U) = \bigcup V_\beta$. Since the V_β 's are disjoint, $f(W_\alpha)$ can intersect at most one of them, so if $f(W_\alpha) \cap V \neq \emptyset$, then $f(W_\alpha) \subset V$, which will then be homeomorphic to V . Therefore, the set $\{W_\alpha\}$ with $f(W_\alpha) \cap V \neq \emptyset$ evenly covers V .

Since the choice of x is arbitrary and there always exists such a V , we conclude that f is a covering map. $\square$

Proposition 14.3. Let $p_1 : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and $p_2 : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ be two pointed covering maps with $\tilde{X}$ simply connected, then $\exists!$ map of covering spaces $f : (\tilde{X}, \tilde{x}_0) \rightarrow (\hat{X}, \hat{x}_0)$.

In particular, if $\hat{X}$ is also simply connected, then f is a homeomorphism.

Proof. Since $\tilde{X}$ is simply connected, $\pi_1(\tilde{X}, \tilde{x}_0)$ is trivial. Therefore, $p_{1*}(\pi_1(\tilde{X}, \tilde{x}_0))$ is a subset of $p_{2*}(\pi_1(\hat{X}, \hat{x}_0))$ so by the lifting criterion there exists a unique lift of p_1 , call it f , s.t. $p_2 \circ f = p_1$, which is a map of covering spaces by definition above. $\square$

Corollary 14.4. If $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ is a covering map with $\hat{X}$ connected and locally path connected, then for any other lift $\hat{x}'_0$ of x_0 , there is at most one deck transformation $(\hat{X}, \hat{x}_0) \rightarrow (\hat{X}, \hat{x}'_0)$.

In other words, the map $v_{\hat{x}_0} : \text{Aut}(\hat{X}/X) \rightarrow p^{-1}(x)$ by $v_{\hat{x}_0}(f) = f(\hat{x}_0)$ is injective.

Proof. This is a direct result from the lifting criterion. $\square$

Definition 14.5. A path connected pointed covering map $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ is **Galois** or **normal** if $p_*(\pi_1(\hat{X}, \hat{x}_0)) \subset \pi_1(X, x_0)$ is a normal subgroup.

Theorem 14.6. If $\hat{X}$ is connected and locally path connected and $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ is a Galois cover, then $\text{Aut}(\hat{X}, \hat{x}_0) \cong \pi_1(X, x_0)/p_*(\pi_1(\hat{X}, \hat{x}_0))$.

To prove this theorem, we have prove the following conclusions first:

Theorem 14.7. *Given a covering space $p : \hat{X} \rightarrow X$ with $\hat{X}$ path-connected, a point $x_0 \in X$, and two lifts $\hat{x}_0$ and $\hat{x}'_0$ of x_0 , the subgroups $p_*(\pi_1(\hat{X}, \hat{x}_0))$ and $p_*(\pi_1(\hat{X}, \hat{x}'_0))$ are conjugate.*

Proof. Because $\hat{X}$ is path connected, we have a path $\gamma_0 : \hat{x}_0 \rightarrow \hat{x}'_0$. As a result, we know there is a group isomorphism from $\pi_1(\hat{X}, \hat{x}_0)$ to $\pi_1(\hat{X}, \hat{x}'_0)$. Thus, for any $p \circ \gamma \in p_*(\pi_1(\hat{X}, \hat{x}_0))$, we have $p \circ (\gamma_0^{-1} \cdot \gamma \cdot \gamma_0) = (p \circ \gamma_0^{-1}) \cdot (p \circ \gamma) \cdot (p \circ \gamma_0) \in p_*(\pi_1(\hat{X}, \hat{x}'_0))$. Since we know $p \circ \gamma_0^{-1}$ is a loop at x_0 , the two subgroups $p_*(\pi_1(\hat{X}, \hat{x}_0))$ and $p_*(\pi_1(\hat{X}, \hat{x}'_0))$ are conjugate to each other with a conjugation by $p \circ \gamma_0$. $\square$

Theorem 14.8. *If $\hat{X}$ is connected, path connected, and locally path connected, and $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ is Galois, then the map $v_{\hat{x}_0} : \text{Aut}(\hat{X}/X) \rightarrow p^{-1}(x)$ by $v_{\hat{x}_0}(f) = f(\hat{x}_0)$ is bijective.*

Proof. We want to show that for any $\hat{x}'_0 \in \hat{X}$ that is a lift of x_0 , there exists a deck transformation f from $(\hat{X}, \hat{x}_0)$ to $(\hat{X}, \hat{x}'_0)$.

By the above theorem, we know that the two subgroups $p_*(\pi_1(\hat{X}, \hat{x}_0))$ and $p_*(\pi_1(\hat{X}, \hat{x}'_0))$ are conjugate to each other.

Since p is Galois at $\hat{x}'_0$, we know $p_*(\pi_1(\hat{X}, \hat{x}_0))$ is normal so the two subgroups are the same. Therefore, the lifting criterion is met and there is a lift (deck transformation) f with $f(\hat{x}_0) = \hat{x}'_0$.

Therefore, $v_{\hat{x}_0}$ is surjective and thus bijective. $\square$

Theorem 14.9. *Given $\hat{X}$ connected and locally path connected, $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ Galois, we are able to define a map*

$$\rho : \pi_1(X, x_0) \rightarrow \text{Aut}(\hat{X}/X),$$

where $\rho(\gamma) = v_{\hat{x}_0}^{-1}(\hat{\gamma})(1)$.

In other words, ρ takes a path homotopy class from $\pi_1(X, x_0)$, gets the unique endpoint, $\hat{x}'_0$, of their unique lift up to homotopy (by the path lifting property), and maps this class to the unique deck transformation (since p is Galois by the last theorem $v_{\hat{x}_0}$ is bijective) that takes $\hat{x}_0$ to $\hat{x}'_0$.

Theorem 14.10. *The map ρ is a surjective homomorphism, and its kernel is $p_*(\pi_1(\hat{X}, \hat{x}_0))$.*

Proof. The surjectivity is proved by the existence of (unique) deck transformation that takes $\hat{x}_0$ to $\hat{x}'_0$ and the path connectedness of $\hat{X}$.

The kernel is clearly p compose with the fundamental group at $\hat{x}_0$ as ρ maps and only maps any element in it to the unique (identity) deck transformation.

To show that ρ is a group homomorphism, notice that we first have $\rho([\beta]) \circ \hat{\alpha} = l_{\hat{\beta}(1)}\alpha$. To prove this, let the endpoint of the lift $\hat{\alpha}$, $\hat{\beta}$ be $\hat{x}_\alpha$ and $\hat{x}_\beta$. Since $\rho([\beta])$ is a deck transformation that takes $\hat{x}_0$ to $\hat{x}_\beta$, we have that $p \circ \rho([\beta]) = p$ so $p \circ \rho([\beta]) \circ \hat{\alpha} = p \circ \hat{\alpha} = \alpha$. Therefore, it is the unique lift of α at $\hat{x}_\beta = \hat{\beta}(1)$ so we have $\rho([\beta]) \circ \hat{\alpha} = l_{\hat{\beta}(1)}\alpha$.

The last step is to show that $\rho([\beta] \cdot [\alpha]) = \rho([\beta]) \circ \rho([\alpha])$. By the uniqueness of deck transformation, we only need to show that $\rho([\beta] \cdot [\alpha])(\hat{x}_0) = \rho([\beta]) \circ \rho([\alpha])(\hat{x}_0)$.

Notice that $\rho([\beta]) \circ \rho([\alpha])(\hat{x}_0) = \rho([\beta])(\hat{\alpha}(1)) = l_{\hat{\beta}(1)}(\alpha)(1)$.

On the other hand, $\rho([\beta] \cdot [\alpha])(\hat{x}_0) = \hat{\gamma}(1)$ where $\gamma = \beta \cdot \alpha$. By the path lifting property, we have a unique lift of α starting at $\hat{\beta}(1)$ which also has to be the second half of $\hat{\gamma}$ so $\hat{\gamma}(1)$ equals $l_{\hat{\beta}(1)}(\hat{\alpha})(1)$.

Thus, the two deck transformations send $\hat{x}_0$ to the same position and is thus same. $\square$

Remark 7. *We are finally able to conclude the proof of Theorem 14.6 by the fundamental theorem of group homomorphism.*

Theorem 14.11. *For any covering map p , we have $\text{Aut}(\hat{X}, \hat{x}_0) \cong N/p_*(\pi_1(\hat{X}, \hat{x}_0))$, where N is the normalizer of $p_*(\pi_1(\hat{X}, \hat{x}_0))$ in $\pi_1(X, x_0)$.*

Proof. The idea is the same since N being the normalizer implies $p_*(\pi_1(\hat{X}, \hat{x}_0))$ is normal in N and any of its conjugate groups is itself. $\square$

Theorem 14.12. *Let G be a group acting freely on a topological space $\hat{X}$. Suppose $q : \hat{X} \rightarrow \hat{X}/G = X$ is a covering map, then we have a bijective group homomorphism:*

$$\varphi : G \rightarrow \text{Aut}(\hat{X}/X)$$

that sends g to the map $f : x \rightarrow g \cdot x$.

Proof. For f to be a deck transformation, we need to have $qf(x) = q(x)$, so $Gx = Gf(x)$. Therefore, x and $f(x)$ are in the same orbit, so $\text{Aut}(\hat{X}/X)$ is the set of all deck transformations that map x to elements in Gx .

φ is clearly injective, since G acts freely on $\hat{X}$. φ is also surjective, as if $f \in \text{Aut}(\hat{X}/X)$ takes x to $g \cdot x \in Gx$, then $\varphi(g) = f$.

Therefore, φ is an isomorphism and thus we have

$$G \cong \text{Aut}(\hat{X}/X).$$

$\square$

Here comes some of the applications from these conclusions about covering spaces.

Theorem 14.13. *Every non-constant complex polynomial $p(z)$ has a root.*

Theorem 14.14. *For every map $f : S^2 \rightarrow \mathbb{R}^2$, there exists some $x \in S^2$ s.t. $f(x) = f(-x)$.*

Theorem 14.15. *Every continuous map $f : D^2 \rightarrow D^2$ has a fixed point.*

Corollary 14.16. *S^1 is not a retract of D^2 .*

Space of The Day (Abstract Simplicial Complex)

Definition 14.17. *Given a set V , an **abstract simplicial complex** is a set K of finite subsets of V s.t.*

- (i) $\{v\} \in K$ for any $v \in V$.
- (ii) $\sigma \in K$ and $\sigma' \subset \sigma$ implies $\sigma' \in K$.

Definition 14.18. *Given a simplicial complex K , we are able to define a abstract simplicial complex $\tilde{K}$ by letting V to be the set of 0-simplices in K (i.e., all the vertices appeared in some simplices in K). Additionally, $S \in \tilde{K}$ iff $\text{Conv}(S) \in K$.*

Definition 14.19. *Given abstract simplicial complexes (V, K) and (V', K') , an **isomorphism** between them is a bijection $f : V \rightarrow V'$ s.t. $S \in K$ iff $f(S) \in K'$*

Corollary 14.20. *Two abstract simplicial complexes K and K' are isomorphic iff there exists a homeomorphism from $|K|$ to $|K'|$ s.t. the restriction on each k -simplex in K is a affine map onto a k -simplex in K' .*

Exercise 14.21. *For any abstract simplicial complex (V, K) , there exists some simplicial complex K' s.t. $K \cong \tilde{K}'$.*

*K' is called a **geometric realization** of K .*

15 The Seifert Van Kampen Theorem

Theorem 15.1. *Let (X, x_0) be a path connected pointed space, $X = X_1 \cup X_2$ be an open cover by path connected spaces containing x_0 s.t. $X_1 \cap X_2 = X_0$ is also path connected. Let $i_k : X_k \rightarrow X$ and $j_k : X_0 \rightarrow X_k$ be inclusion maps.*

Then we have

$$\begin{array}{ccc} \pi_1(X_0, x_0) & \xrightarrow{(j_1)_*} & \pi_1(X_1, x_0) \\ \downarrow (j_2)_* & & \downarrow (i_1)_* \\ \pi_1(X_2, x_0) & \xrightarrow{(i_2)_*} & \pi_1(X, x_0) \end{array}$$

*a pushout square in **Grp**.*

*In other words, $\pi_1(X, x_0) \cong \pi_1(X_1, x_0) *_{\pi_1(X_0, x_0)} \pi_1(X_2, x_0)$.*

Remark 8. *Notice that the diagram of these spaces is a pushout in **Top**, so π_1 is a functor preserving this specific type of pushout square in **Grp**.*

Exercise 15.2. (a) $\pi_1(S^n)$ is trivial.

(b) $\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z}$, the free group on 2 generators.

(c) Given $(X_1, x_1), (X_2, x_2) \in \mathbf{Top}_*$ path connected, if each x_k has a contractible neighborhood then $\pi_1(X_1 \vee X_2) \cong \pi_1(X_1) * \pi_1(X_2)$.

Proof. (a) Let x_1, x_2 be two antipodal points on S^n . We have $S^n - \{x_i\} \cong \mathbb{R}^n$, and $S^n - \{x_1, x_2\}$ clearly path connected. Therefore, Van Kampen applied and we get $\pi_1(S^n) \cong \mathbf{1}$.

(b) Let x be the point of intersection of the wedge of two circles. Choose a small open contractible subset of each S^1 , call them X_1 and X_2 . Notice that $(S^1 \cup X_2) \cap (S^1 \cup X_1) = X_1 \cup X_2$ is path connected and contractible, so it has trivial fundamental group.

Then we have that $\pi_1(S_1 \vee S_1) \cong \pi_1(S_1 \cup X_1) *_{\pi_1(X_1 \cup X_2)} \pi_1(S_1 \cup X_2) \cong \pi_1(S_1 \cup X_1) * \pi_1(S_1 \cup X_2) \cong \pi_1(S^1) * \pi_1(S^1) \cong \mathbb{Z} * \mathbb{Z}$.

(c) The idea is basically the same as what we did above for $S^1 \vee S^1$.

□

Theorem 15.3. *For Groupoid version, we have the following diagram a pushout in **Gpd**.*

$$\begin{array}{ccc} \Pi_1(X_0) & \xrightarrow{(j_1)_*} & \Pi_1(X_1) \\ \downarrow (j_2)_* & & \downarrow (i_1)_* \\ \Pi_1(X_2) & \xrightarrow{(i_2)_*} & \Pi_1(X) \end{array}$$

Theorem 15.4. *Suppose X is a finite cell complex with exactly one 0-cell, and that the 1-skeleton is given by $X_1 \cong \bigvee_{i=1}^n S^1$ and $\pi_1(X) \cong \mathbb{F}_n$.*

Let $\{\varphi_j : D^2 \rightarrow X\}$ be the set of 2-cells, and the attaching maps $\gamma_j : S^1 \rightarrow X$ are all loops at x_0 , so $[\gamma_j] \in \pi_1(X_1) \cong \mathbb{F}_n$.

Then we have

$$\pi_1(X, x_0) \cong \mathbb{F}_n / \langle \gamma_1, \dots, \gamma_m \rangle,$$

where the quotient is the normal subgroups generated by the γ 's.

Therefore, $\pi_1(X)$ only depends on X_2 . We then have that $\pi_1(X_G) \cong G$.

Proposition 15.5. *If $i : A \rightarrow X$ is an inclusion of a contractible subcomplex A of a CW complex X , then the quotient $q : X \rightarrow X/A$ is a homotopy equivalence and $\pi_1(X) \cong \pi_1(X/A)$.*

With this proposition, we are able to prove:

Theorem 15.6. *Given a connected graph Γ , $\pi_1(\Gamma)$ is free on the edges in $\Gamma - T$, where T is the maximal subtree of Γ .*

Proof. Let Γ be finite. Let $T \subset \Gamma$, then T is contractible. We then have that Γ/T is homeomorphic to a wedge of circles, and since $\pi_1(\Gamma) \cong \pi_1(\Gamma/T)$ by the above proposition, we have that $\pi_1(\Gamma)$ is free on the edges in $\Gamma - T$. $\square$

Theorem 15.7. *For any finitely-dimensional CW complex X , $i : X_2 \rightarrow X$ induces an isomorphism $i_* : \pi_1(X_2) \rightarrow \pi_1(X)$.*

We first show the following:

Exercise 15.8. *Fix $n \geq 3$. Suppose Y is obtained from a path connected space X by attaching an n -cell. In other words, there is a pushout:*

$$\begin{array}{ccc} S^{n-1} & \xrightarrow{i} & D^n \\ \downarrow j & & \downarrow i_2 \\ X & \xrightarrow{i_1} & Y \end{array}$$

Then we have that $i_{1} : \pi_1(X, x_0) \rightarrow \pi_1(Y, i_1(x_0))$ is an isomorphism.*

Proof. Let $D' \subset D^n$ be a slightly smaller closed disk. We then have that $\text{Int } D^n$ and $Y - D'$ forming an open cover of Y . Their intersection is homotopic to S^{n-1} by a deformation retract.

Therefore, by VKP, $\pi_1(Y) \cong \pi_1(\text{Int } D^n) *_{\pi_1(S^{n-1})} \pi_1(Y - D')$. Since $\pi_1(\text{Int } D^n)$ and $\pi_1(S^{n-1})$ is trivial and $Y - D'$ can be deformed onto X , we then have $\pi_1(Y) \cong \pi_1(X)$. $\square$

Space of The Day (Affine Algebraic Varieties)

Definition 15.9. *Let k be either $\mathbb{R}$ or $\mathbb{C}$. A **real (complex) affine algebraic variety** is a subspace $V \subset k^n$ which is the vanishing set of some set of polynomials.*

In other words, $V = \{\mathbf{x} \in k^n \mid p_1(\mathbf{x}) = \dots = p_m(\mathbf{x}) = 0\}$, where $n \in \mathbb{N}$ and $p_1, \dots, p_m \in k[x_1, \dots, x_n]$.

Exercise 15.10. *Show that $\{x, y \in \mathbb{C}^2 \mid xy = 0\} \cong \mathbb{R}^2 \vee \mathbb{R}^2$.*

Proof. The intersection is $(0, 0) \in \mathbb{C}^2$. In other cases, one of x and y is 0 and the other one is arbitrary in $\mathbb{C} \cong \mathbb{R}^2$. $\square$

Exercise 15.11. *$\{x, y \in \mathbb{C}^2 \mid x^2 + y^2 = 1\} \cong \mathbb{R}^2 - \{0\}$.*

16 Existence of the Universal Cover

Definition 16.1. A space X is **semi-locally simply connected** if for every $x \in X$, $\exists U \ni x$ open in X s.t. the image of the homomorphism $i_* : \pi_1(U, x) \rightarrow \pi_1(X, x)$ has trivial image, where $i : U \rightarrow X$ is the inclusion map.

In other words, any loop $\gamma : x \rightsquigarrow x$ entirely lying in U is homotopic to the identity map at x in X , but they may not be homotopic in U .

Corollary 16.2. Any manifold or CW complex is semi-locally simply connected.

Theorem 16.3. Any pointed connected and locally connected and semi-locally simply connected space (X, x_0) admits an universal cover $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$, which is unique up to isomorphism of pointed spaces.

Theorem 16.4. Let X be connected and locally connected and semi-locally simply connected.

Let $\text{Cov}(X, x_0) = \{p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)\}$, and $p_1 \sim p_2$ iff the two covering spaces are isomorphic. Let $\text{Sub}(X, x_0) = \{H < \pi_1(X, x_0)\}$.

Now we have $\pi_1 : [p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)] \rightarrow p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ a bijection.

Proof. Injectivity comes from the lifting criterion. If $\pi_1(p_1) = \pi_1(p_2)$, then there is a lift of p_1 along p_2 and vice versa. The two lifts thus form an isomorphism between the two covering spaces so $p_1 \sim p_2$.

To prove surjectivity, we fix an universal cover $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$. Given $H < \pi_1(X, x_0)$, we have an isomorphism

$$\rho : \pi_1(X, x_0) \rightarrow \text{Aut}(\tilde{X}/X)$$

by $\rho([\gamma])(\hat{x}_0) = \hat{\gamma}(1)$, as we proved in the earlier sections. Notice that two points $\tilde{y}_1, \tilde{y}_2$ satisfy $p(\tilde{y}_1) = p(\tilde{y}_2)$ iff they differ by a deck transformation, so therefore $X \cong \tilde{X}/G$, and ρ is therefore a covering space action.

We can then show that $\rho|_H$ is also a covering space action, and hence $q : \tilde{X} \rightarrow \tilde{X}_H = \hat{X}_H$ is a covering map. The map $p_H : \hat{X}_H \rightarrow X$ is a covering map as the inverse of U is $q \circ p^{-1}(U)$, which is a sheet of disjoint sets homeomorphic to U , as p and q are homeomorphisms.

Setting $\hat{x}_H = q(\tilde{x})$, we have that $\pi_1(\hat{X}_H) \cong H$, and we want to show that $(p_H)_*(\pi_1(\hat{X}_H)) \cong H$.

Notice that the isomorphism $\rho_H : \pi_1(\tilde{X}_H, \hat{x}_H) \rightarrow \text{Aut}(\tilde{X}/X_H)$ factors through $(p_H)_*$:

$$\pi_1(\hat{X}_H, \hat{x}_H) \rightarrow \pi_1(X, x_0) \rightarrow \text{Aut}(\tilde{X}/X)$$

as a composition of $(p_H)_*$ and ρ . The claim then follows. □

Exercise 16.5. Let (X, x_0) be path connected and $p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be an covering map.

The associated subgroup $H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ has finite index iff $\tilde{X}$ has a finite number of sheets. In this case the number of sheets is exactly the index.

Proof. Let $[\gamma_1]H$ and $[\gamma_2]H$ be two cosets of H . We claim that they are equal iff their lifts $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ starting at $\tilde{x}_0$ end at the same point, call it $\tilde{x}_j$.

Suppose both lifts end at $\tilde{x}_j$, then we have $\tilde{\gamma}_1 \cdot \overline{\tilde{\gamma}_2}$ a loop at $\tilde{x}_0$. Thus, $p(\tilde{\gamma}_1 \cdot \overline{\tilde{\gamma}_2}) \in H$, so $[\gamma_1 \cdot \overline{\gamma_2}] \in H$, so $[\gamma_1]H = [\gamma_2]H$.

On the other hand, let $[\gamma_1 \cdot \overline{\gamma_2}] \in H$. By the homotopy lifting property, we have $\tilde{\gamma}_1 \cdot \overline{\tilde{\gamma}_2}$ a loop at $\tilde{x}_0$. Therefore, the two lifts $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ have the same endpoints.

Since $\tilde{X}$ is connected, we have a path from $\tilde{x}_0$ to any $\tilde{x}_j$ if $\tilde{x}_j \in p^{-1}(x_0)$. Let it be γ_j . Then $p(\gamma_j)$ is a loop at x_0 and therefore represents a coset of H in $\pi_1(X, x_0)$. Therefore, there is a bijection between the cosets of H and the lifts of x_0 , so the claim holds. □

Theorem 16.6. Any subgroup $H \subset \mathbf{F}_n$ of a free group is free.

Proof. This is proved by noticing that any covering of a graph is a graph.

Assume X is the graph with $\pi_1(X) \cong \mathbf{F}_n$. It has an universal cover and therefore the theorems apply. Any covering is a graph so any subgroup of $\pi_1(X)$ is a free subgroup under p_* , as the image of free groups under group homomorphisms is still free. $\square$

Space of The Day (Projective Varieties)

Definition 16.7. A polynomial $p \in k[x_0, \dots, x_n]$ is **homogeneous** if every monomial of it has the same degree d .

Notice that we then have $p(\lambda a_0, \dots, \lambda a_n) = \lambda^d p(a_0, \dots, a_n)$. Therefore, it makes sense to ask when does $p(a_0, \dots, a_n) = 0$.

Remark 9. Homogeneous polynomials allow us to factor them through kP^n . In other words, the following diagram commutes:

$$\begin{array}{ccc} k^{n+1} & \xrightarrow{q} & kP^n \\ & \searrow p & \downarrow \bar{p} \\ & & k \end{array}$$

Definition 16.8. Given a set of homogeneous polynomials $p_1, \dots, p_m \in k[x_0, \dots, x_n]$, we consider the set of $(a_0 : \dots : a_n) \in k^n$ s.t. $p_i(a_0 : \dots : a_n) = 0$ for any i . Such a space is called a **projective variety**. In some sense, a projective variety to a projective space is similar to an algebraic variety to k^{n+1} .

Exercise 16.9. kP^n is a projective variety itself by taking $p_1 = 0$.

Exercise 16.10. Any projective variety is compact.

Definition 16.11. Any polynomial $p(x_1, \dots, x_n)$ has a **homogenization** by multiplying some power of a new introduced variable, x_0 , to each monomial so that the new polynomial is homogeneous.

Example 10. $x^2 + y^2 = 1$ has a homogenization $x^2 + y^2 = z^2$.

Therefore, any algebraic variety in k^n gives rise to a projective variety in kP^n .

Exercise 16.12. The intersection of $X = \{\tilde{p}_1 = \dots = \tilde{p}_m = 0\} \subset kP^n$ and the affine space $\{x_0 : \dots : x_n \in k \mid x_0 \neq 0\} \cong k^n$ is $\{p_1 = \dots = p_m = 0\}$.

17 The Higher Homotopy Groups

Proposition 17.1. $S^n/E_{n-1} \cong S^n \vee S^n$ where $E_{n-1} = \{x \in S^n, x_0 = 0\}$, which represents the equator of S^n .

Theorem 17.2. Given $f, g : S^n \rightarrow (X, x_0)$, we have an induced map

$$S^n \rightarrow S^n \vee S^n \rightarrow (X, x_0).$$

Remark 10. When $n = 1$, this pinching map defines the composition of loops for $\pi_1(X, x_0)$.

Definition 17.3. Let $\partial[0, 1]^n = \{(x_1, \dots, x_n), x_i = 0 \text{ for some } i\}$. We then have $S^n \cong D^n/S^{n-1} \cong [0, 1]^n/\partial[0, 1]^n$

Thus, a pointed map $f : S^n \rightarrow (X, x_0)$ is represented by a map $f : [0, 1]^n \rightarrow (X, x_0)$ with $f(\partial[0, 1]^n) = x_0$.

Definition 17.4. Generally, a **topological pair** is (X, A) with $X \in \mathbf{Top}$, $A \subset X$.

A **map of pairs** $f : (X, A) \rightarrow (Y, B)$ is a continuous map $f : X \rightarrow Y$ s.t. $f(A) \subset B$.

A **homotopy of maps of pairs** is a map $h : (X, A) \times I \rightarrow (Y, B)$ s.t.

(i) $h_t(A) \subset B, \forall t$.

(ii) $h_0(x) = f(x), h_1(x) = g(x)$.

Notice that a homotopy of maps of pairs with $X = I$, A, B are both single points is exactly a homotopy relative to endpoints.

Definition 17.5. Given $\alpha, \beta : (I^n, \partial I^n) \rightarrow (X, x_0)$, their composite $\alpha \cdot \beta$ is given by

$$\alpha \cdot \beta(x_1, \dots, x_n) = \begin{cases} \alpha(2x_1, x_2, \dots, x_n) & 0 \leq x_1 \leq \frac{1}{2} \\ \beta(2x_1 - 1, x_2, \dots, x_n) & \frac{1}{2} \leq x_1 \leq 1 \end{cases}$$

which is a map of pairs.

Remark 11. Many properties and the corresponding proofs are very similar to the case $n = 1$, so we are not going to mention them again in this section.

Definition 17.6. With the operation defined above, we are able to construct the **n -th homotopy group** $\pi_n(X, x_0)$ to be the set

$$\{\alpha : (I^n, \partial I^n) \rightarrow (X, x_0) / \sim\}$$

where $\alpha \sim \beta$ if they are homotopic. The group operation is still $[\alpha] \cdot [\beta] = [\alpha \cdot \beta]$.

Definition 17.7. For $n = 0$, we define $\pi_0(X, x_0)$ to be the pointed set $\{\alpha : (I^0, \partial I^0) \rightarrow (X, x_0)\} / \sim$.

Here, I^0 is defined as the set with a single point $\{0\}$, and $\partial I^0 = \emptyset$. Therefore, $\pi_0(X, x_0)$ is in bijection with the path components of X , as two maps in $\pi_0(X, x_0)$ are homotopic (equivalent) iff the points they map to are connected by a path, which can be seen as the homotopy between the two maps.

Remark 12. While this seems to violate the pattern that $S^0 \cong I^0/\partial I^0$, it does not because a quotient space X/A is often constructed by first adjoining X with a point, getting $X \sqcup *$, and collapsing $A \cup *$ to a point.

Therefore, $I^0/\partial I^0 = I^0/\emptyset \cong I^0 \sqcup * \cong S^0$.

This definition turns collapsing a subspace into a functor $\mathbf{Top}_2 \rightarrow \mathbf{Top}_*$.

Corollary 17.8. (i) $\gamma : x_0 \rightsquigarrow x_1$ induces an isomorphism $\gamma_* : \pi_n(X, x_0) \rightarrow \pi_n(X, x_1)$.

Hence, $\pi_n(X)$ is well-defined for any path connected X .

(ii) We can thus define $f_* : \pi_n(X, x_0) \rightarrow \pi_n(Y, f(x_0))$ for $f : X \rightarrow Y$, which turns $\pi_n : \mathbf{Top}_* \rightarrow \mathbf{Grp}$ and $\pi_0 : \mathbf{Top}_* \rightarrow \mathbf{Set}_*$ into functors.

- (iii) If f, g are homotopic relative to endpoints, then $f_* = g_* : \pi_n(X, x_0) \rightarrow \pi_n(Y, f x_0)$.
 (iv) Any homotopy equivalence f induces an isomorphism $f_* : \pi_n(X, x_0) \rightarrow \pi_n(Y, f x_0)$.

Theorem 17.9. For any (X, x_0) , $\pi_n(X, x_0)$ is Abelian if $n \geq 2$.

Theorem 17.10. Suppose that $*$, $\otimes$ are two binary operations $M \times M \rightarrow M$, both with a respective unit. If $(a * b) \otimes (x * y) = (a \otimes x) * (b \otimes y)$ for any a, b, x, y , then we have $* = \otimes$, and this operation is associative and commutative.

Proof. Notice that $(1_* * 1_\otimes) \otimes (1_\otimes * 1_*) = (1_* \otimes 1_\otimes) * (1_\otimes * 1_*)$ so $1_\otimes \otimes 1_\otimes = 1_* * 1_*$ so $1_* = 1_\otimes$, which we will denote by 1 in the following proof.

Let $b, x = 1$, we then have $(a * 1) \otimes (1 * y) = a \otimes y = (a \otimes 1) * (1 \otimes y) = a * y$, so the two operations are the same.

Commutativity then follows as $(1 * b) * (x * y) = b * x = (1 * x) * (b * 1) = x * b$. Associativity also follows as $(a * b) * x = (a * b) * (1 * x) = (a * 1) * (b * x) = a * (b * x)$. $\square$

Remark 13. We can thus prove that $\pi_n(X, x_0)$ is Abelian by letting $\cdot_i : \pi_n(X, x_0) \times \pi_n(X, x_0) \rightarrow \pi_n(X, x_0)$ by using the i th index for composition. In other words, we define

$$\alpha \cdot_i \beta(x_1, \dots, x_n) = \begin{cases} \alpha(x_1, x_2, \dots, 2x_i, \dots, x_n) & 0 \leq x_i \leq \frac{1}{2} \\ \beta(x_1, x_2, \dots, 2x_i - 1, \dots, x_n) & \frac{1}{2} \leq x_i \leq 1 \end{cases}$$

Then we have that any two $\cdot_i$ and $\cdot_j$ satisfying the above theorem. Therefore, $\cdot_i$ is commutative and indicates that $\pi_n(X, x_0)$ is Abelian.

Therefore, π_n is actually a functor $\mathbf{Top}_* \rightarrow \mathbf{Ab}$.

Corollary 17.11. $\pi_k(S^n)$ is trivial for any $0 \leq k < n$.

Proof. Due to differential geometry, any map $f : S^k \rightarrow S^n$ is not surjective, so it can be seen as a map from S^k to $S^n - \{*\} \cong \mathbb{R}^n$ which is contractible. Therefore, f is nullhomotopic. $\square$

Theorem 17.12 (Whitehead's Theorem). $f : X \rightarrow Y$ is called a **weak homotopy equivalence** if f induces isomorphisms on π_n for any n .

If X and Y are path connected CW complexes and f is a weak homotopy equivalence, then we have f a homotopy equivalence.

Theorem 17.13. Every space X is weakly equivalent to a CW complex.

18 Semi-Simplicial Set (Δ -Set)

Definition 18.1. Let X be an abstract simplicial complex with a total ordering on the underlying set V . Define X_n to be the set of n -simplices of X .

Any $\sigma \in X_n$ can thus be written as $(v_0, \dots, v_n)$ in increasing order.

Let $\sigma = (v_0, \dots, v_n) \in X_n$, then we obtain an $n-1$ -simplex $\partial_i \sigma$, called the i th face of σ by omitting the i th vertex. (i.e., $\partial_i \sigma = (v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_n)$).

We thus can define a map $\partial_i^{(n)} : X_n \rightarrow X_{n-1}$, for any $0 \leq i \leq n$.

Exercise 18.2. For any $n > 1$, $0 \leq i < j \leq n$, then $\partial_i \circ \partial_j = \partial_{j-1} \circ \partial_i : X_n \rightarrow X_{n-2}$.

Proof. Let $\sigma = (v_0, \dots, v_n)$, then both sides send σ to $(v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_{j-1}, v_{j+1}, \dots, v_n)$. $\square$

Definition 18.3. The *semi-simplicial set* (Δ -set) is thus defined to be

(i) A sequence of sets X_n for $n \geq 0$.

(ii) $\partial_i^n : X_n \rightarrow X_{n-1}$ for $n > 0$, $0 \leq i \leq n$ satisfying:

(iii) $\partial_i \circ \partial_j = \partial_{j-1} \circ \partial_i : X_n \rightarrow X_{n-2}$ for all $n > 2$ and $0 \leq i < j \leq n$.

Definition 18.4. A morphism of Δ -set $f : X \rightarrow Y$ consists of maps

$$f_n : X_n \rightarrow Y_n \text{ for all } n \geq 0 \text{ and } \partial_i \circ f_n = f_{n-1} \circ \partial_i \text{ for any } 0 \leq i \leq n-1.$$

In other words, the following diagram commutes:

$$\begin{array}{ccc} X_n & \xrightarrow{\partial_i} & X_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ Y_n & \xrightarrow{\partial_i} & Y_{n-1} \end{array}$$

Corollary 18.5. The set of semi-simplicial sets form a category, denoted as $\Delta\mathbf{Set}$.

Proof. Everything is clear except for the well-definedness. We need to verify that $g \circ f : X \rightarrow Z$ gives a morphism from X to Z , where $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are morphisms of simplicial sets.

Notice that $\partial_i \circ (gf)_n = \partial_i \circ g_n \circ f_n = g_{n-1} \circ \partial_i \circ f_n = g_{n-1} \circ f_{n-1} \circ \partial_i = (gf)_{n-1} \circ \partial_i$. Therefore, gf is indeed a morphism of simplicial sets, proving that $\Delta\mathbf{Set}$ is indeed a category. $\square$

Without additional remark, $[n]$ will be used to denote the set $\{0, \dots, n\}$.

Definition 18.6. Let $v = (v_0, \dots, v_n) \in X_n$ be an n -simplex, $f : [k] \rightarrow [n]$ strictly increasing. We then have a k -simplex $f^* \sigma = (v_{f(0)}, \dots, v_{f(k)})$. Thus: $f^* : X_n \rightarrow X_k$ is a well-defined map.

Exercise 18.7. $\partial_i = \delta_i^*$ where $\delta_i : [n-1] \rightarrow [n]$ with $i \notin \text{im}(\delta_i)$.

Definition 18.8. Let Δ_{inj} be the subcategory of $\mathbf{Set}$ whose objects are $[n]$ for $n \geq 0$ and morphisms are strictly increasing maps.

Exercise 18.9. Given any Δ -set X , there is a unique functor $X : \Delta_{inj}^{op} \rightarrow \mathbf{Set}$ with $X([n]) = X_n$, $X(\delta_i^{(n)}) = \partial_i^{(n)} : X_n \rightarrow X_{n-1}$ where $\delta_i^{(n)} : [n-1] \rightarrow [n]$ with $i \notin \text{im}(\delta_i^{(n)})$.

Exercise 18.10. Let X be a Δ -set, $\beta_i : [0] \rightarrow [n]$ with $\beta_i(0) = i$. Then X is isomorphic to the one coming from the abstract simplicial complex iff $\langle \beta_0^*, \dots, \beta_n^* \rangle : X_n \rightarrow X_0^{n+1}$ is injective.

In other words, if $\beta_i^*(\sigma) = \beta_i^*(\sigma')$ for all i , then $\sigma = \sigma'$.

Corollary 18.11. $\Delta\mathbf{Set}$ is thus the category of functors $\Delta_{inj}^{op} \rightarrow \mathbf{Set}$.

Exercise 18.12. Given Δ -sets X and Y . A sequence of maps $f_n : X_n \rightarrow Y_n$ is a morphism of Δ -sets iff it defines a natural transformation between X and $Y : \Delta_{inj}^{op} \rightarrow \mathbf{Set}$.

Proof. Notice that the following commutative diagram (defining a natural transformation)

$$\begin{array}{ccc} X_n & \xrightarrow{\partial_i} & X_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ Y_n & \xrightarrow{\partial'_i} & Y_{n-1} \end{array}$$

is exactly the definition of a morphism between X and Y . $\square$

Definition 18.13. For each n , let Δ^n be the standard n -simplex. For each $\alpha : [k] \rightarrow [n]$ in Δ , let $\alpha : \Delta^k \rightarrow \Delta^n$ with $\alpha(e_i) = e_{\alpha(i)}$. This defines a functor $\Delta_{inj} \rightarrow \mathbf{Top}$.

Given a Δ -set X , its **geometric realization** is defined as $|X| = (\bigsqcup X_n \times \Delta^n) / \sim$ where $(\alpha_*(\sigma), x) \sim (\sigma, \alpha(x))$ for $\sigma \in X_n$, $\alpha : [k] \rightarrow [n]$, $x \in \Delta^k$.

In other words, given $\sigma_1, \sigma_2 \in X_{n_1}, X_{n_2}$, $x_1, x_2 \in \Delta^{n_1}, \Delta^{n_2}$, $(\sigma_1, x_1) \sim (\sigma_2, x_2)$ iff $\exists k$ and $(\sigma', x') \in X_k \times \Delta^k$ s.t. $\exists \alpha_i : [k] \rightarrow [n_i]$ s.t. $\alpha'_i(\sigma') = \sigma_i$, $\alpha_i(x') = x_i$.

Exercise 18.14. Given $f : X \rightarrow Y$ in $\mathbf{\Delta Set}$, $\exists$ a unique continuous map $|f| : |X| \rightarrow |Y|$ s.t. $[f][(\sigma, x)] = [f(\sigma), x]$.

This defines a functor $|-| : \mathbf{\Delta Set} \rightarrow \mathbf{Top}$.

Space of The Day (Topological Group)

Definition 18.15. A **topological group** is a topological space with a group structure s.t. the multiplication $G \times G \rightarrow G$ and inverse maps $G \rightarrow G$ are continuous.

Exercise 18.16. (i) Any discrete group (or a group with discrete topology) is a topological group.

(ii) $GL_n(\mathbb{R}) \subset \mathbb{R}^{n \times n}$ and $GL_n(\mathbb{C}) \subset \mathbb{C}^{n \times n}$ are topological group with subspace topology.

(iii) If G is a topological group and $H \subset G$ a subgroup, then H with the subspace topology is a topological group.

Proof. (i) The discrete topology makes sure that the multiplication and inverse maps are continuous, so the result follows.

(ii) We know that addition and inverse on $k^{n \times n}$ is continuous, so the restrictions of them on the subspaces $GL_n(k)$ are also continuous, where k denotes $\mathbb{R}$ or $\mathbb{C}$.

(iii) Similarly, the restriction of the multiplication and inverse maps are continuous on H , so H is also a topological group. $\square$

Definition 18.17. It follows that any subgroups of $GL_n(\mathbb{R})$ and $GL_n(\mathbb{C})$ are topological groups when given subspace topology, including SL_n , O_n , SO_n , etc. These topological groups are called **matrix groups**.

Exercise 18.18. The fundamental group of any connected topological group is Abelian.

Proof. The proof is based on Eckmann-Hilton Theorem. Pick the identity e to be the basepoint. Let $*$: $\pi_1(G) \times \pi_1(G) \rightarrow \pi_1(G)$ with $f * g(t) = f(t) * g(t)$.

We thus have $(f * g) \cdot (h * k) = (f \cdot h) * (g \cdot k)$ where $\cdot$ is the usual concatenation operation in the fundamental group. Therefore, $\cdot$ is commutative so $\pi_1(G)$ is Abelian. $\square$

19 Simplicial Homology

Definition 19.1. Let X be a Δ -set. The n -th chain **chain group** $C_n(X)$ of X is defined to be the free Abelian group generated by the sets X_n of n -simplices of X .

An n -th **chain** is an element in C_n . In other words, any n -th chain can be written as

$$c = \sum_{i=1}^N a_i \sigma_i$$

for $N \in \mathbb{N}$, $a_i \in \mathbb{Z}$, and $\sigma_i \in X_n$.

Additionally, $c \in C_n$ is equivalent to a function $f : X_n \rightarrow \mathbb{Z}$ where $f(\sigma) = 0$ for all but finitely $\sigma \in X_n$.

Definition 19.2. For any $n > 0$, we have the **boundary operator** $\partial : C_n(X) \rightarrow C_{n-1}(X)$ on the generators of $C_n(X)$ by the formula

$$\partial \sigma = \sum_{i=0}^n (-1)^i \partial_i \sigma.$$

Therefore, an arbitrary element in C_n has its boundary computed as

$$\partial c = \sum_{i=0}^n c_i \partial \sigma_i.$$

We also define $C_{-1}(X) = 0$ and $\partial : C_0 \rightarrow C_{-1}$ to be the zero map.

Definition 19.3. An n -th chain is called a **n -cycle** if $\partial c = \sum_{i=0}^n c_i \partial \sigma_i = 0$ ($c \in \text{Ker} \partial^{(n)}$).

c is called a **boundary** if $c \in \text{Im} \partial^{(n+1)}$.

We denote Z_n and B_n to be the subgroups of $C_n(X)$ consisting of all the cycles or boundaries.

By our definition above, we have that $Z_0 = C_0$.

Exercise 19.4. $\partial^{(n)} \circ \partial^{(n+1)} = 0$ (i.e. $B_n \subset Z_n$).

Proof. We directly check this from the definition.

We have

$$\begin{aligned} \partial^{(n)} \circ \partial^{(n+1)}(\sigma) &= \partial^{(n)} \left(\sum_{i=0}^{n+1} (-1)^i \partial_i \sigma \right) \\ &= \left(\sum_{i \leq j} (-1)^{i+j} \partial_j \partial_i \sigma \right) + \left(\sum_{i > j} (-1)^{i+j} \partial_j \partial_i \sigma \right) \\ &= \left(\sum_{i \leq j} (-1)^{i+j} \partial_j \partial_i \sigma \right) + \left(\sum_{i > j} (-1)^{i+j} \partial_{i-1} \partial_j \sigma \right) \\ &= \left(\sum_{i \leq j} (-1)^{i+j} \partial_j \partial_i \sigma \right) + \left(\sum_{i' \leq j'} (-1)^{i'+j'-1} \partial_{j'} \partial_{i'} \sigma \right) \\ &= 0. \end{aligned}$$

□

Definition 19.5. The **n -th homology group** of a Δ -set X is defined to be the quotient $H_n = Z_n/B_n$.

Hence for each $c \in Z_n(X)$, we have its homology class $[c] \in H_n(X)$.

c and c' are called **homologous** if $[c] = [c']$.

Remark 14. If X is a n -dimensional Δ -set, then $H_k(X) \cong 0$ for $k > n$.

Theorem 19.6. For every finitely generated Abelian group G , there are unique primes $p_1 \leq \dots \leq p_k$ and a unique set of non-negative integers $r, s_1, \dots, s_k$ s.t.

$$G \cong \mathbb{Z}^r \bigoplus_{i=1}^k \mathbb{Z}/p_i^{s_i} \mathbb{Z}.$$

r is called the **rank** of G .

Definition 19.7. For a Δ -set X with finitely generated homology groups, the rank of $H_i(X)$ is called the i -th **Betti number** of X , denoted $\beta_i(X)$.

Exercise 19.8. Let X be a Δ -set, H_0 is free Abelian on the set of components of X , where the components of X is defined as the set of 0-simplices up to being connected by a sequence of 1-simplex.

The components are in bijection with the path components of $|X|$.

Proof. By definition, we have $H_0(X) = Z_0(X)/B_0(X)$. Notice that $Z_0(X) = X_0$ and for $x, y \in X_0$, $(x - y) \in B_0(X)$ if $x - y = \partial_0(\sigma) - \partial_1(\sigma)$ for some $\sigma \in X_1$ (i.e., x and y are connected by a 1-simplex).

Therefore, the quotient $H_0(X)$ is as stated. $\square$

Exercise 19.9. If G is a 1- Δ -set (graph) and $T \subset G$ is a maximal sub-tree, then $H_1(G)$ is free Abelian on the edges in the complement of T .

Proof. Notice that by definition, for $c \in C_1(X)$, $\partial c = \sum c_i \partial \sigma_i = 0$ iff c is a cycle, as this sum turns out to be a sequence of addition and subtraction on the vertices in X .

As T is the maximal sub-tree, we know that adding any edge $e \in G \setminus T$ to T we are going to form a cycle, call it c_e .

Now, we claim that $H_1 = Z_1$ is free Abelian on these c_e .

Let's say C is a cycle in G . For every edge $e \in C \cap (G \setminus T)$, define k_e to be the times it appear in the cycle. Then, we are going to have $C - \sum k_e c_e$ equal to some cycle that only has edges from T , which has to be the trivial cycle.

Therefore, we have proved that $H_1(G)$ is free Abelian on the edges in $G \setminus T$. $\square$

Definition 19.10. Given a map $f : X \rightarrow Y$ in $\Delta\mathbf{Set}$, we define $f_* : C_n(X) \rightarrow C_n(Y)$ with $f_*(\sigma) = f(\sigma)$ on the generators (and extended linearly).

We also define $f_* : H_n(X) \rightarrow H_n(Y)$ by $f_*([c]) = [f(c)]$.

This thus defines a functor $H_n : \Delta\mathbf{Set} \rightarrow \mathbf{Ab}$.

Show that everything is well-defined. In other words, the following diagram commutes

$$\begin{array}{ccc} Z_n(X) & \xrightarrow{f_*|_{Z_n}} & Z_n(Y) \\ \downarrow q_X & & \downarrow q_Y \\ H_n(X) & \xrightarrow{f_*} & H_n(Y) \end{array}.$$

Proof. We want to show that for $c, c' \in Z_n(X)$, if $[c] = [c']$, then

- (i) $f(c) \in Z_n(Y)$.
- (ii) $[f(c)] = [f(c')]$.

For (i), by definition of a morphism of Δ -sets, we have $\partial f(c) = f(\partial c) = f(0) = 0$ so $f(c) \in Z_n(Y)$.

For (ii), suppose $c - c' = \partial b$ for some $b \in C_{n+1}(X)$. Since f is a group homomorphism, we have $f(c) - f(c') = f(c - c') = f(\partial b) = \partial f(b)$. Therefore, $f(c) - f(c') \in B_n(Y)$ and $[f(c)] = [f(c')]$.

Therefore, f_* is a well-defined group homomorphism and H_n therefore is indeed a functor. $\square$

Space of The Day (Loop Space)

Definition 19.11. Given a pointed space (X, x_0) , we have defined $\Omega_{x_0} X$ to be the set of loops of X at x_0 , i.e., the set of pointed map $S^1 \rightarrow (X, x_0)$.

Equipped with compact-open topology, $\Omega_{x_0}(X)$ becomes a topological space, called the **loop space** of (X, x_0) .

Remark 15. Given two topological spaces X and Y , the compact-open topology on $C(X, Y)$ is defined by its subbasis: the set $V(K, U)$, which contains $f : X \rightarrow Y$ such that $f(K) \subset U$ for some fixed K compact in X and U open in Y .

We then give Ω_{x_0} the subspace topology relative to $C(S^1, X)$.

Since S^1 is compact Hausdorff, $f : Y \rightarrow \Omega_{x_0}(X)$ is continuous iff the corresponding map $f' : Y \times S^1 \rightarrow X$ is continuous.

Exercise 19.12. Show that the operation $\Omega_{x_0}(X) \times \Omega_{x_0}(X) \rightarrow \Omega_{x_0}(X)$ given by concatenating loops is continuous.

Exercise 19.13. Notice that $\Omega_{x_0}(X)$ has a natural basepoint at c_{x_0} , so Ω can be seen as a functor from $\mathbf{Top}_*$ to $\mathbf{Top}_*$.

Proof. Notice that for any $f : X \rightarrow Y$, $f_* : \Omega_{x_0}(X) \rightarrow \Omega_{f(x_0)}(Y)$ is well defined by sending γ to $f\gamma$. Checking that this turns Ω into a functor is routine. $\square$

Exercise 19.14. Show that $\pi_1(X, x_0) \cong \pi_0(\Omega_{x_0}(X))$.

Proof. Notice that $\gamma_1 \sim \gamma_2$ iff they are connected by a homotopy, i.e., a path in $\Omega_{x_0}(X)$. $\square$

Exercise 19.15. We can now iterate Ω and form $\Omega^2(X, x_0) = \Omega_{c_{x_0}}\Omega_{x_0}(X)$.

Show that $\pi_n(\Omega_{x_0}(X)) \cong \pi_0(\Omega^n(X, x_0))$.

Proof. We prove by induction. Call the basepoint of $\Omega_n(X)$ $c_{x_0}^n$.

Notice that $\Omega^n(X, x_0) \cong \{\gamma : [0, 1] \rightarrow \Omega^{n-1}(X, x_0), \gamma(0) = \gamma(1) = c_{x_0}^{n-1}\} = \dots = \{\gamma : \prod S^1 \rightarrow X, \gamma(\partial \prod S^1) = x_0\}$, so quotient by homotopy we have $\pi_n(\Omega_{x_0}^n(X)) \cong \pi_0(\Omega^n(X, x_0))$. $\square$

20 Singular Homology

Definition 20.1. A **singular n -simplex** of a topological space X is a continuous map $\sigma : \Delta^n \rightarrow X$.

The set of singular n -simplices is denoted as S_n .

Given $\alpha : [k] \rightarrow [n]$ a morphism in Δ_{inj} (and hence an injective affine map $\Delta^k \rightarrow \Delta^n$), we can define $\alpha^* : S_n(X) \rightarrow S_k(X)$ by

$$\alpha^*(\sigma) = \sigma \circ \alpha : \Delta^k \rightarrow X.$$

We thus have defined a functor $S : \Delta_{\text{inj}} \rightarrow \mathbf{Set}$ (which is a Δ -set), called the **singular Δ -set** of X .

Notice that by the definition, $\partial_i \sigma = \sigma \circ \delta'_i$, where δ'_i is the induced map $\Delta^{n-1} \rightarrow \Delta^n$ by $\delta_i : [n-1] \rightarrow [n]$ where $i \notin \text{im}(\delta_i)$.

The n -th chain group of $S(X)$ is denoted as $C_n(X)$ (and thus $Z_n(X)$, $B_n(X)$, $H_n(X)$ by analogous definition) is called the **n -th singular chain group**.

Notice that the boundary map $\partial : C_n(X) \rightarrow C_{n-1}(X)$ in $S(X)$ is therefore defined by

$$\partial \sigma = \sum c_i \partial \sigma_i,$$

where σ_i 's are the generators (elements in $S_n(X)$), and $\partial \sigma_i$ is defined as

$$\partial \sigma_i = \sum (-1)^i \delta'_i \sigma_i = \sum (-1)^i \sigma_i \circ \delta_i.$$

Exercise 20.2. Given $f : X \rightarrow Y$ continuous, we are able to define a map $f_* : S(X) \rightarrow S(Y)$ given by $f_*(\sigma) = f \circ \sigma$ for $\sigma : \Delta^n \rightarrow X$. This thus defines a functor $\mathbf{Top} \rightarrow \Delta\mathbf{Set}$.

Hence singular homology $H_n : \mathbf{Top} \rightarrow \mathbf{Ab}$ is a functor as composition of S and H_n (in simplicial complex).

Corollary 20.3. $H_n(pt) \cong \begin{cases} \mathbb{Z} & n = 0 \\ 0 & n > 0 \end{cases}$

Proof. Let $X = \{x_0\}$ be our one-point space. We have $X_n = S_n(X) = \{c_{x_0}\}$.

If $n = 0$, then $Z_0(X) = C_0(X) \cong \mathbb{Z}$ and $B_0(X) = \mathbb{Z} \partial_{c_{x_0}} = \mathbb{Z}(c_{x_0} \circ \delta_0 - c_{x_0} \circ \delta_1) = \mathbb{Z}(c_{x_0} - c_{x_0}) = 0$. Therefore, $H_0(X) \cong \mathbb{Z}$.

If $n > 0$ and is odd, then $B_n(X) = \partial(c_{x_0}) = C_n(X)$ and $Z_n(X) = C_n(X)$, so $H_n(X) \cong 0$. Otherwise, if n is even, then $B_n(X) \cong 0$ and $Z_n(X) \cong 0$, so $H_n(X) \cong 0$. $\square$

Exercise 20.4. Given a family $\{X_\alpha\}$ of spaces, the maps on homology $(i_\beta)_* : H_n(X_\beta) \rightarrow H_n(\sqcup X_\alpha)$ induced by the inclusions $i_\beta : X_\beta \rightarrow \sqcup X_\alpha$ induce an isomorphism

$$\bigoplus H_n(X_\alpha) \rightarrow H_n(\sqcup X_\alpha).$$

In other words, $H_n : \mathbf{Top} \rightarrow \mathbf{Ab}$ preserves coproducts.

Corollary 20.5. For any discrete space X , $H_n(X) \cong 0$ for $n > 0$ and $H_0(X)$ is free Abelian on X .

21 H_0 and H_1

Corollary 21.1. *For any topological space X , $H_0(X)$ is free Abelian on the path components of X .*

Proof. By definition, let $\gamma : \Delta^1 \rightarrow X$, then $\partial\gamma = \gamma(0) - \gamma(1)$. Therefore, two maps c_{x_0} and c_{x_1} are equivalent in $Z_0(X)/B_0(X)$ iff x_0 and x_1 are connected by a path.

Therefore, $H_0(X)$ is free Abelian on the path components of X . $\square$

Exercise 21.2. *Fix a space X . We have*

- (i) *Regarding a path $[0, 1] \rightarrow X$ as a map $\Delta^1 \rightarrow X$, we then have a path, as a 1-chain, is a 1-cycle iff it is a loop.*
- (ii) *If two loops α and β at x_0 are homotopic relative to endpoints, then they define homologous 1-cycles.*
- (iii) *Show that the resulting map $\eta : \pi_1(X, x_0) \rightarrow H_1(X)$, called the **Hurewitz map**, is a group homomorphism. In other words, given α, β loops at x_0 , $\alpha \cdot \beta$ and $\alpha + \beta$ are homologous, and the trivial loop c_{x_0} is homologous to 0.*

Proof. (i) We have $\partial\gamma = \gamma \circ \delta'_0 - \gamma \circ \delta'_1 = \gamma(0) - \gamma(1)$. Therefore, γ is a 1-cycle iff it is a loop.

- (ii) WTS that $\alpha - \beta \in B_1(X)$. Notice that a homotopy h (as shown below) relative to endpoints can collapse to a map $\Delta^2 \rightarrow X$ as shown. i.e.,

$$\begin{array}{ccc} (0, 1) & \xrightarrow{\beta} & (1, 1) \\ \downarrow c_{x_0} & & \downarrow c_{x_0} \\ (0, 0) & \xrightarrow{\alpha} & (1, 0) \end{array}$$

collapses to

$$\begin{array}{ccc} & 2 & \\ \beta \swarrow & & \downarrow c_{x_0} \\ 0 & \xrightarrow{\alpha} & 1 \end{array}$$

which we call h' .

We then have $\partial(h') = \alpha - \beta + c_{x_0}$ so it suffices to show that c_{x_0} is in $B_1(X)$, which clearly is as $c_{x_0} = \partial(c_{x_0})$.

- (iii) To show that c_{x_0} is homologous to 0, notice that $c_{x_0} = \partial c'_{x_0}$, where c'_{x_0} is the constant map from Δ^2 to X .

Consider the following map from Δ^2 to X : let $c((0, t)) = \alpha \cdot \beta(t)$, and $c(t_1, t_2)$ be $c((0, t))$ if $(0, t)$ is the projection of (t_1, t_2) onto the x -axis.

In other words, c is defined by the following diagram:

$$\begin{array}{ccc} & 1 & \\ \alpha \swarrow & & \searrow \beta \\ 0 & \xrightarrow{\alpha \cdot \beta} & 2 \end{array}$$

Notice that c is continuous by using pasting lemma vertically. We then have $\partial(c) = \alpha + \beta - \alpha \cdot \beta$, proving that $\alpha + \beta$ and $\alpha \cdot \beta$ are homologous. $\square$

Theorem 21.3. *The Hurewitz map $\eta : \pi_1(X, x_0) \rightarrow H_1(X)$ induces an isomorphism $\pi_1(X, x_0)_{ab} \rightarrow H_1(X)$. In other words, η is surjective and the kernel is the commutator group of $\pi_1(X, x_0)$.*

Space of The Day (Mapping Cylinder and Mapping Cone)

Definition 21.4. *Given a continuous $f : A \rightarrow X$, the **mapping cylinder** M_f is the quotient of $A \times [0, 1] \sqcup X$ obtained by identifying $f(a)$ and $(a, 0)$ for every $a \in A$.*

$$\begin{array}{ccc}
 A & \xrightarrow{a \rightarrow (a,1)} & A \times [0, 1] \\
 f \downarrow & & \downarrow i_1 \\
 X & \xrightarrow{i_2} & M_f
 \end{array}$$

Exercise 21.5. Prove that the map $f' : a \rightarrow (a, 0)$ is an embedding and i_2 is a homotopy equivalence, and that the triangle commutes up to homotopy.

$$\begin{array}{ccc}
 & & X \\
 & \nearrow f & \downarrow i_2 \\
 A & \xrightarrow{f'} & M_f
 \end{array}$$

Definition 21.6. The **mapping cone** C_f is the quotient $M_f / i_1(A \times \{1\})$.

When f is the identity map 1_A , the mapping cone is simply called the **cone** $C(A)$.

Exercise 21.7. $C(A)$ is contractible.

Proof. This is trivial by the map as $1_{C(A)} \sim c_{(a,1)}$ by the homotopy $h_t(x, h) = (x, h(1-t) + t)$. □

Definition 21.8. When $i : A \rightarrow X$ is the inclusion map, C_i is called the **homotopy quotient**, denoted as $X // A$.

Exercise 21.9. Show that the composite $q \circ i_2 \circ i : A \rightarrow X // A$ is homotopic to a constant map.

22 Chain Complex

Definition 22.1. A **chain complex** $C = (C, \partial)$ is a $\mathbb{Z}$ -indexed sequence $\{C_n\}$ of Abelian groups and homomorphisms

$$\dots \xrightarrow{\partial^{(n+2)}} C_{n+1} \xrightarrow{\partial^{(n+1)}} C_n \xrightarrow{\partial^{(n)}} C_{n-1} \xrightarrow{\partial^{(n-1)}} \dots$$

satisfying that $\partial^{(n-1)} \circ \partial^{(n)} = 0$ for all n .

Given chain complexes C and D , a **morphism of chain complexes** $f : C \rightarrow D$ is defined as $f_n : C_n \rightarrow D_n$ group homomorphisms s.t. $f \circ \partial^{(n)} = \partial^{(n-1)} \circ f$. (i.e. the following diagram commutes):

$$\begin{array}{ccc} C_n & \xrightarrow{\partial} & C_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ D_n & \xrightarrow{\partial} & D_{n-1} \end{array}$$

Exercise 22.2. This definition turns chain complexes into a category **Ch**.

Proof. We define the composition of morphisms $f : C \rightarrow D$ and $g : D \rightarrow E$ to be $g \circ f : C \rightarrow E$. It is indeed a morphism of chain complexes, as

$$\begin{aligned} \partial \circ (g_n f_n) &= g_{n-1} \partial f_n \\ &= g_{n-1} f_{n-1} \circ \partial. \end{aligned}$$

It is routine to check that with this composition morphism we get a category **Ch**. □

Definition 22.3. Given a Δ -set X , we are able to define the **simplicial chain complex** $C(X)$ associated to X as the chain complex with $(C(X))_n = C_n(X)$ and $\partial^{(n)} : (C(X))_n \rightarrow (C(X))_{n-1} = \partial^{(n)} : C_n(X) \rightarrow C_{n-1}(X)$.

Given a topological space X , the simplicial chain complex $C(S(X))$ is called the **singular chain complex** of X . It is denoted as $C(X)$.

Exercise 22.4. Given a morphism of Δ -set $f : X \rightarrow Y$, the induced map on $f_n : C_n(X) \rightarrow C_n(Y)$ is a morphism of chain complexes.

Proof. We want to show that $f_{n-1} \circ \partial = \partial f_n$, which by definition is satisfied by a morphism of Δ -sets. □

Definition 22.5. Given any chain complex C , we are able to define **n -cycles** $Z_n(C) = \ker \partial^{(n)}$ and **n -boundaries** $B_n(C) = \operatorname{im} \partial^{(n+1)}$. We thus define the **n -homology group** $H_n(C) = Z_n(C)/B_n(C)$.

Exercise 22.6. Given morphism of chain complexes $f : C \rightarrow D$, we have an induced map $f_* : H_n(C) \rightarrow H_n(D)$. Thus, H_n is a functor **Ch** $\rightarrow$ **Ab**.

We thus have factored the singular homology functor $H_n : \mathbf{Top} \rightarrow \mathbf{Ab}$ into three steps:

$$\mathbf{Top} \xrightarrow{S} \Delta\mathbf{Set} \xrightarrow{C} \mathbf{Ch} \xrightarrow{H_n} \mathbf{Ab}.$$

Proof. We first show that given $c \in Z_n(C)$, $f_n(c)$ is in $Z_n(D)$. Notice that $\partial f_n(c) = f_{n-1}(\partial(c)) = 0$ so $f_n(c) \in Z_n(D)$.

Now suppose $c, c' \in Z_n(C)$ with $c = c' + \partial b$ for some $b \in C_{n+1}(C)$. We then have $f(c) = f(c') + f(\partial b) = f(c') + \partial f(b)$ with $f(b) \in C_{n+1}(D)$. Therefore, $[f(c)] = [f(c')]$ so our f_* is well-defined.

It is routine to check that this turns H_n into a functor. □

Definition 22.7. Given two morphisms $f, g : C \rightarrow D$ of chain complexes, a **chain homotopy** $h : f \rightarrow g$ is defined by a collection of homomorphisms $h_n : C_n \rightarrow D_{n+1}$ with

$$\partial h_n + h_{n-1} \partial = g_n - f_n.$$

If such h exists, then we say f and g are **homotopic** with $f \sim g$.

Exercise 22.8. Chain homotopy is an equivalence relation.

Proof. The transitivity is the only non-trivial part. Suppose $h : f \rightarrow g$ and $h' : g \rightarrow k$. By definition, we have

$$\begin{aligned}\partial h_n + h_{n-1} \partial &= g_n - f_n \\ \partial h'_n + h'_{n-1} \partial &= k_n - g_n.\end{aligned}$$

Then adding h and h' we have a new map j from C_n to D_{n+1} which satisfies the homotopy relation. $\square$

Proposition 22.9. If $f, g : X \rightarrow Y$ are homotopic maps of topological spaces, then $f_*, g_* : C(X) \rightarrow C(Y)$ are chain homotopic.

Theorem 22.10. Chain homotopic maps f and g induce the same map on homology: $f_* = g_* : H_n(C) \rightarrow H_n(D)$.

Proof. Suppose $a \in Z_n(C)$. We then have $g_n(a) - f_n(a) = \partial h_n(a) + h_{n-1} \partial(a)$ for some $h_n : C_n \rightarrow D_{n+1}$.

As $\partial(a) = 0$, we have $g_n(a) - f_n(a) = \partial h_n(a)$, so $g_*(a) = f_*(a)$. $\square$

Corollary 22.11. Any homotopy equivalence $f : X \rightarrow Y$ induces isomorphism $f_* : H_n(X) \rightarrow H_n(Y)$ on homology.

Proof. We know $(gf)_* = 1_* : H_n(X) \rightarrow H_n(X)$, so therefore $(gf)_*$ is an isomorphism. Similarly, $(fg)_* : H_n(Y) \rightarrow H_n(Y)$ is also an isomorphism. Therefore, f_* is bijective and thus an isomorphism. $\square$

Corollary 22.12. Any contractible space has the same homology as a point.

Proof. This follows immediately from the above corollary as X contractible is equivalent as saying there is a homotopy equivalence $f : X \rightarrow \{y\}$. $\square$

Definition 22.13. Given a non-empty space X , the **augmentation map** for X is the map $\epsilon : C_0(X) \rightarrow C_0(\text{pt}) \cong \mathbb{Z}$ given by $\epsilon(\sum a_i p_i) = \sum a_i$.

The **reduced singular chain complex** $\tilde{C}(X)$ of X is the chain complex obtained from $C(X)$ by appending ϵ to the end:

$$\dots \xrightarrow{\partial^{(3)}} C_2 \xrightarrow{\partial^{(2)}} C_1 \xrightarrow{\partial^{(1)}} C_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0.$$

The **reduced singular homology** $\tilde{H}_n(X)$ is defined as $H_n(\tilde{C}(X))$.

Exercise 22.14. Show that $\tilde{C}(X)$ is indeed a chain complex.

Show that $\tilde{C}$ defines a functor $\tilde{C} : \mathbf{Top}_{\neq 0} \rightarrow \mathbf{Ch}$.

Proof. We need to verify that $\epsilon \partial^{(1)} = 0$. Notice that for every $\sigma : \Delta^1 \rightarrow X \in S_1(X)$, we have $\partial^{(1)} \sigma = \sigma(0) - \sigma(1)$. Therefore, $\epsilon \partial^{(1)} \sigma = 0$ so by extending linearly on C_1 we have $\epsilon \partial^{(1)} = 0$, making $\tilde{C}(X)$ a chain complex.

It is routine to check that $\tilde{C}$ therefore defines a functor. $\square$

Corollary 22.15. (i) $H_n(X) \cong \tilde{H}_n(X)$ for $n > 0$.

(ii) $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$.

(iii) If X is contractible then $\tilde{H}_n(X) \cong 0$ for any n .

Proof. i) is trivial by definition.

For ii), notice that $B_0(X) \cong \tilde{B}_0(X)$, while $Z_0(X) \cong \tilde{Z}_0(X) \oplus \mathbb{Z}$, as $Z_0(X)$ is free Abelian on $S_0(X)$ while $\tilde{Z}_0(X)$ is free Abelian on $S_0(X - \{x_0\})$ for some $x_0 \in X$, as the coefficient in front of x_0 is uniquely determined by the other coefficients (to make the sum of coefficients 0).

Therefore, we conclude $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$.

iii) comes immediately from i), ii), and **Corollary 22.12**. □

Space of The Day (Suspension)

Definition 22.16. Let $X \in \mathbf{Top}$. The *suspension* of X is a double cone over X . In other word, it is the quotient $(X \times I)/\sim$, where $\sim$ is generated by $(x, 0) \sim (y, 0)$ and $(x, 1) \sim (y, 1)$.

Exercise 22.17. Show that S defines a functor from $\mathbf{Top}$ to $\mathbf{Top}$.

Exercise 22.18. $SS^n \cong S^{n+1}$.

Proof. We define the map $f : S^n \times [-1, 1] \rightarrow S^{n+1}$ by letting

$$f(x_0, \dots, x_n, t) = (\sqrt{1-t^2}x_0, \dots, \sqrt{1-t^2}x_n, t).$$

It is not difficult to see that the map is well-defined and that $f(x_0, \dots, x_n, t) = (0, \dots, 0, t)$ iff $t = \pm 1$. Therefore, the map quotients uniquely through $q : S^n \times [-1, 1] \rightarrow S^n \times [-1, 1]/\sim$ and induces a map $\bar{f}$ s.t. $\bar{f}q = f$.

It is not difficult to see that $\bar{f}$ is bijective and continuous. Since $S^n \times [-1, 1]/\sim$ is compact and S^{n+1} is Hausdorff, we have that $\bar{f}$ is a homeomorphism, so $S^{n+1} \cong S^n \times [-1, 1]/\sim \cong S^n \times I/\sim \cong SS^n$. □

23 Relative Homology

Definition 23.1. A **subcomplex** $C \subset D$ of a chain complex D is a chain complex C s.t. C_n is a subgroup of D_n , and $\partial^{(n)} : C_n \rightarrow C_{n-1}$ is the restriction of $\partial^{(n)} : D_n \rightarrow D_{n-1}$ on C_n .

Exercise 23.2. If $A \subset X$ is a subspace, then $C(A)$ is a subcomplex of $C(X)$.

Proof. Notice that for any $\sigma : \delta^n \rightarrow A$, we can always extend it to a map $\sigma : \delta^n \rightarrow X$. Therefore, $C(A) \subset C(X)$.

Clearly $C(A)$ is also an Abelian group, so $C_n(A)$ is always a subgroup of $C_n(X)$ and $C(A)$ is a subcomplex of $C(X)$. $\square$

Definition 23.3. Given a subcomplex $C \subset D$, the **quotient complex** D/C is defined as $(D/C)_n = D_n/C_n$, and $\partial^{(n)} : D_n/C_n \rightarrow D_{n-1}/C_{n-1}$ is induced by $\partial^{(n)} : D_n \rightarrow D_{n-1}$.

Exercise 23.4. Show that this is well-defined.

Proof. Suppose that $[\sigma] = [\sigma']$, then $\sigma - \sigma' = c$ for some $c \in C_n$.

Therefore, $\partial\sigma - \partial\sigma' = \partial c \in C_{n-1}$ so the map is well-defined. $\square$

Definition 23.5. Given a topological pair (X, A) , the quotient complex $C(X)/C(A)$ is denoted $C(X, A)$ and is called the **relative singular chain complex** of (X, A) .

Exercise 23.6. Show that C defines a functor: $\mathbf{Top}_2 \rightarrow \mathbf{Ch}$.

Proof. To show that the functor $C : \mathbf{Top}_2 \rightarrow \mathbf{Ch}$ is well-defined, we want to first show that if $\sigma \sim \sigma'$ for $\sigma, \sigma' \in S_n(X)$, then $f\sigma \sim f\sigma'$.

Notice that if $\sigma \sim \sigma'$ then $\sigma - \sigma' = 0$ outside A . Therefore, $f\sigma - f\sigma' = 0$ outside B , so $f\sigma \sim f\sigma'$ and the map is well-defined. $\square$

Exercise 23.7. If A is non-empty, show that $C(X, A) \cong \tilde{C}(X)/\tilde{C}(A)$.

Proof. Notice that we always have $C_n(X)/C_n(A) \cong \tilde{C}_n(X)/\tilde{C}_n(A)$.

Therefore, $C(X, A) = C(X)/C(A) \cong \tilde{C}(X)/\tilde{C}(A)$. $\square$

Definition 23.8. The homology $H_n(C(X, A))$ is called the **relative singular homology** of (X, A) , denoted $H_n(X, A)$.

Definition 23.9. A sequence $\{G_n\}$ of groups and homomorphisms $f_{(n)} : G_n \rightarrow G_{n-1}$

$$\dots \xrightarrow{f_{(n+2)}} G_{n+1} \xrightarrow{f_{(n+1)}} G_n \xrightarrow{f_{(n)}} G_{n-1} \xrightarrow{f_{(n-1)}} \dots$$

is **exact** at G_n if $\text{Im } f_{n+1} = \text{Ker } f_n$.

A **finite sequence**

$$G_n \xrightarrow{f_{(n)}} \dots \xrightarrow{f_{(1)}} G_0$$

is **exact** if it is exact at every G_i .

Exercise 23.10. Show that

- (i) $1 \rightarrow G \xrightarrow{f} H$ is exact iff f is injective.
- (ii) $G \xrightarrow{f} H \rightarrow 1$ is exact iff f is surjective.
- (iii) $1 \rightarrow G \xrightarrow{f} H \rightarrow 1$ is exact iff f is an isomorphism.

Definition 23.11. A **short exact sequence (SES)** of groups is an exact sequence of the form

$$1 \rightarrow K \xrightarrow{i} G \xrightarrow{q} Q \rightarrow 1.$$

This is a short exact sequence iff $K \cong \text{Ker } q$ and $Q \cong G/i(K)$.

Definition 23.12. A **short exact sequence of chain complexes** consists of chain complex homomorphisms f, g , s.t.

$$1 \rightarrow C_n \xrightarrow{f_n} D_n \xrightarrow{g_n} E_n \rightarrow 1$$

is a short exact sequence for any n .

Example 11. Given a topological pair (X, A) , the following is a short exact sequence of chain complexes:
 $1 \rightarrow C(A) \xrightarrow{i_*} C(X) \xrightarrow{q_*} C(X, A) \rightarrow 1$.

Theorem 23.13. A short exact sequence of chain complexes

$$1 \rightarrow C \xrightarrow{i} D \xrightarrow{q} E \rightarrow 1.$$

gives rise to a long exact sequence

$$\dots \rightarrow H_n(C) \xrightarrow{i_*} H_n(D) \xrightarrow{q_*} H_n(E) \xrightarrow{\partial} H_{n-1}(C) \xrightarrow{i_*} H_{n-1}(D) \xrightarrow{q_*} H_{n-1}(E) \rightarrow \dots$$

where ∂ is called the **connecting homomorphism** and satisfies that for $c \in Z_n(E)$, $\partial([c]) = [i^{-1}(\partial q^{-1}(c))]$, where $q^{-1}(c) \in D_n$ is any element with image c .

Corollary 23.14. Given a topological pair (X, A) , there is an induced long exact sequence of homology groups

$$\dots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{q_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \xrightarrow{q_*} H_{n-1}(X, A) \rightarrow \dots$$

Similarly, if A is not empty, then

$$\dots \rightarrow \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{q_*} \tilde{H}_n(X, A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \xrightarrow{i_*} \tilde{H}_{n-1}(X) \xrightarrow{q_*} \tilde{H}_{n-1}(X, A) \rightarrow \dots$$

is a long exact sequence of homology groups.

24 The Eilenberg-Steenrod-Milnor Axioms

Definition 24.1. A **homology theory** is a sequence of functors $H_n : \mathbf{Top}_2 \rightarrow \mathbf{Ab}$ and homomorphisms $\partial : H_n(X, A) \rightarrow H_{n-1}(A) = H_n(A, \emptyset)$ for all n and (X, A) , satisfying

(i) For any $(X, A) \in \mathbf{Top}_2$, there is a exact sequence

$$\dots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{q_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \xrightarrow{q_*} H_{n-1}(X, A) \rightarrow \dots$$

(ii) The homomorphisms $\partial : H_n(X, A) \rightarrow H_{n-1}(A)$ constitute a natural transformation of functors $H_n \rightarrow H_{n-1}$:

$$\begin{array}{ccc} H_n(X, A) & \xrightarrow{f_*} & H_n(Y, B) \\ \downarrow \partial & & \downarrow \partial \\ H_{n-1}(A) & \xrightarrow{f_*} & H_{n-1}(B) \end{array}$$

for any $f : (X, A) \rightarrow (Y, B)$.

(iii) For any $f, g : (X, A) \rightarrow (Y, B)$ homotopic, $f_* = g_* : H_n(X, A) \rightarrow H_n(Y, B)$.

(iv) $H_n(\{x_0\}) \cong 0$ for $n > 0$ and $H_0(\{x_0\}) \cong \mathbb{Z}$.

(v) **Excision:** Given (X, A) and $U \subset A$ s.t. $\bar{U} \subset \text{Int } A$, the map induced by $i : (X - U, A - U) \rightarrow (X, A)$

$$i_* : H_n(X - U, A - U) \rightarrow H_n(X, A)$$

is an isomorphism.

(vi) $H_n : \mathbf{Top} \rightarrow \mathbf{Ab}$ preserves coproduct, i.e.,

$$\bigoplus_{\alpha \in I} H_n(X_\alpha) \rightarrow H_n\left(\bigsqcup_{\alpha \in I} X_\alpha\right)$$

is an isomorphism.

Theorem 24.2. Singular homology is a homology theory.

Corollary 24.3. $\tilde{H}_n(X) = \text{Ker } (H_n(X) \rightarrow H_n(\{x_0\}))$.

Definition 24.4. A map $f : (X, A) \rightarrow (Y, B)$ is a **homotopy equivalence** if $\exists g : (Y, B) \rightarrow (X, A)$ s.t. $gf \sim 1_{(X, A)}$, $fg \sim 1_{(Y, B)}$.

Exercise 24.5. If $f : (X, A) \rightarrow (Y, B)$ is a homotopy equivalence, then $H_n(X, A) \cong H_n(Y, B)$.

Theorem 24.6. $H_k(D^n, S^{n-1}) \cong \tilde{H}_{k-1}(S^{n-1})$, for any $k, n \in \mathbb{N}$.

Proof. Notice that we have the following LES:

$$\tilde{H}_k(S^{n-1}) \rightarrow \tilde{H}_k(D^n) \rightarrow H_k(D^n, S^{n-1}) \xrightarrow{\partial} \tilde{H}_{k-1}(S^{n-1}) \rightarrow \tilde{H}_{k-1}(D^n) \rightarrow H_{k-1}(D^n, S^{n-1}).$$

In particular, we have

$$\tilde{H}_k(D^n) \rightarrow H_k(D^n, S^{n-1}) \xrightarrow{\partial} \tilde{H}_{k-1}(S^{n-1}) \rightarrow \tilde{H}_{k-1}(D^n)$$

exact.

As $\tilde{H}_k(D^n)$ and $\tilde{H}_{k-1}(D^n) = 0$, we have $H_k(D^n, S^{n-1}) \cong \tilde{H}_{k-1}(S^{n-1})$. □

Theorem 24.7. $\tilde{H}_k(S^{n-1}) \cong H_k(S^{n-1}, D_-^{n-1})$ where D_-^{n-1} is the lower hemisphere of S^{n-1} .

Proof. Similar to the theorem above, we use the long exact sequence

$$\tilde{H}_k(D_-^n) \rightarrow \tilde{H}_k(S^{n-1}) \rightarrow H_k(S^{n-1}, D_-^{n-1}) \xrightarrow{\partial} \tilde{H}_{k-1}(D_-^{n-1}) \rightarrow \tilde{H}_{k-1}(S^{n-1}) \rightarrow H_{k-1}(S^{n-1}, D_-^{n-1}).$$

In particular, we have

$$\tilde{H}_k(D_-^n) \rightarrow \tilde{H}_k(S^n) \rightarrow H_k(S^n, D_-^n) \xrightarrow{\partial} \tilde{H}_{k-1}(D_-^{n-1}).$$

Since $\tilde{H}_k(D_-^n) = 0$ and $\tilde{H}_k(D_-^{n-1}) = 0$, we have $\tilde{H}_{k-1}(S^{n-1}) \cong H_{k-1}(S^{n-1}, D_-^{n-1})$. □

Theorem 24.8. $H_k(S^n, D_-^n) \cong H_k(D^n, S^{n-1})$.

Proof. Removing a point in D_-^n , we have

$$\begin{aligned} H_k(S^n, D_-^n) &\cong H_k(S^n - \{p\}, D_-^n - \{p\}) \\ &\cong H_k(D_+^n, S^{n-1}) \\ &\cong H_k(D^n, S^{n-1}). \end{aligned}$$

□

Theorem 24.9. $\tilde{H}_k(S^n) \cong \begin{cases} \mathbb{Z} & k = n \\ 0 & \text{otherwise} \end{cases}$

Proof. We induct on n .

When $n = 0$, $S^0 \cong \{v_1\} \sqcup \{v_2\}$ so $\tilde{H}_0(S^0) \cong \mathbb{Z}$ since $H_0(\{v_1\} \sqcup \{v_2\}) \cong \mathbb{Z} \oplus \mathbb{Z}$.

For $k > 0$, we have

$$\begin{aligned} \tilde{H}_k(S^n) &\cong H_k(S^n, D_-^n) \\ &\cong H_k(D^n, S^{n-1}) \\ &\cong \tilde{H}_{k+1}(S^{n+1}). \end{aligned}$$

□

Corollary 24.10. S^n is not contractible for any n .

Proof. This follows directly from the theorem that $\tilde{H}_n(X) \cong 0$ for any n and contractible space X . □

Corollary 24.11. Suppose we have the following morphisms of chain complex SES

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C & \xrightarrow{f} & D & \xrightarrow{g} & E & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C' & \xrightarrow{f'} & D' & \xrightarrow{g'} & E' & \longrightarrow & 0 \end{array}$$

then the following square commutes:

$$\begin{array}{ccc} H_n(E) & \xrightarrow{\partial} & H_{n-1}(C) \\ \downarrow \gamma_* & & \downarrow \alpha_* \\ H_n(E') & \xrightarrow{\partial} & H_{n-1}(C') \end{array}$$

.

Space of The Day (Vector Bundle)

Definition 24.12. A **vector bundle** is a space E over a space X with a map $p : E \rightarrow X$ satisfying that $p^{-1}(x)$ is a vector space in a fixed dimension n and the **local triviality property**:

For any $x \in X$, there is a neighborhood U of x s.t. there is a homeomorphism $\varphi : p^{-1}(U) \rightarrow U \times \mathbb{R}^n$, with $\varphi|_{p^{-1}(y)} : p^{-1}(y) \rightarrow \{y\} \times \mathbb{R}^n$ an isomorphism of vector spaces for any $y \in U$.

Notice that the definition is stronger than saying $p^{-1}(x) \cong \mathbb{R}^n$ for any $x \in X$.

Corollary 24.13. $\pi_1 : X \times \mathbb{R}^n \rightarrow X$ is a vector bundle for any X .

Exercise 24.14. For any vector bundle $p : E \rightarrow X$, p is a homotopy equivalence.

Proof. We have the following commutative diagram:

$$\begin{array}{ccc}
 p^{-1}(U) & \xrightarrow{\varphi} & U \times \mathbb{R}^n \\
 & \searrow p & \downarrow \pi_1 \\
 & & U
 \end{array}$$

Notice that $U \times \mathbb{R}^n \simeq U$ as $\mathbb{R}^n$ is contractible, and thus π_1 as the composition $U \times \mathbb{R}^n \rightarrow U \times \{0\} \rightarrow U$ is an homotopy equivalence. Since φ is a homeomorphism, we know that $p = \pi_1 \circ \varphi$ is also a homotopy equivalence. $\square$

Definition 24.15. Given a manifold $M \subset \mathbb{R}^n$, the **tangent bundle** TM is the space $\{(p, v) \in \mathbb{R}^n \times \mathbb{R}^n \mid p \in M, v \in T_p M\}$, where $T_p M \subset \mathbb{R}^n$ is the set of all the vectors in $\mathbb{R}^n$ tangent to M at p .

The projection is defined as $\pi(p, v) = p$.

Definition 24.16. By replacing $T_p M$ by the normal space $N_p M$, one get the **normal bundle** of $M \subset \mathbb{R}^n$.

Definition 24.17. An **isomorphism of vector bundles** between $p_1 : E_1 \rightarrow X$ and $p_2 : E_2 \rightarrow X$ is a homeomorphism $f : E_1 \rightarrow E_2$ s.t. $p_2 \circ f = p_1$ (the following square commutes) and that $f|_{p_1^{-1}(x)} : p_1^{-1}(x) \rightarrow p_2^{-1}(x)$ is an isomorphism of vector spaces.

$$\begin{array}{ccc}
 E_1 & \xrightarrow{f} & E_2 \\
 & \searrow p_1 & \downarrow p_2 \\
 & & X
 \end{array}$$

Exercise 24.18. $NS^n \subset \mathbb{R}^{n+1}$ is trivial (isomorphic to $S^n \times \mathbb{R}^n$).

25 Homology with Coefficients

Definition 25.1. Let $G \in \mathbf{Ab}$, $X \in \Delta\mathbf{Set}$. We define the **simplicial chain complex with coefficients in G** , $C(X; G)$ the same way as before, except that instead of taking $C_n(X; G) = \bigoplus_{\sigma \in X_n} \mathbb{Z}\sigma$, we define it to

be $\bigoplus_{\sigma \in X_n} G\sigma$.

The boundary map $\partial : C_n(X; G) \rightarrow C_{n-1}(X; G)$ are defined as before, with $\partial^n \circ \partial^{n-1} = 0$. So do Z_n , B_n , H_n .

The **singular chain complex with coefficients in G** is defined as $C(X; G) = C(SX; G)$, and $H_n(X; G) = H_n(C(X; G))$.

Reduced homology and relative homology are analogously defined.

Theorem 25.2. Singular homology with coefficients in G satisfy the ESM axioms with iv) replaced by $H_n(\{x_0\}) \cong G$.

It is also determined uniquely up to isomorphism by the axioms.

Exercise 25.3. $H_0(\mathbb{R}P^2, \mathbb{Z}_2) \cong H_1(\mathbb{R}P^2, \mathbb{Z}_2) \cong H_2(\mathbb{R}P^2, \mathbb{Z}_2) \cong \mathbb{Z}_2$, while $H_2(\mathbb{R}P^2, \mathbb{Z}_2) \cong 0$.

Theorem 25.4. $\mathbb{R}^m \cong \mathbb{R}^n$ iff $m = n$.

Proof. Suppose $\mathbb{R}^m \cong \mathbb{R}^n$. Removing a point 0 from $\mathbb{R}^m$ and $\mathbb{R}^n$, we have $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$ and $\mathbb{R}^m \setminus \{0\} \simeq S^{m-1}$.

Since homotopic spaces have same homology groups, we have $\tilde{H}_k(S^{n-1}) \cong \tilde{H}_k(S^{m-1})$ for any k , which is true iff $n = m$. $\square$

Theorem 25.5. For any $n > 0$, $f : D^n \rightarrow D^n$ has a fixed point.

Exercise 25.6. $\nexists$ retraction $D^n \rightarrow S^{n-1}$.

Proof. Suppose $r : D^n \rightarrow S^{n-1}$ is a retraction. We then have $ri = 1_{S^{n-1}}$.

Therefore, $ri_* : \tilde{H}_n(S^{n-1}) \rightarrow \tilde{H}_n(S^{n-1})$ is an isomorphism from $\mathbb{Z}$ to $\mathbb{Z}$.

However, this is impossible as we must have

$$\tilde{H}_n(S^{n-1}) \xrightarrow{i_*} \tilde{H}_n(D^n) \xrightarrow{r_*} \tilde{H}_n(S^{n-1}).$$

Since $r_* : 0 \rightarrow \mathbb{Z}$ has to be the trivial map, $(ri)_* = r_*i_*$ can never be an isomorphism.

Therefore, no such retraction exists. $\square$

Space of The Day (Smash Product and Reduced Suspension)

Definition 25.7. Given two pointed space (X, x_0) and (Y, y_0) , the **smash product** $(X, x_0) \wedge (Y, y_0)$ is the quotient space $(X \times Y)/(X \times \{y_0\} \cup \{x_0\} \times Y) = (X \times Y)/(X \vee Y)$, as both spaces quotient everything in $X \times Y$ containing x_0 or y_0 to a point. It is again a pointed space.

Definition 25.8. Given another pointed space (Z, z_0) , $f : X \times Y \rightarrow Z$ is **bi-pointed** if $f(x_0, y) = f(x, y_0) = z_0$ for any $x \in X$, $y \in Y$. Therefore, $q : X \times Y \rightarrow X \wedge Y$ is a bi-pointed map, and it has the universal property: for any $f : X \times Y \rightarrow Z$ bi-pointed, we have some unique $\bar{f} : X \wedge Y \rightarrow Z$ such that $\bar{f}q = f$.

$$\begin{array}{ccc} X \times Y & & \\ \downarrow q & \searrow f & \\ X \wedge Y & \xrightarrow{\bar{f}} & Z \end{array}$$

Definition 25.9. The **reduced suspension** of a space (X, x_0) is $\Sigma X = S^1 \wedge X$.

Exercise 25.10. $\sum X \cong SX/(\{x_0\} \times I)$.

Theorem 25.11. *For reasonable space, $q : SX \rightarrow \sum X$ is a homotopy equivalence.*

Corollary 25.12. $\sum S^n \cong S^{n+1}$.

Exercise 25.13. $\sum$ defines a functor $\sum : \mathbf{Top}_* \rightarrow \mathbf{Top}_*$.

Exercise 25.14. *There is a bijection between*

$$\mathrm{Hom}(\sum(X, x_0), (Y, y_0)) \text{ and } \mathrm{Hom}((X, x_0), \Omega(Y, y_0)).$$

26 Degrees of Maps of Spheres

Definition 26.1. Given a continuous map $f : S^n \rightarrow S^n$, its **degree** is an integer defined as $f_*([c]) = \deg(f) \cdot [c]$ for all $[c] \in \tilde{H}_n(S^n)$, where f_* is the induced map $\tilde{H}_n(S^n) \rightarrow \tilde{H}_n(S^n)$.

Proposition 26.2. (i) Given $f, g : S^n \rightarrow S^n$, $\deg(g \circ f) = \deg(f) \cdot \deg(g)$.

(ii) Given a reflection by $v \in \mathbb{R}^{n+1}$, namely $r_v : S^n \rightarrow S^n$ with $r_v(x) = x - \frac{2x \cdot v}{v \cdot v}v$, we have $\deg(r_v) = -1$.

(iii) If $f : S^n \rightarrow S^n$ is the point reflection (i.e., $f(x) = -x$), then $\deg(f) = (-1)^{n+1}$.

(iv) If $f \sim g$ then $\deg(f) = \deg(g)$.

Proof. i) is trivial by definition.

ii) is clear when $n = 0$, as r_v would always be mapping -1 to 1 and 1 to -1. It thus maps $[c] \in H_0(S^0)$ to $[-c] = -[c]$, so $\deg(r_v) = -1$.

Deducing iii) from i) and ii) is also clear, as we need to reverse the sign for each index once, which is the same as composing r_v for $n + 1$ times on each index.

iv) is clear as the induced map by f and g are equal, so their degrees have to be equal. $\square$

Theorem 26.3. Two maps $f, g : S^n \rightarrow S^n$ are homotopic iff they have the same degree.

Corollary 26.4. The antipodal map $f : S^n \rightarrow S^n$ sending x to $-x$ is homotopic to identity iff n is odd.

Exercise 26.5. For any space X , we have $\tilde{H}_k(X) \cong \tilde{H}_{k+1}(SX)$.

Proof. We follow the same pattern as the proof for $\tilde{H}_k(S^n) \cong \tilde{H}_{k+1}(S^{n+1})$.

First, notice that we have the long exact sequence

$$\dots \rightarrow \tilde{H}_k(X) \rightarrow \tilde{H}_k(CX_-) \rightarrow H_k(CX_-, X) \rightarrow \tilde{H}_{k-1}(X) \rightarrow \tilde{H}_{k-1}(CX_-) \rightarrow \dots$$

Notice that $\tilde{H}_k(CX_-)$ is trivial, so $H_k(CX_-, X) \cong \tilde{H}_{k-1}(X)$.

Next, notice that we have the long exact sequence

$$\dots \rightarrow \tilde{H}_k(CX_+) \rightarrow \tilde{H}_k(SX) \rightarrow H_k(SX, CX_+) \rightarrow \tilde{H}_{k-1}(CX_+) \rightarrow \tilde{H}_{k-1}(SX) \rightarrow \dots$$

Since, again, $\tilde{H}_k(CX_+)$ is trivial, we have $\tilde{H}_k(SX) \cong \tilde{H}_k(SX, CX_+)$.

Last but not least, notice that $\tilde{H}_k(SX, CX_+) \cong H_k(CX_-, X)$ by excision. We therefore have $\tilde{H}_k(SX) \cong \tilde{H}_{k-1}(X)$. $\square$

Theorem 26.6. The suspension isomorphism $s : \tilde{H}_k(X) \rightarrow \tilde{H}_{k+1}(SX)$ is natural. In other words, the following diagram commutes:

$$\begin{array}{ccc} \tilde{H}_k(X) & \xrightarrow{s_X} & \tilde{H}_{k+1}(SX) \\ \downarrow f_* & & \downarrow Sf_* \\ \tilde{H}_k(Y) & \xrightarrow{s_Y} & \tilde{H}_{k+1}(SY) \end{array}$$

Theorem 26.7 (Hairy Ball Theorem). S^n admits a continuous non-vanishing vector field iff n is odd.

(Recall that a vector field on S^n is a map $X : S^n \rightarrow \mathbb{R}^{n+1}$ such that $X(p) \cdot p = 0$ for any $p \in S^n$).

Proof. When n is odd, we can construct $X : (x_1, \dots, x_{2k}) \rightarrow (x_2, -x_1, \dots, x_{2k}, -x_{2k-1})$, which is clearly a vector field meeting the conditions.

When n is even, we define the map $\tilde{X} : S^n \rightarrow S^n$ by sending p to $X(p)/|X(p)|$. Consider the following map $F : S^n \times I \rightarrow S^n$:

$$h_t(p) = p \cos(\pi t) + \tilde{X}(p) \sin(\pi t).$$

Notice that F is a homotopy between 1_{S^n} and the antipodal map f , contradicting that 1_{S^n} and f are not homotopic when n is even. $\square$

Corollary 26.8. *The tangent bundle of S^n is not trivial when n is even.*

27 The Mayer-Vietoris Sequence

Theorem 27.1. *Let $X = A \cup B$ be a cover by subspaces A and B s.t. $X = \text{Int } A \cup \text{Int } B$.*

Let $i_A : A \rightarrow X$, $j_A : A \cap B \rightarrow A$ be inclusion maps, and i_B, j_B defined analogously.

We then have a long exact sequence:

$$\dots \rightarrow H_{n+1}(X) \rightarrow H_n(A \cap B) \xrightarrow{\langle j_{A*}, -j_{B*} \rangle} H_n(A) \oplus H_n(B) \xrightarrow{i_{A \circ \pi_1} + i_{B \circ \pi_2}} H_n(X) \rightarrow H_{n-1}(A \cap B) \rightarrow \dots$$

If $A \cap B$ is path connected, then we also have the following exact sequence:

$$\dots \rightarrow \tilde{H}_{n+1}(X) \rightarrow \tilde{H}_n(A \cap B) \xrightarrow{\langle j_{A*}, -j_{B*} \rangle} \tilde{H}_n(A) \oplus \tilde{H}_n(B) \xrightarrow{i_A \circ \pi_1 + i_B \circ \pi_2} \tilde{H}_n(X) \rightarrow \tilde{H}_{n-1}(A \cap B) \rightarrow \dots$$

Exercise 27.2. *The Mayer Vietoris sequence preserves a pushout in some sense (with an adjustment $H_{n+1}(X) \rightarrow H_n(A \cap B)$), as a pushout in **Ab** happens*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow g & & \downarrow h \\ C & \xrightarrow{k} & D \end{array}$$

iff the sequence

$$A \xrightarrow{\langle f, -g \rangle} B \oplus C \xrightarrow{h\pi_1 + k\pi_2} D \rightarrow 0$$

is exact.

Proof. Suppose that the sequence is exact. We know that $h(B) + k(C)$ generate D , so we can define i as $i(d) = p_1(b) + p_2(c)$ if $d = h(b) + k(c)$. Notice that i is well-defined, as if $d = h(b_1) + k(c_1) = h(b_2) + k(c_2)$, then we have $(b_1 - b_2, c_1 - c_2) \in \text{Ker}(h\pi_1 + k\pi_2) = \text{Im}\langle f, -g \rangle$, so there exists $a \in A$ s.t. $f(a) = b_1 - b_2$, $g(a) = c_2 - c_1$.

Since $p_1 f(a) = p_1(b_1) - p_1(b_2)$ and $p_2 g(a) = p_2(c_1) - p_2(c_2)$, we have that $p_1(b_1) + p_2(c_1) = p_1(b_2) + p_2(c_2)$, proving that i is well-defined, so the square is indeed a pushout.

We thus have the following commutative square:

$$\begin{array}{ccc}
 A & \xrightarrow{f} & B \\
 \downarrow g & & \downarrow h \\
 C & \xrightarrow{k} & D
 \end{array}
 \begin{array}{c}
 \nearrow p_1 \\
 \nearrow p_2 \\
 \nearrow i
 \end{array}
 F$$

so the square is a pushout.

On the other hand, notice that if the square is a pushout, then the square commutes so $hf = kg$. Therefore, the sequence is exact at $B \oplus C$. It is clear that $B \oplus C \rightarrow D$ has to be surjective, as D is also isomorphic to $B * C / \langle hf(a) \cdot kg(a)^{-1} \rangle$. $\square$

To prove Mayer-Vietoris, we have the following theorem:

Theorem 27.3. *Given the commutative diagram below where the rows are exact and the morphisms γ_n are isomorphisms,*

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial^{n+1}} & A_n & \xrightarrow{i_n} & B_n & \xrightarrow{q_n} & C_n & \longrightarrow & \cdots \\ & & \downarrow \gamma_{n+1} & & \downarrow \alpha_n & & \downarrow \beta_n & & \downarrow \gamma_n & & \\ \cdots & \longrightarrow & C'_{n+1} & \xrightarrow{\partial'^{n+1}} & A'_n & \xrightarrow{i'_n} & B'_n & \xrightarrow{q'_n} & C'_n & \longrightarrow & \cdots \end{array}$$

the sequence

$$\dots \longrightarrow B_{n+1}^{\rho^{n+1} \circ \gamma_{n+1}^{-1} \circ q^{n+1}} A_n^{\langle \alpha_n, -i_n \rangle} A'_n \oplus B_n^{i'_n \pi_1 + \beta_n \pi_2} B_n^{\rho^n \circ \gamma_n^{-1} \circ q^n} A_{n-1} \longrightarrow \dots$$

is exact.

Lemma 27.4. $H_n(A, A \cap B) \cong H_n(X, B)$, where A, B is a good cover of X .

Proof. This follows from the excision axiom by removing $\text{Int}(X - A)$ from X . $\square$

Remark 16 (Mayer-Vietoris). *The Mayer-Vietoris sequence thus follows if we replace the top row in the above theorem by the long exact sequence of pairs (X, B) and the bottom row by the long exact sequence of pairs $(A, A \cap B)$.*

Corollary 27.5. *Let $(X_1, p_1), \dots, (X_k, p_k) \in \mathbf{Top}_*$ s.t. each $p_i \in X_i$ has a contractible neighborhood.*

We then have

$$\tilde{H}_n(\bigvee X_i) \cong \bigoplus \tilde{H}_n(X_i).$$

Proof. We prove the statement by induction. Let $Y_1 = X_1 \vee U$ and $Y_2 = X_2 \vee V$, where U, V are the contractible neighborhoods of p_1 and p_2 .

We then have $X_1 \vee X_2$ covered by $Y_1 \cup Y_2$, so we have

$$\tilde{H}_n(Y_1 \cap Y_2) \rightarrow \tilde{H}_n(Y_1) \oplus \tilde{H}_n(Y_2) \rightarrow \tilde{H}_n(X_1 \vee X_2) \rightarrow \tilde{H}_{n-1}(Y_1 \cap Y_2)$$

exact.

Notice that we have $\tilde{H}_n(Y_1 \cap Y_2) \cong 0$ for any n , and that $\tilde{H}_n(Y_i) \cong \tilde{H}_n(X_i)$. Therefore, we have $0 \rightarrow \tilde{H}_n(X_1) \oplus \tilde{H}_n(X_2) \rightarrow \tilde{H}_n(X_1 \vee X_2) \rightarrow 0$ exact, implying that $\tilde{H}_n(X_1) \oplus \tilde{H}_n(X_2) \cong \tilde{H}_n(X_1 \vee X_2)$.

Therefore, the claim holds for any k . $\square$

Corollary 27.6. $\tilde{H}_n(X) \cong \tilde{H}_{n+1}(SX)$.

Proof. We have the following exact sequence:

$$\dots \rightarrow \tilde{H}_{n+1}(CX) \oplus \tilde{H}_{n+1}(CX) \rightarrow \tilde{H}_{n+1}(SX) \rightarrow \tilde{H}_n(X) \rightarrow \tilde{H}_n(CX) \oplus \tilde{H}_n(CX) \rightarrow \dots$$

since we have $\tilde{H}_n(CX) \cong 0$, we always have that $\tilde{H}_{n+1}(SX) \cong \tilde{H}_n(X)$, proving the statement. $\square$

Corollary 27.7. *Suppose $X = \bigcup U_\alpha$ is an open cover and all finite intersections $U_{i_1} \cup \dots \cup U_{i_k}$ are empty or have trivial homology group, then $H_k(X)$ is finitely generated and $H_k(X) \cong 0$ for $k > n$ if the cover is finite with n open subsets.*

Proof. We induct on n . The case when $n = 1$ is trivial. Suppose that the assumption holds for n and we now have $U_1, \dots, U_{n+1}$ being a good cover of X .

Let $V = U_1 \cup \dots \cup U_n$, then $X = V \cup U_{n+1}$.

By assumption we have that $H_k(V) \cong 0$ if $k > n$. Additionally, $U_{n+1} \cap U_1, \dots, U_{n+1} \cap U_n$ forms a good cover of $V \cap U_{n+1}$, so $V \cap U_{n+1}$ also admits a good cover and $H_k(V \cap U_{n+1}) \cong 0$ if $k > n$.

By Mayer-Vietoris we are able to construct an exact sequence

$$\dots \rightarrow H_k(V) \oplus H_k(U_{n+1}) \rightarrow H_{k+1}(X) \rightarrow H_k(V \cap U_{n+1}) \rightarrow \dots,$$

since the terms on the left and right of $H_k(X)$ are trivial when $k > n + 1$, we conclude that $H_k(X) \cong 0$ when $k > n + 1$. $\square$

28 The Euler Characteristic

Definition 28.1. Let X be a space with finitely generated homology groups s.t. $\beta_i = 0$ for all but finitely many i .

The **Euler characteristic** of X is defined as

$$\chi(X) = \sum (-1)^i \beta_i(X).$$

Proposition 28.2. If $X = A \cup B$ is covering such that $X = \text{Int}A \cup \text{Int}B$, then

$$\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B),$$

assuming that all the spaces have well-defined Euler characteristics.

Exercise 28.3. $\chi(\sum_g) = 2 - 2g$.

Proof. We prove the claim by induction. We know that $\chi(\mathbb{T}^2) = 0$ as we computed earlier.

We first show that $\chi(\sum_g - D^2) = 2 - 2g - 1$, assuming that $\chi(\sum_g) = 2 - 2g$. By the above proposition, we have that $\chi(\sum_g) = \chi(\sum_g - D^2) + \chi(D^2) - \chi(S^1) = \chi(\sum_g - D^2) - 1$ so $\chi(\sum_g - D^2) = 2 - 2g - 1$.

Now, we prove by induction that $\chi(\sum_g) = 2 - 2g$. Notice that $\sum_g = (\sum_{g-1} - D^2) \sqcup_{S^1} (\mathbb{T}^2 - D^2)$, so $\chi(\sum_g) = \chi(\sum_{g-1} - D^2) + \chi(\mathbb{T}^2 - D^2) - \chi(S^1) = (2 - 2g - 1) - 1 = 2 - 2(g + 1)$, as the statement claims. $\square$

To prove the above proposition, we need the following lemma:

Lemma 28.4. Let $0 \rightarrow A_1 \rightarrow \dots \rightarrow A_n \rightarrow 0$ be an exact sequence of finitely generated Abelian groups, then

$$\sum (-1)^i \text{rank}(A_i) = 0.$$

Notice that the vector space version of the lemma is quite easy to prove, as we have the Rank-Nullity theorem.

Therefore, we try to take the problem from **Ab** to **Vect**.

Remark 17. Assuming this lemma, we are able to prove the above proposition by plugging in the definition of the Euler characteristic into the Mayer-Vietoris sequence:

$$\dots \rightarrow H_n(A \cap B) \rightarrow H_n(A) \oplus H_n(B) \rightarrow H_n(X) \rightarrow \dots$$

Definition 28.5. Given G an Abelian group, its **rationalization** $\mathbb{Q} \otimes G = G \times (\mathbb{Z} - \{0\}) / \sim$, where $(g, q) \sim (g', q')$ iff $gq' = qg'$. The corresponding equivalence class is denoted as $\frac{g}{q}$.

This is a $\mathbb{Q}$ -vector space via

$$\begin{aligned} \frac{g}{q} + \frac{g'}{q'} &= \frac{q'g + g'q}{qq'} \\ \frac{p}{q} \cdot \frac{g}{q'} &= \frac{pg}{qq'} \text{ for } \frac{p}{q} \in \mathbb{Q}. \end{aligned}$$

Exercise 28.6. (i) Prove that $\mathbb{Q} \otimes G$ has the following universal property: there is a group homomorphism $i : G \rightarrow \mathbb{Q} \otimes G$ s.t. for any homomorphism $f : G \rightarrow V$, where V is a $\mathbb{Q}$ -vector space, there is a unique $\mathbb{Q}$ -linear map $\tilde{f} : \mathbb{Q} \otimes G \rightarrow V$ s.t. $\tilde{f}i = f$.

(ii) Show that the rationalization defines a functor $\mathbb{Q} \otimes - : \mathbf{Ab} \rightarrow \mathbf{Vect}_{\mathbb{Q}}$.

Proof. (i) Let $i(g) = (g, 1)$ for any $g \in G$.

Given a homomorphism $f : G \rightarrow V$, we define $\tilde{f} : \mathbb{Q} \otimes G \rightarrow V$ by $\tilde{f}(g, q) = \frac{1}{q}f(g)$.

This is indeed a linear map, as

$$\begin{aligned}
 \tilde{f}(g, q) + \tilde{f}(g', q') &= \frac{1}{q}f(g) + \frac{1}{q'}f(g') \\
 &= \frac{qf(g') + q'f(g)}{qq'} \\
 &= \frac{f(qg') + f(q'g)}{qq'} \\
 &= \tilde{f}(gq' + qg', qq') \\
 &= \tilde{f}((g, q) + (g', q'))
 \end{aligned}$$

and

$$\begin{aligned}
 \frac{p}{r}\tilde{f}(g, q) &= \frac{p}{rq}f(g) \\
 &= \tilde{f}(pg, rq) \\
 &= \tilde{f}\left(\frac{p}{r}(g, q)\right)
 \end{aligned}$$

and it clearly satisfies $\tilde{f}i = f$.

- (ii) For any Abelian group homomorphism $f : G \rightarrow H$, we define $\mathbb{Q} \otimes f : \mathbb{Q} \otimes G \rightarrow \mathbb{Q} \otimes H$ by $\mathbb{Q} \otimes f(g, q) = (f(g), q)$.

This is indeed a linear map as

$$\begin{aligned}
 \mathbb{Q} \otimes f\left(\frac{p}{r}(g, q)\right) &= \mathbb{Q} \otimes f(pg, rq) \\
 &= (f(pg), rq) \\
 &= \frac{p}{r}(f(g), q) \\
 &= \frac{p}{r}\mathbb{Q} \otimes f(g, q)
 \end{aligned}$$

and

$$\begin{aligned}
 \mathbb{Q} \otimes f(g, q) + \mathbb{Q} \otimes f(g', q') &= (f(g), q) + (f(g'), q') \\
 &= (f(g)q' + f(g')q, qq') \\
 &= (f(gq' + g'q), qq') \\
 &= \mathbb{Q} \otimes f(gq' + g'q, qq') \\
 &= \mathbb{Q} \otimes f((g, q) + (g', q'))
 \end{aligned}$$

□

Exercise 28.7. (i) $\mathbb{Q} \otimes \mathbb{Z}^r \cong \mathbb{Q}^r$.

(ii) If G is finite, then $\mathbb{Q} \otimes G \cong 0$.

(iii) $\mathbb{Q} \otimes (G \oplus H) \cong \mathbb{Q} \otimes G \oplus \mathbb{Q} \otimes H$.

(iv) If G is finitely generated then $\text{rank}(G) = \dim(\mathbb{Q} \otimes G)$.

(v) If $A \rightarrow B \rightarrow C$ is exact in **Ab**, then $\mathbb{Q} \otimes A \rightarrow \mathbb{Q} \otimes B \rightarrow \mathbb{Q} \otimes C$ is exact in **Vect** $_{\mathbb{Q}}$.

Remark 18. More generally, one can define $\chi(C) = \sum (-1)^i \text{rank}(H_i(C))$, if $H_i(C)$ is finitely generated and 0 for all but finitely many i .

Exercise 28.8. $\chi(C) = \sum (-1)^i \text{rank}(C_i)$.

Proof. Notice that we have the following exact sequence:

$$\begin{aligned} 0 &\rightarrow B_i(C) \rightarrow Z_i(C) \rightarrow H_i(C) \rightarrow 0 \\ 0 &\rightarrow Z_{i+1}(C) \rightarrow C_{i+1} \rightarrow B_i(C) \rightarrow 0. \end{aligned}$$

As a result, we have $\text{rank}(Z_i(C)) = \text{rank}(B_i(C)) + \text{rank}(H_i(C))$ and $\text{rank}(C_{i+1}) = \text{rank}(Z_{i+1}(C)) + \text{rank}(B_i(C))$.

Therefore, we have

$$\begin{aligned} \sum (-1)^i \text{rank}(C_i) &= \sum (-1)^i (\text{rank}(Z_i(C)) + \text{rank}(B_{i-1}(C))) \\ &= \sum (-1)^i (\text{rank}(B_i(C)) + \text{rank}(H_i(C)) + \text{rank}(B_{i-1}(C))) \\ &= \sum (-1)^i \text{rank}(H_i(C)) \\ &= \chi(C). \end{aligned}$$

□

29 More Homotopy Groups

Definition 29.1. Let $[S^1] \in H_1(S^1)$ be the generator. By suspension isomorphism $H_n(S^n) \cong H_{n+1}(S^{n+1})$, we have $[S^n] \in H_n(S^n) \cong \mathbb{Z}$ a generator.

For any $X \in \mathbf{Top}_*$, we define the **Hurewicz map** η to be

$$\eta : \pi_n(X) \rightarrow H_n(X)$$

with $\eta([\alpha]) = \alpha_*[S^n]$, where $\alpha_* : H_n(S^n) \rightarrow H_n(X)$ induced by α .

Exercise 29.2. The Hurewicz map is a group homomorphism.

Proof. Let $[\alpha], [\beta]$ be elements in $\pi_n(X, x_0)$.

We then have $\eta([\alpha + \beta]) = (\alpha + \beta)_*[S^n] = [(\alpha + \beta)S^n] = [\alpha S^n] + [\beta S^n] = \eta([\alpha]) + \eta([\beta])$, so η is indeed a group homomorphism.

Notice that S^n in the computation here represents a map $\Delta^n \rightarrow S^n$, instead of the n -sphere. $\square$

Theorem 29.3. Let X be a simply connected topological space. Fix $n > 1$ s.t. $\tilde{H}_k(X) = 0$ for all $k < n$, then $\eta : \pi_n(X) \rightarrow H_n(X)$ is an isomorphism.

Corollary 29.4. Hurewicz's theorem is equivalent to the following statement: Fix $n > 1$ and (X, x_0) a path connected space. If $\pi_k(X, x_0) \cong 0$ for any $k < n$, then $\eta : \pi_n(X, x_0) \rightarrow H_n(X)$ is an isomorphism.

Proof. Assume that Hurewicz's Theorem is true. If $\pi_1(X) \cong 0$, then $H_1(X) \cong 0$ as it is the abelianization of $\pi_1(X)$.

Now suppose that the $\pi_k(X, x_0) \cong 0$ for any $k < n$, then $\tilde{H}_k(X, x_0) \cong 0$ as well by repetitively using Hurewicz's theorem. Therefore, $\eta : \pi_n(X) \rightarrow H_n(X)$ is also an isomorphism by Hurewicz's theorem.

On the other hand, suppose that the statement is true and that X is simply connected, then $\pi_1(X) \cong 0$. Suppose that $\tilde{H}_k(X) \cong 0$ for any $k < n$, then $\pi_k(X, x_0) \cong H_k(X) \cong 0$ for any $1 < k < n$ by repetitively using the statement. Therefore, η is also an isomorphism by the statement.

Therefore, the statement is equivalent to the Hurewicz's Theorem. $\square$

Exercise 29.5. (i) $\pi_n(S^n) \cong \mathbb{Z}$ and $\pi_k(S^n) \cong 0$ for any $k < n$.

(ii) For any $n \geq 1$, $f, g : S^n \rightarrow S^n$ are homotopic iff $\deg(f) = \deg(g)$.

Proof. (i) This is proved by using the fact that $\tilde{H}_k(S^n) \cong 0$ for any $k < n$.

(ii) Notice that the Hurewicz map $\eta : \pi_n(S^n) \rightarrow H_n(S^n)$ is an isomorphism. Consider the following diagram:

$$\begin{array}{ccc} \pi_n(S^n) & \xrightarrow{f_*} & \pi_n(S^n) \\ \downarrow \eta & & \downarrow \eta \\ \tilde{H}_n(S^n) & \xrightarrow{f_* = g_*} & \tilde{H}_n(S^n) \\ \downarrow \eta^{-1} & & \downarrow \eta^{-1} \\ \pi_n(S^n) & \xrightarrow{g_*} & \pi_n(S^n) \end{array}$$

The top square commutes because ηf_* maps $[\gamma]$ to $[f\gamma]$ and then to $f\gamma[S^n]$, which is the same as where $f_*\eta$ maps $[\gamma]$ to. Similarly the square on the bottom commutes so the diagram commutes.

Therefore, $f_* = g_*$ and f is homotopic to g . $\square$

Theorem 29.6. Let $p : E \rightarrow B$ be a fibre bundle with fibre F . Fix $b \in B$ and $i : F \rightarrow p^{-1}(b)$ homeomorphism and $f \in F$ and $e = i(f)$. We then have the following exact sequence:

$$\dots \rightarrow \pi_1(F, f) \xrightarrow{i_*} \pi_1(E, e) \xrightarrow{p_*} \pi_1(B, b) \rightarrow \pi_0(F, f) \xrightarrow{i_*} \pi_0(E, e) \xrightarrow{p_*} \pi_0(B, b).$$

30 Cellular Homology

Definition 30.1. Let X be a CW-complex with a collection of maps $\{\varphi_\alpha : D^{n_\alpha} \rightarrow X\}_{\alpha \in I}$ satisfying some conditions.

Let $I_n = \{\alpha \in I, n_\alpha = n\}$, and let $C_n^{CW}(X) = \bigoplus_{\alpha \in I_n} \mathbb{Z}$.

Theorem 30.2. There is a chain complex $C^{CW}(X)$ with $(C^{CW}(X))_n = C_n^{CW}(X)$ and homology groups $H_n^{CW}(X)$ satisfying $H_n^{CW}(X) \cong H_n(X)$.

Moreover, such a chain complex exists and is independent of the choice of CW decomposition of X .

Exercise 30.3. $H_k(\mathbb{C}P^n) \cong \begin{cases} \mathbb{Z} & k = 0, 2, \dots, 2n \\ 0 & \text{otherwise} \end{cases}$

Proof. We know that $\mathbb{C}P^n \cong \mathbb{R}^0 \sqcup \mathbb{R}^2 \sqcup \dots \sqcup \mathbb{R}^{2n}$.

We therefore have the following cellular chain complex:

$$0 \rightarrow C_{2n} \cong \mathbb{Z} \rightarrow C_{2n-1} \cong 0 \rightarrow \dots \rightarrow C_1^{CW} \cong 0 \rightarrow C_0^{CW} \cong \mathbb{Z} \rightarrow 0.$$

Notice that the homology groups at C_{2k} is always $\mathbb{Z}/0 \cong \mathbb{Z}$ and the homology groups at C_{2k+1} is always $0/0 \cong 0$.

Therefore, by our definition of homology group of chain complexes and the above theorem, we have

$$H_k(\mathbb{C}P^n) \cong \begin{cases} \mathbb{Z} & k = 0, 2, \dots, 2n \\ 0 & \text{otherwise} \end{cases}$$

□

Corollary 30.4. If a space X admits an n -dimensional CW structure, then $H_k(X) \cong 0$ for any $k > n$.

Corollary 30.5. A finite CW complex X with n_i i -cells satisfies $\chi(X) = \sum (-1)^i n_i$.

Remark 19. The basic idea for defining the boundary maps in $C^{CW}(X)$ is by taking $\partial\alpha$ as the number of times each $(n-1)$ -cell appear in the boundary of α , where α here is an n -cell of X .

For example, the 2-cell of $\mathbb{T}^2$, α , would satisfy $\partial\alpha = a + b - a - b = 0$. This implies $H_2(\mathbb{T}^2) \cong \mathbb{Z}$.

We now try to define the boundary maps with precise language.

Lemma 30.6. Let $y : D^n/S^{n-1} \rightarrow S^n$ be a fixed homeomorphism for each $n > 0$. Recall that by definition we have $\varphi_\alpha : D^n \rightarrow X_n$ with $\varphi_\alpha(S^{n-1}) \subset X_{n-1}$. Therefore, we can define:

$$i_\alpha : S^n \xrightarrow{y^{-1}} D^n/S^{n-1} \xrightarrow{\bar{\varphi}_\alpha} X_n/X_{n-1}$$

For any $n > 0$, the quotient X_n/X_{n-1} is homeomorphic to $\bigvee_{\alpha \in I_n} S^n$. More precisely, the maps i_α induce an homeomorphism from $\bigvee_{\alpha \in I_n} S^n$ to X_n/X_{n-1} .

Proof. This is done by considering X_n/X_{n-1} as the coproduct of D^n/S^{n-1} . □

Definition 30.7. For each n -cell $\varphi_\alpha : D^n \rightarrow X_n$ with $n > 0$, we define the projection map $p_\alpha : X_n/X_{n-1} \rightarrow S^n$ to be the unique map with

$$p_\alpha \circ i_{\alpha'} \cong \begin{cases} 1_{S^n} & \alpha = \alpha' \\ c & \text{otherwise} \end{cases}$$

where c is a constant pointed map from S^n to S^n .

Definition 30.8. For each $\alpha \in I_n$ and $\beta \in I_{n-1}$ with $n \geq 2$, we define the **incidence number** $[\alpha : \beta]$ to be the degree of the map

$$S^{n-1} \xrightarrow{\varphi_\alpha|_{S^{n-1}}} X_{n-1} \xrightarrow{q} X_{n-1}/X_{n-2} \xrightarrow{p_\beta} S^{n-1}$$

where $q : X_{n-1} \rightarrow X_{n-1}/X_{n-2}$ is the quotient map.

Basically, $[\alpha : \beta]$ counts how many times β occurs in the boundary of α , as we can view X_{n-1}/X_{n-2} as a wedge of S^{n-1} .

For $n = 1$, we define $p = \varphi_\beta(D^0)$ and

$$[\alpha : \beta] = \begin{cases} 1 & \varphi_\alpha(1) = p \text{ and } \varphi_\alpha(-1) \neq p \\ -1 & \varphi_\alpha(1) \neq p \text{ and } \varphi_\alpha(-1) = p \\ 0 & \text{otherwise} \end{cases}$$

Lemma 30.9. For a fixed $\alpha \in I_n$, there are only finitely many $\beta \in I_{n-1}$ such that $[\alpha : \beta] \neq 0$.

Proof. Suppose that $U \subset \bigvee S^n$ intersects infinitely many summands of the wedge sum, then for each S^n that U intersects, we can find an open cover of $U \cap S^n$ for each S^n , and clearly the disjoint union of the coverings cover U but do not have finite subcover that covers U .

Therefore, U intersects at finitely many summands of $\bigvee S^n$.

Now, if $\varphi_\alpha(D^n) \cap \varphi_\beta(D^{n-1}) = \emptyset$, then by the above definition of p_β we have the composite of the map in **Definition 30.8** being a constant map, and hence have degree 0. $\square$

Definition 30.10. For $n > 0$, the boundary map $\partial = \partial^{(n)} : C_n^{CW}(X) \rightarrow C_{n-1}^{CW}(X)$ on generators $\alpha \in I_n$ by the formula:

$$\partial\alpha = \sum [\alpha : \beta]\beta.$$

Remark 20. It is difficult to show that this indeed defines a chain complex, but the idea is to first notice that given a Δ -set K , its geometric realization $|K|$ has a CW structure in which the n -cells are indexed by K_n . Consequently, we have $C_n(K) \cong C_n^{CW}(|K|)$, as both are free Abelian on K_n , and we can show that the cellular homology of $|K|$ is the same as the simplicial homology of K .

Corollary 30.11. Suppose that X is homeomorphic to the geometric realization of a Δ -set K , then we have

$$H_n(K) \cong H_n^{CW}(|K|) \cong H_n(X).$$

Proposition 30.12.

$$H_k(X_n, X_{n-1}) \cong \begin{cases} \bigoplus_{\alpha \in I_n} \mathbb{Z} & k = n \\ 0 & \text{else} \end{cases}$$

Exercise 30.13. (i) $H_n(X_k) = 0$ for $n > k$.

(ii) $i_* : H_n(X_k) \rightarrow H_n(X_{k+1})$ is an isomorphism for any $n < k$, where i_* is induced by the inclusion map $i : X_k \rightarrow X_{k+1}$.

(iii) If X is finite dimensional, then $i_* : H_n(X_k) \rightarrow H_n(X)$ is an isomorphism for any $n < k$, where i_* is induced by $i : X_k \rightarrow X$.

Proof. Both (i) and (ii) come directly by induction and the following long exact sequence:

$$\dots \longrightarrow H_n(X_k) \xrightarrow{i_*} H_n(X_{k+1}) \xrightarrow{q_*} H_n(X_{k+1}, X_k) \xrightarrow{\partial} H_{n-1}(X_k) \longrightarrow \dots$$

(iii) comes by induction on (ii), as $H_n(X_m) \cong H_n(X)$ if X is m -dimensional. $\square$

Proposition 30.14. Define $G_n : H_n(X_n, X_{n-1}) \rightarrow C_n^{CW}(X)$ to be the isomorphism. We then have the following diagram commutes:

$$\begin{array}{ccccc}
 H_n(X_n, X_{n-1}) & \xrightarrow{\partial} & H_{n-1}(X_{n-1}) & \xrightarrow{q_*} & H_{n-1}(X_{n-1}, X_{n-2}) \\
 \downarrow G_n & & & & \downarrow G_{n-1} \\
 C_n^{CW}(X) & \xrightarrow{\quad \partial \quad} & C_{n-1}^{CW}(X) & &
 \end{array}$$

Theorem 30.15. *Let C be the chain complex $C_n = H_n(X_n, X_{n-1})$ and the boundary map given in the above proposition, then C is indeed a chain complex.*

Proof. We can show that $\partial \circ \partial = 0$ by considering the following diagram:

$$\begin{array}{ccccccc}
 & & & & & & 0 \\
 & & & & & \nearrow & \\
 & & & & H_{n-1}(X_n) & & \\
 & & \nearrow & & \searrow & & \\
 0 & & & H_{n-1}(X_{n-1}) & & & \\
 \nearrow & & \searrow & & \nearrow & & \\
 H_n(X_n, X_{n-1}) & & H_{n-1}(X_{n-1}, X_{n-2}) & & H_{n-2}(X_{n-2}, X_{n-3}) & & \\
 \nearrow & & \searrow & & \nearrow & & \\
 \dots & & & H_{n-2}(X_{n-2}) & & & \\
 & & \nearrow & & \searrow & & \\
 & & 0 & & & &
 \end{array}$$

where all the diagonals are exact, as we have shown. Therefore, if we plug C_n 's in we have $\partial \circ \partial = 0$. $\square$

Exercise 30.16. *Show that*

$$H_k(\mathbb{R}P^n, \mathbb{Z}/2\mathbb{Z}) \cong \begin{cases} \mathbb{Z}/2\mathbb{Z} & 0 \leq k \leq n \\ 0 & \text{otherwise} \end{cases}$$

Proof. Notice that $\mathbb{R}P^n$ has a cell-structure with exactly 1 k -cell for any $0 \leq k \leq n$. Therefore, we have the following exact sequence:

$$0 \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow \dots \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

and by induction we can show the homology groups of $\mathbb{R}P^n$ with coefficients in $\mathbb{Z}/2\mathbb{Z}$ are as stated. $\square$

31 Cohomology & Cup Product

Definition 31.1. Fix an Abelian group G , a **cohomology theory with coefficients in G** consists of a sequence of functors $H^n : \mathbf{Top}_2^{\text{op}} \rightarrow \mathbf{Ab}$ and homomorphisms $\delta_{(X,A)}^{(n)} : H^n(A) \rightarrow H^{n+1}(X, A)$, such that

(i) For any $(X, A) \in \mathbf{Top}_2$, the following sequence is exact:

$$\dots \rightarrow H^n(X, A) \rightarrow H^n(X) \rightarrow H^n(A) \xrightarrow{\delta} H^{n+1}(X, A) \rightarrow \dots,$$

where $H^n(X) \rightarrow H^n(A)$ and $H^n(X, A) \rightarrow H^n(X)$ are induced by inclusions $i : A \rightarrow X$ and $j : (X, \emptyset) \rightarrow (X, A)$.

(ii) The following square commutes for any $f : (X, A) \rightarrow (Y, B)$ and $n \in \mathbb{Z}$:

$$\begin{array}{ccc} H^n(B) & \xrightarrow{\delta} & H^{n+1}(Y, B) \\ \downarrow f^* & & \downarrow f^* \\ H^n(A) & \xrightarrow{\delta} & H^{n+1}(X, A) \end{array}$$

(iii) $H^n(\{x\}) \cong G$ if $n = 0$ and 0 if $n \neq 0$.

(iv) If $f, g : (X, A) \rightarrow (Y, B)$ are homotopic, then $f^* = g^* : H^n(Y, B) \rightarrow H^n(X, A)$ for any $n \in \mathbb{Z}$.

(v) For any $U \subset A$ such that $\bar{U} \subset \text{Int}(A)$, we have an isomorphism $H^n(X - U, A - U) \cong H^n(X, A)$ induced by $i : (X - U, A - U) \rightarrow (X, A)$.

(vi) $H^n(\bigsqcup X_\alpha) \cong \prod H^n(X_\alpha)$.

Remark 21. If one replaces homotopy invariance by weak homotopy invariance, then the cohomology theory is unique up to isomorphism.

In other words, if we replace homotopy invariance by the statement that if $f, g : X \rightarrow Y$ are weak homotopy equivalent, (i.e., $f_*, g_* : \pi_n(X, x_0) \cong \pi_n(Y, y_0)$ is an isomorphism for any $n \geq 1$), then $f^* = g^* : H^n(X, A) \rightarrow H^n(Y, B)$.

Definition 31.2. Given an Abelian group G , its **dual** $G^\vee$ is defined as $\text{Hom}(G, \mathbb{Z})$ under pointwise addition.

Clearly any homomorphism $G \rightarrow H$ induces a homomorphism $H^\vee \rightarrow G^\vee$.

Exercise 31.3. This defines a functor $(-)^\vee : \mathbf{Ab}^{\text{op}} \rightarrow \mathbf{Ab}$.

Remark 22. For a chain complex C , we wish to define its **dual chain complex** $C^\vee$ where $(C^\vee)_n = (C_n)^\vee$. However, this causes problem as the boundary map goes in the wrong direction $(C^\vee)_n = (C_n)^\vee \rightarrow (C_{n+1})^\vee = (C^\vee)_{n+1}$.

One way to fix the problem is to introduce the **cochain complex** where the boundary maps go from C^n to C^{n+1} . $C^\vee$ is thus a cochain complex.

Another way to fix it is simply by claiming that $D^n = D_{-n}$ for any chain complex D . Therefore, if we let $(C^\vee)^n = (C_n)^\vee$, then $C^\vee$ becomes a well-defined chain complex.

In a chain complex D , the **cocycles**, **coboundaries**, **cohomology classes** $Z^n(D)$, $B^n(D)$, $H^n(D)$ are defined as $Z_{-n}(D)$, $B_{-n}(D)$, and $H_{-n}(D)$.

Definition 31.4. Given a chain complex C , its **dual chain complex** $C^\vee$ is defined as $(C^\vee)^n = (C_n)^\vee$. The boundary maps are

$$(\partial^{(n)})^\vee = \delta^{(n)} : (C^\vee)^n = (C_n)^\vee \rightarrow (C_{n+1})^\vee = (C^\vee)^{n+1}$$

Exercise 31.5. Given a chain map $f : C \rightarrow D$, we have the induced chain map $f^\vee : D^\vee \rightarrow C^\vee$. Therefore, $(-)^\vee$ is a functor from $\mathbf{Ch}^{\text{op}} \rightarrow \mathbf{Ch}$.

Proof. We define $(f^\vee)^n : (D^\vee)^n \rightarrow (C^\vee)^n$ as the dual map of $f_n : C_n \rightarrow D_n$.

We thus only need to check that the following diagram commute, which comes due to the fact that $(-)^\vee$ is a contravariant from $\mathbf{Ab}$ to itself, so $f\partial = \partial f$ implies $\delta f^\vee = \partial^\vee f^\vee = f^\vee \partial^\vee = f^\vee \delta$.

$$\begin{array}{ccc} (D^\vee)^n & \xrightarrow{(f^\vee)^n} & (C^\vee)^n \\ \downarrow \delta^{(n)} & & \downarrow \delta^{(n)} \\ (D^\vee)^{n+1} & \xrightarrow{(f^\vee)^{n+1}} & (C^\vee)^{n+1} \end{array}$$

□

Definition 31.6. For a topological space X , we define its **group of singular n -cochains**, $C^n(X)$, as the dual group $(C_n(X))^\vee$, and the **singular cochain complex** is the dual chain complex $C^\vee$.

The n th **singular cohomology group** is defined as $H^n(X) = H^n(C(X)^\vee)$. Note that this defines a contravariant functor as follows

$$\mathbf{Top}^{op} \rightarrow \mathbf{Ch}^{op} \rightarrow \mathbf{Ab}.$$

Everything is defined analogously for a topological pair (X, A) .

Remark 23. As $C_n(X)$ is the free Abelian group generated by $S_n(X)$, its dual $C^n(X)$ consists of all maps from $S_n(X)$ to $\mathbb{Z}$.

Remark 24. Notice that since the boundary map in singular chains are defined as

$$\partial\sigma = \sum (-1)^i \partial_i \sigma,$$

where $\partial_i \sigma = \sigma \circ \delta_i$.

Therefore, by our definition, we have $\delta : C^n(X) \rightarrow C^{n+1}(X)$ defined as by sending every map $f : S_n(X) \rightarrow \mathbb{Z}$ to $f^* : S_{n+1}(X) \rightarrow \mathbb{Z}$, with f^* defined as

$$f^*(\sigma) = f(\sum (-1)^i \partial_i \sigma) = \sum (-1)^i f(\sigma| [x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n]).$$

Remark 25. Given an Abelian group G , we can consider $H^\vee = \text{Hom}(H, G)$ instead of $\text{Hom}(H, \mathbb{Z})$. If $G = k$ is a field, then $\text{Hom}(H, k)$ is a vector space over k .

By repeating everything above for $H^\vee = \text{Hom}(H, G)$, we have **singular cohomology with coefficients in G** , $H^n(X, A; G)$. If k is a field, this defines a functor $H^n : \mathbf{Top}_2^{op} \rightarrow \mathbf{Vect}_k$.

Theorem 31.7. Singular cohomology with coefficients in G is a cohomology theory with coefficients in G .

Theorem 31.8. There is also a Mayer-Vietoris sequence in cohomology:

$$\dots \rightarrow H^{n-1}(A \cap B) \rightarrow H^n(X) \rightarrow H^n(A) \oplus H^n(B) \rightarrow H^n(A \cup B) \rightarrow H^{n+1}(X) \rightarrow \dots$$

Definition 31.9. By definition, we have the following evaluation $\mathbb{Z}$ -bilinear map:

$$C^n(X, G) \times C_n(X) \rightarrow G,$$

where (f, c) is mapped to $f(c)$.

Exercise 31.10. Prove that the evaluation map induces a well-defined $\mathbb{Z}$ -bilinear map

$$H^n(X; G) \times H_n(X) \rightarrow G,$$

where $([f], [c])$ is mapped to $f(c)$. Show that this is indeed well-defined and gives a homomorphism

$$H^n(X; G) \rightarrow \text{Hom}(H_n(X), G).$$

Theorem 31.11. The map $H^n(X; \mathbb{Q}) \rightarrow \text{Hom}(H_n(X), \mathbb{Q})$ is an isomorphism of $\mathbb{Q}$ -vector spaces.

In particular, if $H_n(X)$ is finitely generated Abelian group of rank r , then $H^n(X; \mathbb{Q}) \cong \mathbb{Q}^r$, and hence $\dim H^n(X; \mathbb{Q}) = \beta_r$.

Proof. The special case is proved by noticing that the set of group homomorphisms from $H_n(X) \cong \bigoplus \mathbb{Z}$ to $\mathbb{Q}$ is the same as a $\bigoplus \text{Hom}(\mathbb{Z}, \mathbb{Q})$, which is $\bigoplus \mathbb{Q} = \mathbb{Q}^r$. □

Definition 31.12. Given a singular n -simplex $\sigma : \Delta^n \rightarrow X$ and any $0 \leq p \leq n$, the **front face** $p \rfloor \sigma : \Delta^p \rightarrow X$ is defined to be the restriction of σ onto Δ^p along the map $\alpha : [p] \rightarrow [n]$ in $\Delta_{in,j}$, where $\alpha(i) = i$. In other words, $p \rfloor \sigma = \sigma i_* : \Delta^p \rightarrow X$.

The **back face** $\sigma \rfloor_p : \Delta^p \rightarrow X$ is defined as σj_* , where $j : [p] \rightarrow [n]$ is defined by $j(i) = n + i - p$.

Given singular cochains $f \in C^k(X)$, $g \in C^l(X)$, their **cup product** $f \cup g \in C^{k+l}$ is defined as

$$(f \cup g)(\sigma) = f(k \rfloor \sigma) \cdot g(\sigma \rfloor_l) \in \mathbb{Z}$$

where $\sigma \in S_{k+l}(X)$.

Lemma 31.13. Given $f \in C^k(X)$ and $g \in C^l(X)$, show that $\delta(f \cup g) = (\delta f) \cup g + (-1)^k f \cup (\delta g)$.

Proof. Notice that $\delta : C^k(X) \rightarrow C^{k+1}(X)$ is the map where $f \in C^k(X)$ is mapped to $f\partial \in C^{k+1}(X)$. Therefore, $\sigma \in C_{k+l}(X)$ is mapped to $f\partial\sigma$.

$$\begin{aligned} \text{We have } (\delta f) \cup g &= \sum_{i=0}^{k+1} (-1)^i f(\sigma \rfloor [v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_{k+1}]) \cdot g(\sigma \rfloor [v_{k+1}, \dots, v_{k+l+1}]). \text{ On the other hand,} \\ (-1)^k f \cup (\delta g) &= \sum_{i=k}^{k+l+1} (-1)^i f(\sigma \rfloor [v_0, \dots, v_k]) \cdot g(\sigma \rfloor [v_k, \dots, v_{i-1}, v_{i+1}, \dots, v_{k+l+1}]). \end{aligned}$$

When we add these two expressions, the last term of the first sum and the first term of the second sum cancel out. The resulted sum is the same as $\delta(f \cup g)(\sigma)$ as $\delta(f \cup g)(\sigma) = (f \cup g)(\partial\sigma) = (f \cup g)(\sum_{i=0}^{k+l+1} (-1)^i \sigma \rfloor [v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_{k+l+1}])$. $\square$

Corollary 31.14. The cup product induces a $\mathbb{Z}$ -bilinear map $H^k(X) \times H^l(X) \rightarrow H^{k+l}(X)$, where $([f], [g])$ is mapped to $[f \cup g]$.

Theorem 31.15. The resulting operation makes $\bigoplus H^k(X)$ into a ring.

Theorem 31.16. Notice that for a compact oriented manifold, we can define its fundamental class $[M] \in H_n(M; \mathbb{Q})$. Therefore, this gives maps $H^n(M; \mathbb{Q}) \rightarrow \mathbb{Q}$ by evaluating on $[M]$.

Composing this with cup product, we have a bilinear map:

$$H^k(X; \mathbb{Q}) \times H^{n-k}(X; \mathbb{Q}) \rightarrow \mathbb{Q}$$

We have this map non-degenerative, i.e., the induced map

$$H^k(M; \mathbb{Q}) \rightarrow \text{Hom}_{\mathbf{Vect}_{\mathbb{Q}}}(H^l(X; \mathbb{Q}), \mathbb{Q})$$

is an isomorphism.

In particular, $\beta_k = \dim H^k(M; \mathbb{Q}) = \dim H^{n-k}(M; \mathbb{Q}) = \beta_{n-k}$, hence indicating Poincaré's Duality.

Theorem 31.17. For any spaces X and Y , we have

$$H_n(X \times Y) = \bigoplus_{k+l=n} H_k(X; \mathbb{Q}) \otimes_{\mathbb{Q}} H_l(Y; \mathbb{Q})$$

for any $n \geq 0$. In particular, if X and Y both have finitely generated homology groups, then so does $X \times Y$, and $\beta_n(X \times Y) = \sum_{k+l=n} \beta_k(X) \cdot \beta_l(Y)$.

Corollary 31.18. $\beta_k(\mathbb{T}^n) = \binom{n}{k}$.

Proof. This comes directly from the above theorem and the fact that $\mathbb{T}^n = \prod_{i=1}^n S^1$. $\square$